BloodLedger: An Iot-Enabled Consortium Blockchain Framework for Inter-Hospital Blood Inventory and Redistribution Management in Lipa City

A Thesis Presented to
The Faculty of the College of Information Technology and Engineering
De La Salle Lipa

In fulfillment
of the requirements for the Degree
Bachelor of Science in Computer Science

by

Buno, Mikayla Loera Marie M.
Jopia, Luis Antonio M.
Lat, Yuri Benjamin Y.

April 2026

Table of Contents

List of Figures

List of Tables

CHAPTER 1

BACKGROUND OF THE STUDY

Introduction

Blood banks have a crucial role in healthcare systems because they act as the lifeline for patients during emergencies, surgeries, trauma care, or life-threatening conditions (World Health Organization [WHO], 2023). These facilities do not only store blood components; they conduct rigorous screening, cross-matching, continuous inventory monitoring, and urgent transferring of blood bags occurring several times a day. The efficiency ibn blood bank operations determines how fast the transfusion can be administered in a hospital. Accordingly, having compatible blood on hand is as critical as the medical intervention itself, as recent data on hemorrhaging trauma and surgical emergencies demonstrates that survival probability drops continuously with mortality increasing by 2% for every single minute a Whole Blood transfusion is delayed (Hosseinpour et al., 2023; Torres et al., 2024). Therefore, a coordinated, transparent, and responsive blood management system across institutions is vital in ensuring equal and balanced access to these perishable, life-saving resources.

Understanding the complexity of this supply chain demands recognizing that blood is rarely treated as a uniform commodity. Upon collection, whole blood is immediately processed and separated into specialized biological components, primarily Packed Red Blood Cells (PRBCs), platelets, and fresh frozen plasma to maximize its therapeutic utility for different patient needs. Each component needs different storage environments and strict expiration lifespans. For instance, while frozen plasma can be stored for up to a year, refrigerated red blood cells expire within 35 to 42 days, and room-temperature platelets perish in only 5 days (American Association of Blood Banks [AABB], 2024). Consequently, the daily inventory process is highly complicated since it requires recording the exact receipt of units from main hubs, continuously monitoring storage, and updating statuses when units are reserved for patient cross-matching. The inventory system must flawlessly register the endpoint of every single bag, whether it is dispatched for an internal hospital transfusion, transferred to an outside facility during an emergency, or disposed of due to expiration. Evidently, the consequences of poor tracking are critical, with global estimates showing that 5 million of the 96 million annual blood donations are completely wasted due to expiration, supply chain gaps, and human error (ISBT Working Party on Blood Supply Management, 2023; World Health Organization [WHO], 2023). Since these life-saving products are highly perishable, this multi-step lifecycle makes blood inventory more complex than standard hospital logistics, so any delay in recording these transactions directly leads to critical wastage.

Globally, the practices for managing blood inventory range from highly vulnerable manual methods to advanced, decentralized digital systems. In many developing regions, hospitals still operate at a basic level while relying entirely on paper-based logbooks or standalone spreadsheets that are only updated periodically. For example, a 2024 report by WHO evaluating national blood transfusion centers in Yemen highlighted that the reliance on manual ledgers causes critical delays in sourcing blood during emergencies. Similarly, a 2024 cross-sectional study of hospitals in Abia State, Nigeria, revealed that 100% of the assessed healthcare institutions still maintain their blood bank records manually, so they lack computerized tracking systems (Uche et al., 2024). These manual systems pose a high risk of human error and create “ghost stock,” leading to supply-demand mismatches and critical emergency delays.

As an upgrade, well-funded institutions in developed countries adopt localized Blood Bank Information Systems (BBIS) to reduce internal transcription errors. Current commercial software used in first-world healthcare systems like SCC SoftBank (used by Stanford Health Care) and Haemonetics SafeTrace Tx (widely integrated across the Epic Electronic Health Record network, a leading comprehensive, cloud-based platform for large hospitals that covers over 325 million patients) provide highly efficient features such as expiration monitoring, barcode-driven stock tracking, and inventory sharing across hospitals within the same corporate network (Epic Systems Corporation, n.d.; Haemonetics Corporation, 2024; SCC Soft Computer, n.d.). However, they prevent real-time inventory visibility and automated transfer management with outside, unaffiliated hospitals because these databases remain isolated in the enterprises.

Consequently, effectively managing the blood supply requires moving away from isolated systems toward a shared network that links different hospitals together.

To address this inter-institutional fragmentation, developed healthcare networks have implemented centralized national web platforms such as Australia's BloodNet (National Blood Authority, n.d.) and India's e-RaktKosh (Centre for Development of Advanced Computing, 2016) which successfully aggregate supply data across hundreds of districts. The Kundu et al. (2024) study analyzed how e-RaktKosh maps the blood supply for 660 million people across Northern India using geospatial evaluations. The results demonstrated massive inter-hospital compliance, with 806 out of 824 regional blood banks successfully reporting their inventory data to the centralized public database. By connecting different hospitals into a single system, these platforms provide a unified view of available blood to significantly reduce the time spent searching for compatible units during emergencies. Furthermore, to enhance the tracking capabilities within the large networks, hospitals rely on the global regulatory standard of ISBT 128 barcoding to give every blood bag a universally readable identity (International Council for Commonality in Blood Banking Automation [ICCBBA], 2020). This standardization acts as a massive aid in tracking, ensuring that blood types and expiration dates are accurately categorized across different facilities.

As a strategy to the advancement of inventory management, the transition away from manual data entry in healthcare research is being driven by the integration of Internet of Things (IoT) and blockchain technology. Physical barcodes act as the link between physical blood bags and the digital ledger. By utilizing IoT barcode scanners to automatically register the movement of blood bags, facilities can guarantee real-time inventory updates without burdening the medical staff with manual data entry (Varghese et al., 2024). Furthermore, integrating these automated scanners with a blockchain ensures that the scanned barcode transaction is instantly and securely shared across a decentralized network. For example, the Government of India's National Informatics Centre (NIC) initiated a blockchain-based pilot that utilizes a permissioned distributed ledger where barcode scans track blood properties, such as blood group and storage temperature, directly from regional collection centers to government hospitals (Centre of Excellence in Blockchain Technology, n.d.). Similarly, in Nigeria, the health technology enterprise LifeBank successfully deployed SmartBag, a blockchain-enabled tagging system. This innovation attaches a unique digital barcode identity to each blood unit, allowing IoT sensors to automatically log its safety history, transit path, and cold-chain temperature onto a tamper-proof ledger as it travels to the receiving hospital (Faheem and Dutta, 2023). However, while these real-world innovations successfully secure vertical distribution (i.e., tracking blood from the main hub or collection center to the hospital), they do not open the sharing of inventory between independent, unaffiliated hospitals. Ultimately, achieving a truly reliable inventory system requires extending this barcode-driven, tamper-proof network to empower direct, inter-institutional coordination.

In the Philippine context, the Department of Health (DOH) and the Philippine Red Cross (PRC) are mandated to ensure a safe, adequate, and accessible blood supply under the National Blood Services Act of 1994 (Republic Act No. 7719). However, this national directive is heavily undermined by severe logistical fragmentation across the country's healthcare network. While the World Health Organization (WHO) mandates a minimum blood collection rate equivalent to 1% of the total population to maintain an adequate national supply, the Philippines continually struggles to meet this baseline, leading to critical regional shortages (Mappala et al., 2023; World Health Organization [WHO], 2023). Compounding this physical scarcity is a continuous reliance on manual, isolated hospital databases. The recent filing of House Bill No. 1283 in 2025 explicitly acknowledged this systemic flaw, noting that Philippine hospitals continue to struggle with delayed deliveries and critical wastage due to the severe lack of centralized tracking and real-time inventory visibility (House of Representatives, 2025). Research by Yap and Fedoc-Minguito (2023), who analyzed blood supply chain vulnerabilities in the country, emphasizes that these systemic inefficiencies in monitoring and redistribution directly lead to preventable wastage and unlocated supply issues. Furthermore, this technological gap actively shifts the burden of inventory searching onto the patients' families. Highlighting this systemic failure specifically within the Calabarzon region, the DOH-Center for Health Development recently issued Circular 2024-0406 to address rising complaints. Medical personnel noted that due to the lack of inter-hospital coordination, relatives of patients are frequently forced to physically travel between unconnected facilities searching for compatible blood, a dangerous practice that risks blood spoilage and fatal delays (Philippine Information Agency, 2025). Consequently, the lack of a unified, real-time platform among Philippine hospitals severely limits their ability to proactively redistribute near-expiry units and coordinate emergency transfusions.

Operating as the primary driving force behind the national mandate, the Philippine Red Cross (PRC) serves as the largest and most critical pillar of the country's blood supply network. To sustain the medical demands of the country, the PRC provides life-saving blood products to hundreds of patients daily (Manila Bulletin, 2024; Philippine Red Cross, 2023). Through its National Blood Services division, the organization manages an expansive infrastructure of regional blood centers and collection units. Recognizing the severe inefficiencies of manual record-keeping, various regional chapters across the Philippines have actively begun transitioning toward digital blood bank management systems to handle donor registration, laboratory screening, and internal inventory (Eder, 2024; Pabellano et al., 2023). However, regardless of the specific software deployed by these regional chapters, the PRC's overall technological capabilities remain inherently bounded by its own enterprise walls. While their internal digital networks efficiently track inventory from the point of collection to the point of dispatch, they lack an interoperable connection with the independent databases of the diverse hospitals they supply. Consequently, once a blood bag leaves any regional PRC facility, it drops out of their system's visibility and enters the fragmented hospital network. This architectural disconnect severs the supply chain's audit trail, preventing the national hub from monitoring real-time hospital utilization, tracking inter-hospital transfers, or rescuing near-expiry units before they are wasted.

Narrowing down from this national landscape, the Philippine Red Cross (PRC) - Lipa City Chapter operates as a primary blood collection and distribution hub for local healthcare institutions. Mirroring the national push for digitalization, PRC Lipa recently modernized its operations. As of the previous year, the branch transitioned away from traditional paper forms and localized Google Sheets, adopting a new, nationwide PRC information system. This centralized platform effectively manages donor registration, laboratory screening, internal inventory, and outbound distribution. It utilizes a role-based access architecture where local branch personnel manage daily logistical inputs, while national administrators retain access to broader statistical data. Because of this streamlined internal processing and the region's high volume of requests, preliminary consultations with PRC Lipa confirm that blood expiration is rarely an issue at their facility; rather, their primary operational hurdle is chronic supply depletion. However, while this newly deployed system optimizes internal workflows, it suffers from the exact same critical post-release blind spot observed at the national level. Once a blood unit is dispatched from the PRC hub to a requesting Lipa City hospital, digital tracking completely ceases. The PRC system lacks an interoperable mechanism to trace the lifecycle of the blood bag beyond its physical release, leaving the hub entirely unaware of whether the unit was successfully transfused to a patient, allowed to expire at the receiving facility, or informally transferred to another hospital. Consequently, despite having a modernized internal database, the absence of an end-to-end tracking integration perpetuates a fragmented audit trail across the local healthcare network (Conjares et al., 2025).

Once blood units transition from the regional PRC hub into the localized control of Lipa City's healthcare institutions, the tracking network completely degrades into manual and fragmented processes. Initial consultations with primary local suppliers, the hospitals equipped with their own internal blood banks, reveal that inventory management remains heavily reliant on vulnerable, paper-based methodologies. As noted by the Blood Bank Section Head of Mary Mediatrix Medical Center, the absence of an automated local barcode scanning system forces laboratory personnel to maintain inventory through manual logbooks. Consequently, during peak operational hours or critical emergencies, staff frequently fail to manually deduct utilized units from the ledger, instantly creating dangerous discrepancies between the physical stock and the recorded inventory. Furthermore, while these hospitals attempt to digitize their regulatory reporting to the DOH using cloud-based spreadsheets, this data entry is performed asynchronously at static intervals (e.g., 9:00 AM and 4:00 PM). This periodic batch-processing completely fails to capture real-time supply fluctuations, leaving the digital ledger persistently outdated. Compounding this internal blindness is the severe lack of formalized inter-hospital coordination. According to the Chief Medical Technologist at N.L. Villa Memorial Center, hospitals are forced to rely on informal, consumer-grade communication tools such as phone calls or Messenger group chats to broadcast emergency blood requests to other facilities. Because laboratory personnel are actively engaged in critical medical tasks, checking mobile devices for external chat notifications requires disruptive context switching, resulting in urgent supply requests frequently going completely unnoticed. As a direct result of these localized tracking failures and reactive redistribution methods, staff confirmed instances of preventable blood wastage, where life-saving units inevitably expire within the vaults due to low internal demand and an inability to locate external buyers in time. Evidently, the absence of an automated, interoperable tracking system leaves Lipa City's hospital blood network highly susceptible to inventory mismanagement, communication breakdowns, and fatal emergency delays.

The logistical bottlenecks observed in local supplier hospitals create cascading failures for the end consumers of this supply chain: smaller Lipa City healthcare facilities that do not maintain their own internal blood banks. Because these institutions lack the infrastructure and regulatory capacity to store blood long-term or process donations, they are entirely dependent on immediate, ad-hoc transfers from the Lipa PRC hub or larger tertiary hospitals. However, without access to an interoperable, inter-hospital digital dashboard, these consumer facilities operate completely blind to the real-time stock levels of their potential local suppliers. When a critical patient requires a transfusion, medical personnel are forced into a desperate, trial-and-error procurement process. Instead of digitally reserving a unit, staff, or frequently, the patients' distressed relatives, must manually contact multiple independent facilities via phone calls, or physically travel between hospitals simply to inquire about blood availability. This fragmented sourcing method not only consumes critical hours but also severely delays urgent medical interventions. In emergencies where every minute continuously increases mortality risk, the inability to instantly locate and secure compatible blood components translates directly to compromised patient care. Ultimately, for hospitals without dedicated blood banks, the absence of a transparent, real-time tracking network transforms what should be a seamless inter-institutional transfer into a chaotic, life-threatening race against time.

Despite the gradual push for digitalization in healthcare, a critical architectural gap persists in the way blood inventory is managed across independent institutions. Current technological interventions, whether they are localized Blood Bank Information Systems (BBIS) or the DOH’s centralized cloud spreadsheets, fundamentally operate as isolated data systems. Because these traditional databases are strictly designed for internal enterprise management, they lack the inherent interoperability required to foster trust and data-sharing between unaffiliated hospitals (Access Partnership, 2025). Furthermore, as observed in both global innovations and the Philippine setting, existing platforms prioritize vertical supply chains—tracking blood from a main hub down to a receiving hospital. They fail to provide a decentralized framework that automates the direct, lateral transfer of physical inventory between peer hospitals. Compounding this software limitation is a severe hardware-to-digital disconnect. Because current systems rely heavily on manual data encoding performed asynchronously by laboratory staff, the databases are perpetually vulnerable to human error and delayed updates, resulting in the life-threatening "ghost stock" phenomenon. Without the integration of Internet of Things (IoT) devices, such as automated barcode scanners acting as a real-time bridge between the physical blood bag and the digital ledger, inventory records will always lag behind physical realities (Suganya et al., 2026). Therefore, the major technological gap lies in the absence of an integrated, trustless network. Resolving the fragmented blood supply chain requires moving beyond vulnerable, centralized databases and deploying a decentralized consortium architecture, one that securely links independent healthcare entities, automates real-time inventory visibility through IoT scanning, and maintains an immutable, end-to-end audit trail for every inter-hospital transfer.

To directly resolve these architectural vulnerabilities, this study proposes the development of an IoT-Enabled Consortium Blockchain Framework designed specifically for inter-hospital blood inventory and transfer management in Lipa

City. By shifting away from isolated data silos, this framework leverages a permissioned blockchain network (e.g., Hyperledger Fabric) to establish a decentralized, synchronized ledger shared equally among the Philippine Red Cross (PRC), primary hospital suppliers, and local clinics (Kotian et al., 2024). The primary IT benefit of this architecture is the establishment of real-time, trustless visibility. Instead of relying on asynchronous spreadsheets and informal messaging apps, the consortium network grants all participating facilities instant access to the region's available blood supply, drastically accelerating emergency sourcing. Furthermore, by integrating IoT barcode and QR scanners at the edge of the network, the system automates physical-to-digital data entry. This entirely eliminates the reliance on manual logbooks, prevents "ghost stock" discrepancies during peak hours, and ensures that the digital inventory perfectly mirrors the physical reality. Additionally, the framework utilizes programmable smart contracts integrated with a Blood-Redistribution Optimization Algorithm to automate complex logistical decisions, such as flagging near-expiry units and recommending proactive, lateral redistribution to nearby facilities to minimize wastage. By requiring IoT scan confirmations equipped with geo-tagging capabilities at both the dispatch and receiving points, the blockchain permanently records every transaction, completely eliminating the PRC's post-release blind spot (Ajayi et al., 2024). Ultimately, this integration of IoT edge devices and immutable blockchain technology provides a secure, end-to-end audit trail, transforming Lipa City's fragmented blood network into a highly interoperable, transparent, and automated supply chain capable of saving lives.

Problem Statement

The main problem that this study will address is the inefficient and disconnected blood inventory management among Lipa City Hospitals. Specifically, the following problems will be addressed:

1. How will the integration of an IoT-based scanning system address recording discrepancies and improve inventory monitoring to instantly identify out-of-stock and overstock scenarios?

2. How will a unified, real-time dashboard improve the communication aspect of transactions between hospitals, specifically in streamlining blood requests, coordinating operations, and automating report generation.

3. How will a consortium blockchain framework establish a tamper-proof audit trail and utilize geo-tagging to resolve the fragmentation of custody records and lack of verification in inter-hospital blood transfers?

4. How will the implementation of a Blood Redistribution Optimization Algorithm in a proactive notification mechanism minimize preventable blood wastage by automatically detecting and recommending the redistribution of near-expiry blood units?

Objectives of the Study

To develop an IoT-Enabled Consortium Blockchain Framework that synchronizes blood inventory in real-time and secures inter-hospital transfers in Lipa City.

1. To develop an IoT-based scanning system that will enforce immediate digital recording of blood units and provide real-time inventory monitoring to track out-of-stock and overstock levels.

2. To create a centralized dashboard that will enhance the communication aspect of blood transactions, allowing hospitals to seamlessly initiate requests, manage operations, and automate report generation such as daily summaries to the DOH CALABARZON.

3. To implement a Consortium Blockchain with smart contracts that will secure inter-hospital blood transfers by recording every step of the transit process using geo-tagging, ensuring a tamper-proof and traceable location history.

4. To design the implementation of a Blood Redistribution Optimization Algorithm that will proactively detect near-expiry units and suggest optimal redistribution routes to other hospitals, ensuring resource optimization and reducing wastage.

Scopes and Limitation

This study focuses on the design and development of a prototype IoT-Enabled Consortium Blockchain Framework for improving inter-hospital blood inventory monitoring and transfer tracking among all hospitals in Lipa City. The proposed framework connects all hospitals within Lipa City and the Philippine Red Cross Lipa Chapter as members of a permissioned consortium network. Within this network, the Philippine Red Cross Lipa Chapter functions as the primary collection and distribution hub. Mary Mediatrix Medical Center, Medix Medical Center, and N.L. Villa Memorial Medical Center operates as participating hospitals with active blood banks, while the remaining hospitals in Lipa City function as recipient institutions that request and receive blood units through the shared system. The system incorporates QR/barcode-based blood unit scanning for real-time digital recording, enabling continuous physical-to-digital synchronization of inventory movement. It features a unified consortium-wide inventory dashboard that provides real-time visibility of stock levels, automated monitoring of out-of-stock and overstock conditions, and proactive near-expiry detection. A Blood-Redistribution Optimization Algorithm is integrated to analyze inventory levels, expiration timelines, and hospital demand patterns in order to generate redistribution recommendations that minimize preventable wastage and improve resource utilization. The framework utilizes a permissioned Hyperledger Fabric consortium blockchain architecture to log all blood transfer transactions. Smart contracts are implemented to enforce transfer validation, maintain a tamper-proof custody trail, and synchronize inventory updates across participating institutions. Geo-tagging functionality is incorporated during dispatch and receipt scanning to record verified transaction locations, establishing a traceable location history for inter-hospital transfers. The system manages inventory-level data such as blood type, component classification, unit identification number, expiration date, transfer status, and transaction timestamps. The system also includes a membership module, where hospitals may apply either as a blood bank or a requesting institution. Membership applications are submitted through the system and evaluated based on defined requirements. Approval of membership is managed within the network, where authorized institutions review submitted requirements and determine eligibility before granting access to system features. The study covers system architecture design, smart contract development, dashboard creation, backend integration, and implementation of real-time inventory synchronization within the consortium environment.

This study does not extend to provincial, regional, or national blood service networks. Expansion to other cities or integration into a nationwide infrastructure is beyond the scope of this research. The system does not collect, store, or process patient medical records or personal health information. The system will not be developed as a mobile application and is accessible only through a web-based interface. It will not record or manage the disposal process of blood units. Access to the system is not automatically granted. Participation is controlled through a membership approval process, where institutions must apply and meet defined requirements before gaining access to system functionalities. Furthermore, the study does not cover long-term clinical outcomes, nationwide implementation, or full regulatory accreditation processes, as these require broader institutional coordination beyond the scope of this research.

Purpose and Description

The primary purpose of this study is to develop an IoT-Enabled Consortium Blockchain Framework for blood inventory and transfer management that fundamentally aligns with the United Nations Sustainable Development Goals (SDGs). Specifically, the proposed system contributes to SDG 3: Good Health and Well-Being by reducing preventable deaths and ensuring timely access to essential healthcare services during emergencies. Furthermore, it supports SDG 12: Responsible Consumption and Production by minimizing biological waste and ensuring optimal resource utilization through proactive redistribution. The framework also strongly embodies SDG 17: Partnerships for the Goals by replacing institutional competition with cooperation and unifying standards across private and public health sectors.

Driven by these overarching global goals, the primary beneficiaries of this study are the patients and the community of Lipa City who are in need of urgent blood transfusions. By establishing a system that ensures real-time visibility of blood units, the waiting time for sourcing compatible blood is significantly reduced, providing the immediacy that is crucial for survival in trauma and emergency cases. Furthermore, the system's ability to track expiration ensures that these patients consistently receive safe, high-quality blood products.

Beyond direct patient care, the framework extends its advantages to the partner hospitals. Hospital administrators will benefit from a strategic tool that minimizes financial losses caused by blood wastage. Transitioning from isolated manual logbooks to a shared consortium network allows hospitals to proactively redistribute near-expiry units, ensuring cost-efficiency and fostering better inter-institutional collaboration during shortages. On an operational level, this transition directly addresses the operational fatigue faced by medical technologists and laboratory staff. By automating data entry through IoT scanning and replacing manual phone inquiries with a real-time dashboard, the system significantly reduces administrative workload. Consequently, this allows laboratory personnel to focus on their core clinical duties such as testing and processing rather than consuming time on clerical tasks and manual coordination.

At a broader regulatory scale, the Department of Health (DOH) and the Philippine Red Cross (PRC), as the regulatory bodies mandated by RA 7719, will substantially benefit from the study's insights on digital transformation. The proposed framework serves as a feasibility model for modernizing national blood surveillance, wherein blockchain's capacity to provide a tamper-proof audit trail can guide future policy-making for a transparent, data-driven blood supply chain management.

Finally, to the broader research community, this study provides a significant contribution to the existing body of knowledge regarding the convergence of IoT and Consortium Blockchain in healthcare supply chains. While current literature heavily focuses on theoretical blockchain models or implementations tailored for highly digitized, first-world settings, this research provides empirical data and a localized, practical architectural model using Hyperledger Fabric in the context of a developing country's hospital network. By addressing specific technological and infrastructural constraints in Lipa City, this study bridges the critical gap between theoretical blockchain utility and real-world clinical application. It offers academic scholars, health informatics researchers, and future student investigators a validated, scalable framework that can be analyzed, critiqued, or adapted for larger-scale national implementations and alternative medical supply chain studies.

Definition of Terms

To facilitate a clear understanding of the technical and operational concepts used in this research, the following terms are defined

Audit Trail. A chronological record of all transactions and system activities. In this study, it refers to the sequential log of every blood unit movement including dispatch, transit, and receipt that is permanently stored on the blockchain and cannot be altered or deleted.

Barcode/QR Code Scanning. A digital data capture method using optical readers to decode information embedded in barcodes or QR codes. In this study, it refers to the IoT-enabled scanning devices used at hospital blood banks to instantly record blood bag entries and exits, serving as the primary physical-to-digital bridge for inventory synchronization.

Blockchain. A decentralized, distributed ledger technology that records transactions across a network of computers, ensuring that the record cannot be altered retroactively without the alteration of all subsequent blocks. In this study, it serves as the foundational security layer that guarantees the integrity, transparency, and immutability of blood inventory data shared among the consortium hospitals.

Blood Component. A specific fraction of whole blood that has been separated for therapeutic use, including red blood cells, platelets, plasma, and cryoprecipitate. In this study, it refers to the distinct blood products tracked individually by the system, each with its own unit identification number, blood type classification, and expiration date.

Blood Inventory Management. The systematic process of monitoring the quantity, location, blood type, and expiration dates of blood units. In this study, it refers to the transition from manual logbooks to a digitized, real-time tracking system to prevent "ghost stock" and wastage.

Blood Redistribution Optimization Algorithm. A computational logic embedded in the proposed system that analyzes current inventory levels, expiration timelines, and hospital demand patterns to automatically generate redistribution recommendations. In this study, it serves as the proactive decision-support mechanism that flags near-expiry units and suggests the optimal receiving hospital to minimize preventable wastage.

Blood Type. A classification of blood based on the presence or absence of specific antigens on the surface of red blood cells, commonly categorized under the ABO and Rh systems (e.g., A+, O−, AB+). In this study, it is a primary data field used by the system to match available blood units with hospital demand and ensure compatibility during redistribution.

Chain of Custody. The documented and unbroken sequence of possession, transfer, and handling of a blood unit from its source to its final recipient. In this study, it refers to the traceable record maintained by the blockchain to verify that every handoff during inter-hospital transfer is authenticated and logged.

Consensus Mechanism. The protocol by which all participating nodes in a blockchain network agree on the validity and order of transactions before they are permanently recorded. In this study, it refers to the agreement process used within the Hyperledger Fabric framework that ensures all consortium hospitals maintain a consistent and verified copy of the blood inventory ledger.

Consortium Blockchain. A type of blockchain network where the consensus process is controlled by a pre-selected set of nodes rather than the general public. In this study, it refers to the permissioned network connecting Mary Mediatrix, Medix, N.L. Villa Memorial Medical Center, other hospitals, and the Red Cross Chapter involved allowing them to share data securely without a central authority.

Data Latency. The delay between when a real-world event occurs and when it is reflected in a data system. In this study, it refers to the critical time gap introduced by the current practice of updating blood inventory only at fixed reporting intervals (9:00 AM and 4:00 PM), which prevents hospitals from accessing accurate, real-time stock information during emergencies.

Distributed Ledger. A digital record-keeping system in which data is simultaneously stored, shared, and synchronized across multiple locations or nodes without a central administrator. In this study, it serves as the underlying data architecture of the consortium blockchain, ensuring that all participating hospitals maintain an identical and up-to-date copy of the blood inventory records.

First Expired, First Out (FEFO). An inventory management method where blood units with the nearest expiration date are used or released first before newer stocks. In this study, FEFO ensures proper stock rotation, minimizes expired blood products, and promotes efficient blood inventory handling.

Geo-Tagging. The process of attaching verified geographic coordinates to a digital record or transaction at the moment it is created. In this study, it refers to the location-stamping feature activated during barcode/QR code scanning at both the dispatch and receiving points of an inter-hospital transfer, establishing a verifiable location history on the blockchain.

Ghost Stock. A discrepancy in inventory records where the system indicates a blood unit is available, but it is physically missing or has already been used. This error typically arises from missed manual deductions during peak hospital shifts.

Hyperledger Fabric. An open-source, permissioned blockchain framework used in this study to build the underlying ledger for the blood inventory system. It allows for private channels and smart contracts suited for enterprise healthcare applications.

Immutability. The property of blockchain data whereby once a transaction is recorded and confirmed by the network, it cannot be altered, deleted, or overwritten. In this study, immutability ensures the permanent integrity of all blood inventory entries and transfer records, providing a trustworthy historical log that is resistant to tampering or manipulation.

Inter-Hospital Transfer. The operational process of relocating a blood unit from a source hospital with surplus supply to a recipient hospital with urgent demand. This study focuses on securing and tracking this specific workflow to ensure traceability.

Internet of Things (IoT). A network of physical devices that exchange data with other systems over the internet. In this study, IoT specifically refers to the Barcode/QR code scanning devices used to digitally capture blood bag entries and exits, serving as the physical-to-digital bridge.

Interoperability. The ability of different information systems, institutions, or devices to connect, communicate, and exchange data in a coordinated and meaningful way. In this study, it refers to the capacity of the proposed framework to unify the previously isolated blood bank systems of all Lipa City hospitals into a single, coherent consortium network.

Inventory Synchronization. The real-time, automatic updating of blood stock records across all nodes of the consortium network whenever a transaction is recorded. In this study, it ensures that every participating hospital and the Philippine Red Cross Lipa Chapter simultaneously reflects the most current inventory status without manual reporting.

Isolated Systems. Refers to the current state of hospital blood banks where inventory data is stored in standalone manual logbooks or unconnected local databases, preventing real-time sharing of information with other institutions.

Manual Inventory. The traditional method of recording blood stock using pen and paper or static spreadsheets. This practice is identified in the study as the primary cause of human error, delayed reporting, and data inconsistencies in Lipa City hospitals.

Near-Expiry Units. Blood products that are approaching the end of their shelf life (e.g., 35 days for whole blood). The proposed system defines these as units that trigger a proactive notification for potential redistribution to prevent wastage.

Node. An individual participant or institution connected to and maintaining a copy of the blockchain network. In this study, it refers to each consortium member — namely, the Philippine Red Cross Lipa Chapter, Mary Mediatrix Medical Center, Medix Medical Center, N.L. Villa Memorial Medical Center, and other Lipa City hospitals that collectively validates and stores blood inventory transactions on the shared ledger.

Out-of-Stock/Overstock. Inventory threshold conditions wherein out-of-stock refers to the depletion of a specific blood type or component below a minimum safe level, and overstock refers to an excess supply that increases the risk of expiration and wastage. In this study, the real-time dashboard automatically monitors and flags both conditions across all consortium hospitals to enable timely corrective action.

Periodic Reporting. The current practice of updating blood inventory data at fixed intervals (specifically 9:00 AM and 4:00 PM via Google Sheets), which fails to reflect real-time supply changes and creates data latency during emergencies.

Permissioned Network. A blockchain architecture in which participation, access, and transaction authority are restricted to pre-approved and authenticated entities. In this study, it refers to the controlled consortium environment where only verified Lipa City hospitals and the Philippine Red Cross Lipa Chapter are granted access to view and record blood inventory data, distinguishing it from open, public blockchains.

Philippine Red Cross (PRC). The national humanitarian organization mandated under Republic Act No. 7719 as the primary authority for blood collection, processing, and distribution in the Philippines. In this study, it specifically refers to the PRC Lipa Chapter, which functions as the central blood collection and distribution hub within the proposed consortium network.

Primary Nodes. An inventory management method where blood units with the nearest expiration date are used or released first before newer stocks. In this study, FEFO ensures proper stock rotation, minimizes expired blood products, and promotes efficient blood inventory handling.

Proactive Expiry Alert. An automated notification feature that detects blood units approaching their expiration date and immediately alerts authorized users or hospitals. In this study, it helps staff take timely action such as prioritizing usage or redistributing near-expiry blood units to reduce wastage.

RA 7719 (National Blood Services Act of 1994). The governing Philippine legislation that mandates the establishment of a national voluntary blood services program, regulates blood banks, and designates the Philippine Red Cross as the lead agency for blood supply management. In this study, it serves as the primary legal and regulatory framework within which the proposed system is designed to operate and support compliance.

Real-Time Dashboard. A web-based user interface developed in this study that aggregates live inventory data from all consortium members, allowing authorized staff to instantly view available blood stocks across the network without phone coordination.

Secondary Nodes. Supporting institutions connected to the consortium blockchain network that can access shared information and request blood units but have limited validation or governance roles compared to primary nodes. In this study, these refer to hospitals or healthcare facilities without active blood banks that mainly function as recipient institutions.

Smart Contract Trigger. The automated event-driven execution of a pre-programmed rule embedded within a smart contract, activated when specific conditions are met without requiring human intervention. In this study, it refers to the automatic actions initiated by the system — such as sending near-expiry alerts, validating inter-hospital transfer requests, and updating inventory records upon the detection of defined inventory thresholds or transaction events.

Smart Contracts. Self-executing digital contracts with the terms of the agreement directly written into code. In this system, smart contracts automatically validate transactions (such as blood transfers) and trigger alerts (such as expiration warnings) without human intervention.

Tamper-Proof Audit Trail. A chronological and immutable record of all transactions (dispatch, transit, receipt) stored on the blockchain. It ensures that once a blood unit's status is updated, the history cannot be altered or deleted, guaranteeing data integrity.

Transaction Timestamp. A digitally recorded date and time marker automatically generated and permanently attached to each blockchain entry at the exact moment a blood inventory event occurs. In this study, it serves as a verifiable time reference for every scan, transfer, and update within the system, ensuring an accurate and sequenced chronological history of all blood unit movements.

Wastage. The loss of viable blood units due to expiration, improper handling, or failure to redistribute surplus stock in time. Minimizing this metric is a key objective of the proposed notification algorithm.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

Technical Background

Following the establishment of healthcare and supply chain challenges discussed in the previous chapter, this section outlines the core technical concepts that will strengthen the proposed inter-hospital blood inventory and transfer management system. Each technology, algorithm, and tool described here will form a distinct layer of the system's architecture, collectively enabling automated, trustless, and data-driven supply chain operations.

The proposed system will be built upon a consortium blockchain architecture, a permissioned distributed ledger model in which a predefined partnership of authorized organizations will jointly govern a shared, cryptographically linked chain of transaction records. Unlike public blockchains that allow anonymous participation, consortium blockchains restrict membership to verified institutional participants; within this system, each will maintain a synchronized ledger copy that will preserve decentralization while enforcing the access control and accountability required in regulated industries such as healthcare and logistics (Saad et al., 2025; Alqarni et al., 2023).

The specific platform implementing this architecture will be Hyperledger Fabric, an enterprise-grade open-source framework maintained by the Linux Foundation. Its core components such as peers that maintain ledger copies and execute chaincode, orderers that sequence transactions through a Raft-based crash fault-tolerant consensus algorithm, Membership Service Providers (MSPs) that issue and verify X.509 cryptographic identities, and channels that partition data into private sub-ledgers, will collectively enable high-throughput, deterministic transaction processing within a fully permissioned multi-institutional network (Saad et al., 2025).

Deployed on this fabric will be smart contracts, called chaincode, which are self-executing programs that will encode business rules as programmatic conditions and autonomously record their outcomes as immutable ledger transactions without human intermediation (Yigit & Dag, 2024). Governed by endorsement policies requiring a defined quorum of peers to independently validate each transaction, chaincode will enforce data validation, trigger conditional events, manage multi-party approval workflows, and generate verifiable audit trails to reduce the risk of data manipulation and process inconsistency across consortium institutions (Ajayi et al., 2024).

Bridging the physical asset layer to this digital ledger will be the Internet of Things, specifically through barcode and QR code optical scanning. Barcodes encode data as one-dimensional patterns of variable-width parallel lines, while QR codes extend this into a two-dimensional matrix capable of encoding hundreds of characters of structured metadata including unique unit identifiers, collection dates, and expiration timestamps governed by international standards

ISO/IEC 15415 and ISO/IEC 18004. Upon scanning, the encoded data will be parsed into a time-stamped digital record, eliminating manual transcription errors and generating a verifiable event log for every physical asset (De Nerol et al., 2024; Girish et al., 2025). Compared to infrastructure-intensive modalities such as RFID, barcode and QR scanning provide a lightweight, cost-effective, and globally standardized data acquisition layer validated by Ismail et al. (2022) and Heeres et al. (2023) to substantially improve inventory accuracy in healthcare supply chain contexts.

Scan events generated at the edge will be stabilized in PostgreSQL, an open-source object-relational database management system renowned for its full SQL:2016 compliance, ACID transactional guarantees, and native JSONB column support. JSONB stores JSON documents in a decomposed binary format that supports indexing and field-level querying, enabling PostgreSQL to handle both structured relational data and semi-structured IoT scan payloads within a single database instance. Its declarative foreign key constraints, window functions, and common table expressions make it a well-established foundation for enterprise-grade, critical data management. The entire database, along with other system services, will be containerized using Docker, a platform that packages applications with all their runtime dependencies into isolated, portable containers through Linux namespaces and cgroups, and managed through Docker Compose, which will deploy the full multi-service stack from a single declarative YAML configuration file, ensuring reproducibility and operational consistency across deployment environments.

With the physical-to-digital data pipeline established, the system will enforce First Expired, First Out (FEFO) as the governing unit-selection discipline for all blood dispatches. FEFO mandates that the unit with the earliest expiration date is always prioritized for outbound transfer regardless of when it was received, distinguishing it fundamentally from First In, First Out (FIFO), which sequences items only by arrival order (Dillon et al., 2021; Prastacos, 1984). When implemented as a hard constraint, FEFO will query available inventory in ascending order of expiration date and reject any transaction that would violate this ordering, a mechanism validated by Kandasamy et al. (2025) and Jiménez-Delgado et al. (2025) to reduce biological waste in perishable supply chains.

Determining where a FEFO-selected unit should be sent will require a multi-criteria evaluation framework. Simple Additive Weighting (SAW), originally formalized by Fishburn (1967), will address this by normalizing each criterion value to a dimensionless scale, multiplying each by a weight w_i reflecting its relative importance, and summing the products into a composite score expressed as Score = $\Sigma(w_i \times x_i\_normalize)$. Because every component of the score, the raw value, normalization transformation, and assigned weight, remains explicitly traceable, SAW produces auditable, adjustable rankings applicable across healthcare resource allocation, supplier evaluation, and logistics optimization contexts, as documented in Mardani et al.'s (2015) two-decade review of MCDM applications.

These two methods will be integrated into the Blood Redistribution Optimization Algorithm (BROA), a composite two-tier decision pipeline designed to minimize biological waste across the consortium network. The first tier will apply FEFO as a hard unit-selection constraint, while the second tier will apply SAW across three weighted criteria to rank candidate recipient hospitals according to the formula Score = (w_1 × Urgency) + (w_2 × StockShortage) − (w_3 × Distance), directing each unit to the facility with the greatest verified need and proximity. This layered approach, biologically constrained at the unit level, multi-criteria rational at the routing level, has been validated in the literature as effective for proactive lateral supply management across multi-node perishable networks (Pangestika et al., 2024; Jiménez-Delgado et al., 2025).

When multiple requestor hospitals compete for the same limited stock, contention will be resolved through a Request Priority Scoring (RPS) mechanism that will rank competing requests by a composite formula: Score = (w_1 × ClinicalUrgency) + (w_2 × WaitTime). ClinicalUrgency will capture the declared medical severity of the request while WaitTime will enforce a fairness constraint that will prevent indefinite deprioritization, transforming otherwise ad hoc contention resolution into a systematic, auditable allocation process suited to emergency-driven healthcare supply environments.

Beyond managing which units are moved and to where, the system will also record where each movement physically occurs through geo-tagging. the process of appending WGS 84 latitude and longitude coordinates to a digital record at the precise moment an event takes place, cryptographically binding verified location evidence to each ledger transaction (Chafiq et al., 2024). Rather than continuous satellite-based GPS tracking, which is constrained by high energy consumption and the inability of satellite signals to penetrate enclosed hospital environments (Wagner & Postel, 2022; Asdecker & Felch, 2026), the system will employ checkpoint-based geo-tagging that will capture spatial data only at designated logistical control points, specifically outbound dispatch and inbound receipt. Herrera et al. (2025) confirmed that embedded geographic coordinates substantially improve supply chain oversight by enabling precise, objective verification of event locations without requiring uninterrupted satellite connectivity.

The user-facing layer of the system will be built on React, an open-source JavaScript library maintained by Meta that follows a declarative, component-based programming paradigm. React minimizes rendering overhead through a virtual DOM, an in-memory representation of the browser DOM, against which it performs a diffing algorithm to compute the minimal set of actual

DOM mutations needed when application state changes, ensuring low-latency rendering under high data throughput. Its unidirectional data flow model, TypeScript compatibility, and extensive ecosystem have established it as a standard for enterprise-grade web application interfaces. Serving as the runtime for the backend middleware will be Node.js, a cross-platform JavaScript environment built on Google's V8 engine that employs a non-blocking, event-driven I/O model. By processing asynchronous operations through an event loop rather than blocking threads, Node.js efficiently handles concurrent connections from databases, blockchain networks, and IoT device streams simultaneously. The Hyperledger Fabric SDK for Node.js complements this by abstracting the low-level gRPC communications and cryptographic lifecycle management required to interact with Fabric peer and orderer nodes, exposing high-level functions for transaction submission, endorsement collection, event listening, and ledger querying. Communication between the frontend client and backend middleware will follow RESTful API conventions, an architectural style formalized by Roy Fielding in 2000 that structures services around resource-oriented HTTP endpoints responding to GET, POST, PUT, and DELETE methods, with a stateless constraint requiring each request to be fully self-contained. This stateless, uniform interface decouples the client from the backend implementation, enabling web browsers, mobile applications, and IoT firmware to consume the API independently of the underlying blockchain or database architecture.

The development environment will be supported by three tools. Visual Studio Code will provide syntax highlighting, IntelliSense code completion, real-time linting, an integrated terminal, and an extensive extension marketplace covering Dockerfile authoring, Hyperledger Fabric chaincode development, and database querying. Postman will enable developers to construct, send, and inspect HTTP requests, organize them into automated test collections, and validate response codes and payloads across all API endpoints before connecting frontend or IoT clients substantially accelerating integration error detection. Git, the distributed version control system originally developed by Linus Torvalds in 2005, will track the complete history of codebase changes as cryptographically identified commit objects, support parallel development through branching and merging, and provide an immutable, auditable development record essential for safety-critical software engineering contexts.

Theoretical Framework

The development of BloodLedger: An IoT-Enabled Consortium Blockchain Framework for Inter-Hospital Blood Inventory and Redistribution Management in Lipa City is anchored in two complementary frameworks that justify the system's deployment context and core technological architecture. The Technology-Organization-Environment (TOE) Framework explains the institutional conditions that enable the system's adoption across Lipa City hospitals, while the Blockchain Technology Acceptance Model (BTAM) justifies why blockchain's specific architectural properties constitute the appropriate technical response to the documented supply chain failures.

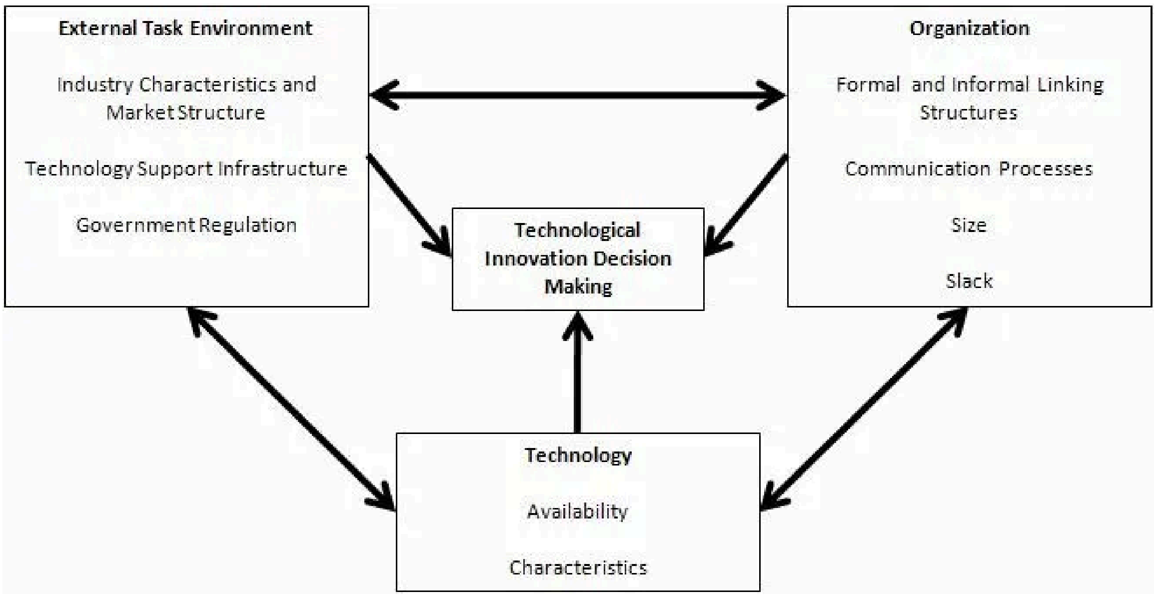

Figure 01. Technology-Organization-Environment (TOE) Framework.

Note: From Tornatzky, L. G., & Fleischer, M. (1990). The processes of technological innovation. Lexington Books.

The Technology-Organization-Environment (TOE) Framework, originally proposed by Tornatzky and Fleischer (1990), posits that technology adoption within an organization is shaped by three interconnected dimensions: the technological context, which refers to the characteristics and availability of relevant technologies; the organizational context, which encompasses internal resources, structures, and workflows; and the environmental context, which captures external regulatory pressures and industry demands. In this study, the technological context is defined by BloodLedger's use of Hyperledger Fabric, IoT barcode and QR scanning, and smart contracts, which were selected for their compatibility with Lipa City hospitals' existing infrastructure and resource constraints. The organizational context is reflected in the consortium structure, where three hospitals with active blood banks and the Philippine Red Cross Lipa Chapter are networked as co-equal ledger nodes, distributing governance and eliminating the single-point-of-failure vulnerabilities observed in prior Philippine health system integration efforts (Aguila et al., 2025; Zuniga et al., 2026). The environmental context is defined by the mandates of Republic Act No. 7719, DOH-CHD Circular 2024-0406, and House Bill No. 1283, which together create a regulatory imperative for real-time, interoperable blood inventory management. Collectively, the three TOE dimensions confirm that BloodLedger is institutionally appropriate, technically feasible, and externally mandated within the Lipa City healthcare network.

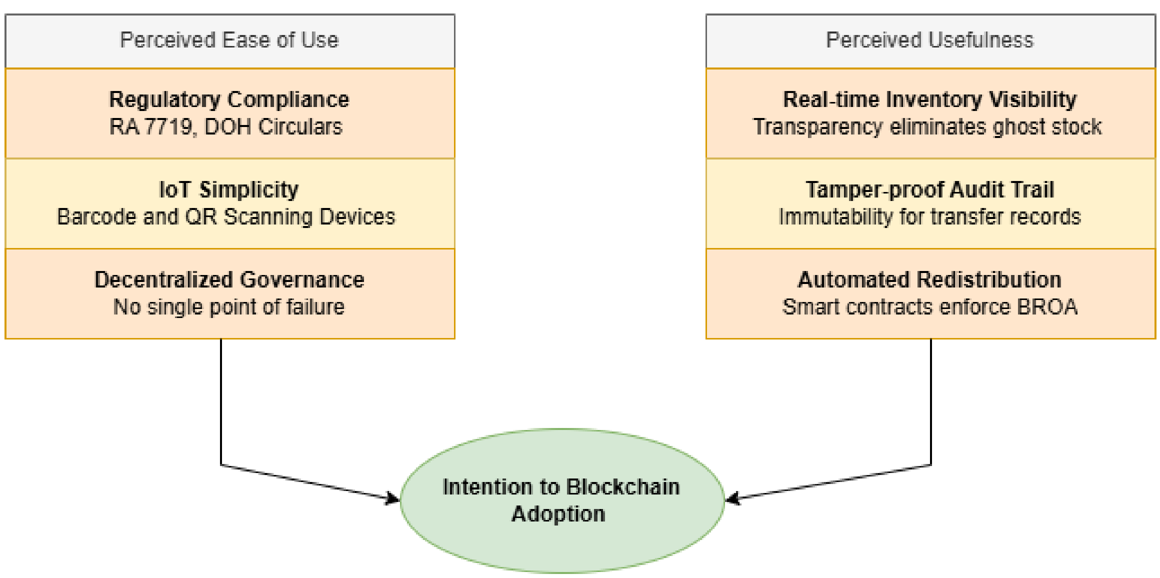

Figure 02. Blockchain Technology Acceptance Model (BTAM).

Note: Adapted from Yli-Huumo et al. (2016) and Clohessy and Acton (2019).

Complementing TOE at the system-architecture level is the Blockchain Technology Acceptance Model (BTAM), which builds on the foundational work of Davis (1989) and was further developed by Yli-Huumo et al. (2016) and Clohessy and Acton (2019) to identify four core blockchain properties as the key variables driving adoption and performance outcomes in multi-institutional environments: immutability, transparency, decentralization, and smart contract automation. Each property addresses a specific operational failure. Immutability ensures that every transfer event scanned by IoT devices is permanently recorded on the Hyperledger Fabric ledger, resolving the post-dispatch tracking blind spot where the Philippine Red Cross Lipa Chapter loses visibility of blood units once released. Transparency ensures all authorized consortium nodes access the same synchronized inventory ledger in real time, eliminating the ghost stock discrepancies caused by delayed manual logbook entries. Decentralization distributes ledger governance equally among participating institutions, removing the political and administrative vulnerability that has historically caused centralized health systems in the Philippines to collapse during government transitions (Aguila et al., 2025). Smart contract automation, implemented as Hyperledger Fabric chaincode, executes the Blood Redistribution Optimization Algorithm (BROA), combining First Expired, First Out (FEFO) prioritization with Simple Additive Weighting (SAW), to autonomously detect near-expiry units and trigger redistribution without human intervention. BTAM thus provides the theoretical basis for why a consortium blockchain is not merely an alternative to a centralized database but the architecturally necessary solution for the trust, traceability, and coordination failures specific to this study.

Together, TOE and BTAM form a cohesive and non-redundant theoretical foundation for this study. TOE establishes that BloodLedger's deployment is grounded in the actual institutional, technological, and regulatory conditions of Lipa City's healthcare network, ensuring the system is designed for viable adoption. BTAM then specifies why the consortium blockchain architecture is the technically correct response to the four documented operational failures. By anchoring the proposed framework in these two models, this study demonstrates that BloodLedger is both institutionally viable and architecturally justified, enabling the transformation of a fragmented, manual, and reactive blood network into a transparent, automated, and resilient supply chain.

Review of Related Literature

The management of highly perishable medical resources, particularly blood, presents a critical logistical challenge within modern healthcare systems. Traditional inventory practices often rely on manual record-keeping and isolated databases, leading to fragmented communication across hospital networks. This lack of real-time coordination frequently results in severe supply shortages during emergencies and the preventable expiration of life-saving biological assets.

Consequently, there is an urgent operational need to transition from reactive, siloed procurement methods toward integrated, technology-driven supply chain management.

To address these systemic vulnerabilities, emerging digital architectures offer robust frameworks for secure and transparent data exchange. Decentralized ledger technologies, specifically consortium blockchains, provide a trustless environment where multiple independent institutions can share synchronized, immutable records without relying on a vulnerable centralized authority. Furthermore, pairing these digital ledgers with Internet of Things (IoT) edge devices effectively bridges the gap between physical medical assets and virtual databases. By utilizing automated scanning and geospatial tracking checkpoints, hospitals can establish a tamper-proof audit trail that mitigates human reporting errors and drastically enhances overall supply chain visibility.

While capturing accurate physical data secures the transit of medical assets, maximizing the utility of this network information requires advanced decision-making logic. Mathematical optimization algorithms are essential for evaluating real-time inventory levels and autonomously routing resources to facilities with the highest clinical demand. By applying strict queuing disciplines based on expiration dates alongside multi-criteria scoring models, healthcare networks can actively execute lateral transshipments and prevent biological wastage. Ultimately, synthesizing decentralized infrastructure, automated tracking, and intelligent redistribution algorithms establishes the foundational architecture necessary for a modernized inter-hospital logistics system.

In the Philippines, blood inventory management is governed by the National Blood Services Act of 1994 (Republic Act No. 7719), which mandates voluntary, non-profit blood donation to ensure a safe and accessible supply (National Blood Services Act, 1994). While this law established strict regulatory standards for patient safety, it also introduced a significant operational burden for hospitals. Because blood components are highly perishable, relying on inefficient management systems makes these voluntarily donated units susceptible to expiration and wastage, directly contradicting the law's core objectives.

DOH Administrative Order No. 2005-0002 establishes the Rules and Regulations for the Philippine National Blood Services, implementing RA No. 7719. The order further requires that each hospital blood bank and health facility maintain a proper audit and inventory system of blood and blood products to ensure adequate and safe supply for patients, including during emergencies. Critically, the order also envisions a Blood Services Network, an organized system composed of blood centers, hospital blood banks, blood stations, and end-user hospitals operating within a specific geographical area as the structural mechanism through which coordinated blood service delivery should be achieved.

Despite this well-established regulatory framework, the actual operationalization of a networked, coordinated blood inventory system across Philippine hospitals has remained largely unrealized in practice. The order's goal of strengthening a nationally coordinated network of Blood Service Facilities through improved blood collection and distribution systems has yet to be fulfilled at the inter-hospital level, particularly in provincial healthcare networks where manual logbook-based inventory practices still predominate. This persistent gap between regulatory mandate and operational reality directly motivates the development of BloodLedger, which operationalizes the very network structure envisioned by DOH AO No. 2005-0002 by connecting hospital blood banks in Lipa City through a consortium blockchain that enforces real-time, automated, and auditable blood inventory management across all participating institutions.

To prevent blood wastage and uphold national standards, hospitals must adopt highly efficient inventory practices. Alghamdi (2023) emphasized that the extreme perishability of blood requires continuous monitoring and strict adherence to the First-Expire, First-Out (FEFO) method. The study noted that manual inventory practices often fail in dynamic healthcare environments, leading to critical supply shortages or high discard rates. Overcoming these challenges requires shifting from isolated tracking to integrated, technology-driven supply chains. This operational gap highlights the necessity of the proposed study. By implementing an IoT-enabled consortium blockchain, the framework provides the automated, real-time tracking needed to efficiently manage the blood lifecycle, ultimately fulfilling the safety and adequacy mandates of Republic Act No. 7719 across the Lipa City hospital network.

The Philippine healthcare landscape is characterized by a high degree of digital adoption that is ironically limited by widespread disconnection. Tinam-isan and Naga (2024) conducted a systematic review of health information systems in the Philippines, identifying that the primary barrier to effective digital health is the common use of disconnected systems using incompatible formats. Their analysis of 28 peer-reviewed studies confirmed that a lack of unified data leads to delayed access to reliable patient records and compromised data quality nationwide. This national fragmentation is further detailed in a policy review by Conjares et al. (2025), which found that while 91% of primary care facilities have transitioned to electronic records, these systems operate as isolated silos. The researchers argued that current systems are built primarily for insurance claims rather than integrated patient care, and they noted that building a traditional centralized data warehouse to solve this is difficult due to the high resource requirements and governance complexity.

Recognizing the severity of these workflow delays, national legislative efforts are now pushing for mandatory technical integration. House Bill No. 1283, introduced in the House of Representatives (Vargas), seeks to legally mandate the interoperability of Health Information Systems (HIS) across the country. The bill's explanatory note emphasizes that critical health data currently remains scattered across different clinics and hospitals, a fragmentation that leads to repetitive diagnostic tests, delayed medical interventions, and systemic inefficiencies. By requiring both public and private healthcare providers to adopt seamless data-sharing mechanisms, this proposed legislation highlights an urgent national demand for unified health networks. However, fulfilling this legal mandate through traditional centralized infrastructure remains technically and politically too difficult to sustain, as previously noted. This legislative pressure further validates the necessity of exploring alternative, decentralized architectures such as a consortium blockchain that can achieve the mandated interoperability without placing the heavy burden of a single, centralized server on local government units.

Existing research highlights blockchain technology as a critical enabler of transparency, traceability, and coordination in supply chain systems, particularly in multi-stakeholder environments. Traditional supply chains are often constrained by fragmented data systems, manual recordkeeping, and reliance on centralized authorities, which lead to inefficiencies, delays, and risks of data manipulation. Studies such as Alqarni et al. (2023) and Ajayi et al. (2024) explain that blockchain addresses these limitations through a decentralized and immutable ledger, allowing multiple participants to securely access and verify a shared, synchronized dataset. This distributed structure enhances trust and removes the need for intermediaries while ensuring that all transactions are permanently recorded and tamper-proof.

In addition to improving transparency, the integration of smart contracts introduces automation into supply chain processes by enabling rule-based execution of transactions, validation, and tracking. Alqarni et al. (2023) note that smart contracts reduce paperwork, minimize human error, and support real-time monitoring of goods, while Ajayi et al. (2024) emphasize their role in strengthening data security and interoperability in healthcare systems. Despite these advantages, studies also point out challenges related to scalability and system complexity, as public blockchain networks may face performance limitations when handling large volumes of transactions. Furthermore, many proposed frameworks remain conceptual and lack real-world implementation. To address these concerns, research increasingly supports the use of consortium blockchain models, where participation is limited to authorized entities, improving efficiency while maintaining security and data integrity. This approach is particularly suitable for sectors such as healthcare that require controlled access and high levels of trust.

Consortium blockchains provide a highly secure digital infrastructure for healthcare, but current research emphasizes that maximizing their potential requires the Internet of Things (IoT) to bridge the gap between physical medical assets and these digital platforms. Studies such as Varghese et al. (2024) and Taj et al. (2023) show that technologies like sensors, RFID, and connected devices enable real-world items to be monitored and represented digitally. This supports physical-to-digital standardization, where data from physical processes is accurately captured and reflected in digital systems. As a result, organizations can reduce human error, improve inventory accuracy, and make more informed decisions based on real-time data. In healthcare environments, IoT also contributes to safety and quality by monitoring the storage, handling, and movement of sensitive resources such as blood products.

However, the literature also identifies several challenges in implementing IoT systems. Concerns related to data security, privacy, and system integration remain significant, especially given the large volume of data generated by connected devices. High implementation costs and the lack of standardization across technologies further complicate adoption. In addition, continuous monitoring through sensor-heavy systems can increase complexity and resource consumption, which may not always be practical in real-world settings (Taj et al., 2023; Varghese et al., 2024). These limitations suggest that while IoT plays a vital role in enabling reliable physical-to-digital integration, effective solutions should focus on approaches that are simple, secure, and scalable, capturing essential data without introducing unnecessary system burden.

Directly addressing this need to capture essential data without the heavy burden of continuous IoT monitoring, global literature emphasizes the critical role of targeted geospatial tracking in achieving supply chain visibility. Wagner and Postel (2022) categorize tracking systems (e.g., GPS, RFID, cellular) based on transmission range, energy consumption, and cost, noting that continuous satellite tracking is often financially and energetically unfeasible for everyday, high-volume logistics. Supporting this, Asdecker and Felch (2026) differentiate between satellite-based continuous trackers for outdoor environments and proximity-based systems (like RFID and NFC), which are vastly superior for indoor, infrastructure-reliant settings due to their cost-efficiency. Because of these power and cost limitations, industry standards dictate that physical cargo tracking is rarely conducted in continuous real-time; instead, it relies heavily on strict logistical checkpoints—specifically capturing spatial data upon dispatch at one node and arrival at another.

Establishing these automated checkpoints successfully resolves the technical and financial limitations of continuous monitoring. Furthermore, current literature emphasizes that integrating Location Tracking Devices (LTDs) is equally essential for mitigating human error and bias in supply chain reporting. Relying on manual or self-reported data leaves supply chains highly vulnerable to manipulation, non-response, and social desirability biases (Asdecker & Felch, 2026). However, transitioning to automated geospatial tracking introduces significant ethical limitations, particularly regarding the perceived surveillance of personnel and the involuntary exposure of confidential commercial routing structures. To responsibly harness the transparency and predictive value of LTDs, tracking frameworks must embed strict data protection boundaries that shield sensitive business information from public misuse while still ensuring verifiable operational accountability.

Secure geo-tagging establishes a verifiable and ethical foundation for tracking medical assets; nevertheless, maximizing the utility of this spatial data requires advanced decision-making logic. Global literature underscores that because blood is highly perishable and clinical demand is fundamentally uncertain, the continuous application of mathematical optimization algorithms is critical to minimizing biological waste. Pangestika et al. (2024) synthesized 45 empirical studies on blood management, finding that implementing a lateral supply mechanism which allows hospitals to fulfill demand using peer hospitals' surplus supplies rather than relying solely on central blood banks is a primary indicator of strong network performance. However, despite the availability of advanced optimization models such as Genetic Algorithms and the Multi-Objective Gray Wolf Optimizer, the review highlights a critical recurring vulnerability—severe shortages and biological wastage persistently occur whenever there is a lack of real-time coordination between inventory management and distribution networks.

Ultimately, the reviewed literature shapes the architectural design of the proposed study. It establishes that while IoT and targeted geo-tagging efficiently bridge the physical-to-digital divide, and consortium blockchains secure interoperable trust, these technologies only succeed when actively paired with advanced redistribution logic. By synthesizing these established concepts, this research is directly informed to construct a decentralized, checkpoint-based framework that integrates real-time inventory management with lateral distribution algorithms, effectively addressing the systemic coordination vulnerabilities identified in modern healthcare supply chains.

Review of Related Studies

Recent studies emphasize that modernizing blood banks requires reliable digital systems to address severe supply shortages and inventory mismanagement. Mishra et al. (2024) proposed an innovative blood bank management system aimed at addressing the alarming shortage of blood supply in India. To develop this platform, the researchers employed an iterative development methodology, utilizing Python, JavaScript, and a Database Management System (DBMS) to build a centralized web application. Their design structured operations across three defined user roles (e.g., Admin, Donor, and Receiver) and incorporated specific modules for secure authentication, donor engagement, and real-time inventory monitoring. The results demonstrated that this structured database approach successfully streamlined the blood request process during emergencies and provided administrators with early warnings regarding low supplies and product expiry dates. However, while the study effectively optimized the immediate donor-to-bank relationship and internal tracking, it depended heavily on a traditional centralized database. BloodLedger expands upon this foundation; instead of merely notifying a central administrator of an expiring unit, it utilizes a consortium blockchain to automate peer-to-peer, direct hospital-to-hospital transfers, ensuring that excess blood supplies in one facility can be securely moved to another before it expires.

To bridge the gap between digital databases and physical blood units, other researchers have turned to connecting physical devices. Ismail et al. (2022) developed the "Lifeline" system to improve blood bank management in developing regions by combining web technologies with the Internet of Things (IoT). The researchers utilized a mixed-methods evaluation, designing a system equipped with an ESP32 microcontroller, DHT-11 sensors for real-time temperature monitoring, and RFID technology for tracking individual blood bags. To validate the system, they conducted black-box functionality testing and administered a structured questionnaire measuring satisfaction, effectiveness, and efficiency to 22 blood bank professionals in Iraq. The results indicated that the IoT integration successfully reduced the time required to find compatible blood types and significantly minimized errors associated with manual data entry, leading to high user acceptance. Nevertheless, the study acknowledged that large-scale implementation is limited by the need for stable internet connectivity and the high cost of attaching an RFID tag to every single blood bag. BloodLedger directly addresses these practical limitations by replacing expensive, continuous RFID tracking with targeted barcode and QR code scanning at designated hospital checkpoints (dispatch and receipt). Furthermore, while Ismail et al. focused on tracking bags within a single facility, BloodLedger pairs edge scanning with a decentralized ledger to ensure that the tracking data is securely and permanently shared across the entire Lipa City hospital network.

Beyond local management challenges, international studies reveal that the lack of unified data standards is a main roadblock to health system integration. Chuma (2025) investigated these roadblocks in South African public hospitals using a convergent parallel mixed-methods design with 144 clinical and administrative personnel. The results demonstrated that inconsistent standards and unclear guidelines heavily restricted data exchange, proving that without strictly followed rules, interoperability is practically impossible. Locally, attempts to bridge these gaps have highlighted the technical and financial costs of enforcing such integration. Zuniga et al. (2026) documented the implementation of the Smarter and Integrated Local Health Information System (SMILHIS) across three Philippine pilot sites. Utilizing a proof-of-concept methodology, the researchers deployed a centralized Enterprise Service Bus (ESB) architecture mapped to HL7 FHIR standards. While results showed a successful reduction in data duplication, the study exposed the huge cost and effort of maintaining a main server at the local government level.

These operational and centralized risks are made much worse by the human element of local governance. Investigating this barrier, Aguila et al. (2025) utilized a descriptive qualitative design, conducting Key Informant Interviews (KIIs) with 10 SMILHIS implementers. Their thematic analysis revealed that political transitions and shifting administrative priorities frequently led to the reallocation of funds or the early cancellation of centralized digital health projects. BloodLedger directly resolves the technical, standardization, and political vulnerabilities identified in these studies. By utilizing a consortium blockchain architecture instead of a traditional centralized ESB server, the system enforces uniform data entry through unchangeable smart contracts and distributes the responsibility of hosting the digital ledger collectively among participating hospitals in Lipa City. This decentralized approach eliminates the single point of failure, making the inter-hospital network politically stable and ensuring the blood transshipment system remains permanently operational regardless of changes in local administration agendas.

Building directly on the necessity of these decentralized architectures, Kotian et al. (2024) proposed a blockchain-based system called Blood Chain, developed using a design and implementation approach that integrates blockchain, smart contracts, and QR code-based tracking across donor, blood bank, and hospital workflows. The system records donor information, blood group, and transaction data on a distributed ledger, while incorporating physical blood inspection and an algorithm that selects the most optimal blood bank based on proximity and availability, supported by Ethereum, MetaMask for transaction verification, and Google Maps API for location-based decision-making. The results showed that the system enhances transparency, traceability, and coordination through secure and immutable data sharing, while improving efficiency via automated donor-recipient matching, QR-based verification, and optimized blood distribution, reducing delays and preventing fraudulent activities. It also enables real-time tracking of blood from donation to transfusion and improves availability during emergencies. However, the system remains conceptual and lacks real-world deployment and integration with existing healthcare infrastructure. In contrast, BloodLedger addresses these limitations by implementing a consortium blockchain (Hyperledger Fabric) within a hospital network, combined with QR and barcode-based IoT scanning and a shared dashboard to support practical, real-time, and secure blood supply management.

Further exploring the automation of coordination and resource sharing in supply chains, Agrawal et al. (2022) developed a blockchain-based framework using a design science approach, beginning with a systematic literature review to identify key smart contract applications, followed by the design of a collaborative network architecture, rule-based smart contract algorithms, and UML interaction models. The framework was implemented by coding smart contracts in Solidity and deploying them on an Ethereum blockchain, where interactions between buyers and suppliers were simulated using test data to validate functionality. The results showed that the framework successfully enabled automated order distribution, real-time data sharing, and secure coordination among participants through a shared ledger, while smart contracts effectively enforced predefined rules such as supplier prioritization and capacity-based allocation. Additionally, the study demonstrated that smart contracts can support collaboration while maintaining partner independence, improving resource utilization and reducing risks such as overbooking and data manipulation. However, the system was tested in a controlled environment and relied on a public blockchain, which presents limitations in scalability, cost, and enterprise integration. In contrast, BloodLedger addresses these constraints by using a consortium blockchain (Hyperledger Fabric) to enhance scalability and efficiency, while integrating IoT-based scanning and real-time dashboards to support practical deployment in hospital environments, enabling secure, automated, and coordinated supply chain operations.

Validating the structural scalability of these restricted consortium deployments across multi-hospital environments, Thamrin (2022) proposed a hierarchical cloud-based consortium blockchain framework to address the challenge of securely storing and sharing electronic health records across distributed hospital networks. The system organized hospitals into multiple blockchain tiers and applied role-based access control with cryptographic identity verification to ensure that only authorized institutions could access or update records. Results showed improved scalability, secure data exchange, and efficient record retrieval, demonstrating that consortium blockchain is suitable for large-scale healthcare collaboration. However, the study mainly focused on health record management rather than the real-time movement of medical resources between institutions.

Similarly, Yang et al. (2023) demonstrated that consortium blockchain can establish secure and trusted medical information exchange among hospitals through controlled access and digital signature mechanisms that prevent transaction forgery. Supporting this, Sembiring et al. (2026) found that consortium blockchain frameworks reduce single points of failure and provide tamper-resistant, synchronized records across participating healthcare organizations. Collectively, these studies confirm that consortium blockchain improves security, interoperability, traceability, decentralized control, and operational efficiency in healthcare environments. These findings strongly support the development of BloodLedger, a consortium blockchain-based blood bank and blood transfer management system that enables secure, transparent, and real-time coordination among hospitals in Lipa City.

In the field of healthcare data management, safeguarding information requires advanced cryptographic frameworks to ensure security and integrity. Richard (2024) investigated the application of blockchain technology in healthcare, concluding that its decentralized architecture inherently ensures data integrity and protects sensitive medical information from unauthorized alterations. Advancing this concept, Kunal et al. (2024) proposed a blockchain-driven protocol utilizing advanced encryption to further secure patient data within the healthcare industry. Their findings demonstrated that integrating advanced cryptographic techniques with blockchain networks significantly fortifies data privacy against cyber threats and unauthorized access. Consequently, BloodLedger adapts these robust data security principles but shifts the operational focus; instead of storing highly sensitive patient records, the system strictly enforces PHI data isolation by exclusively tracking the physical lifecycle of blood units using cryptographic dual-signatures, thereby eliminating patient data breach risks entirely while ensuring the integrity of the supply chain.

Beyond specific cryptographic protocols, Conduah et al. (2025) evaluated global challenges concerning healthcare data privacy, highlighting that traditional, centralized health information systems are increasingly susceptible to data breaches and compliance failures. Their findings emphasized the urgent need for modernized, privacy-preserving technological frameworks to restore institutional trust across fragmented networks. Despite effectively outlining these global privacy challenges and theoretical solutions, the study lacks a concrete, localized implementation architecture designed to comply with specific national data residency laws. This aligns with the proposed study's objective to secure healthcare data, but differs by remaining broad and conceptual rather than deploying an operational system. To address this gap, BloodLedger provides a practical solution by deploying an on-premise PostgreSQL database containerized via Docker alongside the Hyperledger Fabric blockchain; by keeping all operational data within the hospital's secure Local Area Network (LAN), the system directly fulfills the strict data residency and minimization requirements of the Philippine Data Privacy Act of 2012 (RA 10173).

Evaluating the real-world feasibility of this essential IoT integration, Heeres et al. (2023) employed a mixed-methods multicase study combining semi-structured interviews and quantitative surveys across multiple hospitals, involving supply chain managers, IT personnel, and healthcare professionals. The study found that IoT implementation significantly improves real-time visibility, inventory accuracy, and workflow efficiency, enabling hospitals to better track medical supplies and reduce stockouts through continuous digital monitoring of physical assets. However, the findings also revealed critical barriers, including high implementation costs, lack of technical expertise, staff resistance to change, and cybersecurity risks. Additionally, integration challenges with existing hospital systems and the lack of standardized protocols were identified as major constraints to seamless adoption. While the study strongly validates the role of IoT in enabling accurate physical-to-digital mapping and improving operational efficiency, it differs by emphasizing organizational and adoption challenges rather than proposing a simplified or optimized implementation model. Consequently, BloodLedger addresses these barriers by utilizing a lightweight IoT approach through QR and barcode-based scanning, minimizing infrastructure complexity while still capturing real-time inventory events, and integrating these inputs into a consortium blockchain to ensure secure, standardized, and tamper-proof data management across participating hospitals.

Providing a broader theoretical foundation for these digital integrations, Khan et al. (2022) conducted a systematic literature review and conceptual analysis of smart technologies such as IoT, blockchain, artificial intelligence, and cloud computing. The study demonstrated that IoT enables real-time data capture from physical assets, allowing organizations to digitally represent inventory, logistics activities, and resource usage, thereby improving traceability, responsiveness, and decision-making. Furthermore, the integration of IoT with sustainability practices was shown to reduce waste, optimize resource allocation, and enhance overall supply chain efficiency. Despite these benefits, the study identified key limitations, particularly the high cost of implementation and the need for skilled personnel, as well as the absence of empirical validation due to its conceptual nature. While this supports the importance of IoT in achieving physical-to-digital standardization and data-driven supply chain optimization, it differs by lacking a concrete system implementation within a specific operational context. To address this gap, BloodLedger provides a practical application by transforming physical blood units into digitally traceable assets through QR and barcode-based IoT scanning, combined with blockchain integration to ensure secure, real-time, and standardized data exchange, demonstrating how these technologies can be effectively applied in a real-world healthcare supply chain.

Addressing the necessity of verifying the exact physical transit paths of these digitally tracked assets, Garcia et al. (2025) developed an integrated agri-tech solution utilizing GPS hardware modules and drone surveillance to embed geographic metadata directly into digital distribution media. The integration of location-based monitoring successfully established a verifiable audit trail, minimizing resource wastage and significantly enhancing distribution planning by geographically verifying that distributed inventory matched actual field utilization. However, the proposed hardware relies heavily on satellite-based continuous GPS tracking, a system that the authors note suffers from severe delays and a complete inability to receive satellite signals within enclosed indoor environments like warehouses. While this validates the use of geographic metadata for supply chain transparency, it differs by relying on continuous outdoor satellite tracking which is technologically not feasible for enclosed medical logistics. Consequently, BloodLedger addresses this hardware limitation by abandoning continuous satellite signals in favor of discrete, fixed-coordinate geo-tagging; by capturing spatial data strictly via IoT scans at designated hospital checkpoints (i.e., outbound dispatch and inbound receipt), the system securely binds timestamps to exact locations without the risk of indoor signal loss.

Further emphasizing the administrative value of such location-specific metadata, Herrera et al. (2025) utilized a quantitative-descriptive research design to evaluate the effectiveness of spatial data in monitoring rural infrastructure, collecting empirical data from 30 purposely selected stakeholders regarding the integration of embedded geographic coordinates and timestamps into digital workflow records. The results demonstrated that geo-tagging fundamentally improved project oversight, with 90% of respondents confirming that embedded GPS metadata enabled precise, objective verification of work progress while drastically reducing the need for resource-intensive physical site visits. Despite its high verification accuracy, the framework structurally relies on field personnel using mobile devices to capture manual photo documentation, leaving the workflow susceptible to human error, delayed reporting, and governance risks associated with manual initiation. This strongly aligns with the proposed study's premise that immutable, location-specific data is a practical necessity for establishing supply chain accountability, but differs by relying heavily on human-initiated manual documentation. To close this operational gap, BloodLedger completely automates the geo-tagging process; rather than relying on manual photographic uploads, the system utilizes IoT-linked smart contracts to autonomously capture and immutably record precise GPS coordinates the exact millisecond a blood bag is scanned at any designated transit checkpoint.

Focusing specifically on the optimization of such perishable medical inventories, Jiménez-Delgado et al. (2025) developed a three-phase hybrid methodology combining data analytics with Multi-Criteria Decision Making (MCDM) techniques. Specifically, they utilized F-AHP and TOPSIS for the multicriteria classification of supplies, alongside an exponential smoothing forecasting model and a First-Expired, First-Out (FEFO) queuing discipline to calculate precise reorder points. The integrated MCDM approach successfully categorized supplies based on rotation, criticality, and availability, while the FEFO-driven model accurately established dynamic reordering schedules to prevent clinic stockouts. Despite its effectiveness, the framework remains structurally restricted to centralized demand forecasting for external purchasing, entirely lacking an automated mechanism for the real-time, peer-to-peer redistribution of existing near-expiry stock. While this explicitly aligns with the proposed study's use of MCDM and FEFO for managing perishable medical inventory, it differs by focusing exclusively on static classification and centralized reordering. Consequently, the proposed system addresses this operational gap via the Blood Redistribution Optimization Algorithm (BROA), which pairs FEFO with Simple Additive Weighting (SAW) to autonomously evaluate and execute immediate lateral transshipments of near-expiry blood across consortium hospitals.

Investigating the operational hurdles of actually deploying this First-Expired, First-Out (FEFO) queuing discipline, Kandasamy et al. (2025) utilized a mixed-methods approach combining a PRISMA literature review with MCDM tools like DEMATEL and TISM. By collecting empirical data from a panel of 10 industry experts, the researchers systematically evaluated and ranked 13 core challenges to FEFO adoption. The DEMATEL analysis, applying an alpha threshold value of 0.368, revealed that effective storage management and real-time expiration tracking are the most critical, foundational barriers across a seven-level partition chart, ultimately concluding that integrating IoT and blockchain technologies is essential to overcome these hurdles and minimize biological waste. The study successfully models the technological barriers to adopting FEFO; however, it structurally treats the discipline merely as a localized, intra-facility dispatch policy, lacking an active mathematical mechanism to dynamically redistribute near-expiry goods across a multi-node network. This strongly validates the proposed study's integration of a FEFO queuing discipline, IoT, and blockchain to secure perishable medical inventory, yet it differs by focusing solely on evaluating implementation barriers rather than automating logistics. Consequently, the proposed system advances this framework via the Blood Redistribution Optimization Algorithm (BROA), which not only enforces FEFO at the blockchain layer but actively pairs it with Simple Additive Weighting (SAW) to autonomously calculate and execute immediate lateral transshipments of near-expiry stock across consortium hospitals.

Synthesis

The reviewed literature and studies collectively establish a comprehensive set of technical, regulatory, and operational gaps in current blood supply management. Literature confirms that Philippine health systems suffer from fragmented digital infrastructure, with electronic records operating as isolated silos incapable of supporting coordinated, multi-institutional workflows, while DOH Administrative Order No. 2005-0002 has long mandated a networked Blood Services structure across facilities that has yet to be operationalized at the inter-hospital level. Related studies further reveal that existing digital solutions, whether centralized blood bank platforms, single-facility IoT systems, or Philippine health integration pilots, consistently fail due to data isolation or centralization risks. On the data security front, while blockchain's capacity to enforce data integrity and privacy in healthcare has been validated, these frameworks remain focused on managing electronic health records rather than securing the physical logistics of perishable medical assets. On the algorithmic side, studies demonstrate that FEFO and MCDM methods have proven effective for perishable inventory management, yet none extend these methods into an automated, cross-facility redistribution mechanism. Similarly, while geo-tagging has been validated as a transparency tool, existing implementations rely on either continuous satellite tracking or manual documentation, both unsuitable for enclosed medical logistics.

These gaps collectively point to a single, unaddressed need for a deployable, decentralized system that integrates IoT-based physical tracking, automated redistribution logic, verifiable location data, and privacy-preserving data architecture within a multi-hospital framework in the Philippine context. Research confirms that consortium blockchains are architecturally suited for exactly this kind of secure, interoperable, multi-institutional coordination. BloodLedger directly addresses all identified gaps by being the first local study to combine a Hyperledger Fabric consortium blockchain with QR/barcode IoT scanning, BROA-driven lateral transshipment logic, automated geo-tagging, and on-premise PHI-isolated data architecture into a unified blood supply management system for Lipa City. Rather than proposing yet another centralized platform vulnerable to political and technical failure, this study distributes both data governance and operational responsibility across participating institutions—operationalizing the Blood Services Network envisioned by DOH AO No. 2005-0002, fulfilling the interoperability mandate of national health legislation, and protecting the integrity of a blood supply that directly sustains patient lives.

CHAPTER 3

DESIGN AND METHODOLOGY

Software Development Paradigm

The development of BloodLedger will follow the Agile Scrum methodology, a structured iterative framework that organizes development into time-boxed cycles called Sprints, each governed by four formal ceremonies: Sprint Planning, Daily Scrum, Sprint Review, and Sprint Retrospective. Scrum operationalizes the broader Agile principles of continuous stakeholder collaboration, adaptive planning, and rapid response to evolving requirements, properties that are essential for a system designed to operate across multiple healthcare institutions with distinct workflows, regulatory mandates, and technological baselines.

Three structural characteristics of BloodLedger make Agile Scrum the most appropriate development paradigm over alternatives such as Waterfall, the V-Model, or less structured Agile approaches.

First, multi-hospital feedback cycles are integral to the system's validity. BloodLedger's operational requirements were derived from direct consultations with blood bank personnel across participating institutions in Lipa City. As noted by Ms. Hannah Mae Atienza, the former Section Head of the Blood Bank at Mary Mediatrix Medical Center, inventory discrepancies arise from the pressures of peak-hour operation, a constraint that cannot be fully specified in advance without iterative validation from actual laboratory staff. Scrum accommodates this through its formal Sprint Review ceremony at the close of every Sprint, where the development team demonstrates a working system increment to hospital stakeholders, blood bank heads, medical technologists, and PRC Lipa personnel, and collects structured feedback that directly reshapes the next Sprint's backlog priorities.

Second, the blockchain smart contract layer requires iterative refinement. The chaincode governing the Blood Redistribution Optimization Algorithm (BROA), the Request Priority Scoring (RPS) mechanism, and the transfer state machine must be validated against simulated hospital scenarios before network-wide deployment. Unlike standard software modules, smart contracts are immutable once deployed to the ledger, meaning errors are costly to correct post-production. Scrum's Sprint-by-Sprint delivery structure allows each chaincode function to be incrementally developed, tested in isolation, and validated against integrated system behavior within its Sprint before being committed to the consortium network, with the Sprint Retrospective capturing any lessons learned for the next chaincode Sprint.

Third, the system's integration requirements evolve across hospital nodes. Because each participating institution operates with slightly different communication tools and manual workflows, as observed during consultations at N.L. Villa Memorial Medical Center and Mary Mediatrix Medical Center, the API layer and local PostgreSQL buffer must be adaptable to node-specific operational environments. Scrum's defined Sprint cadence enables the development team to deploy and test integration at one hospital node, present findings at the Sprint Review, and carry forward verified configurations to subsequent Sprints.

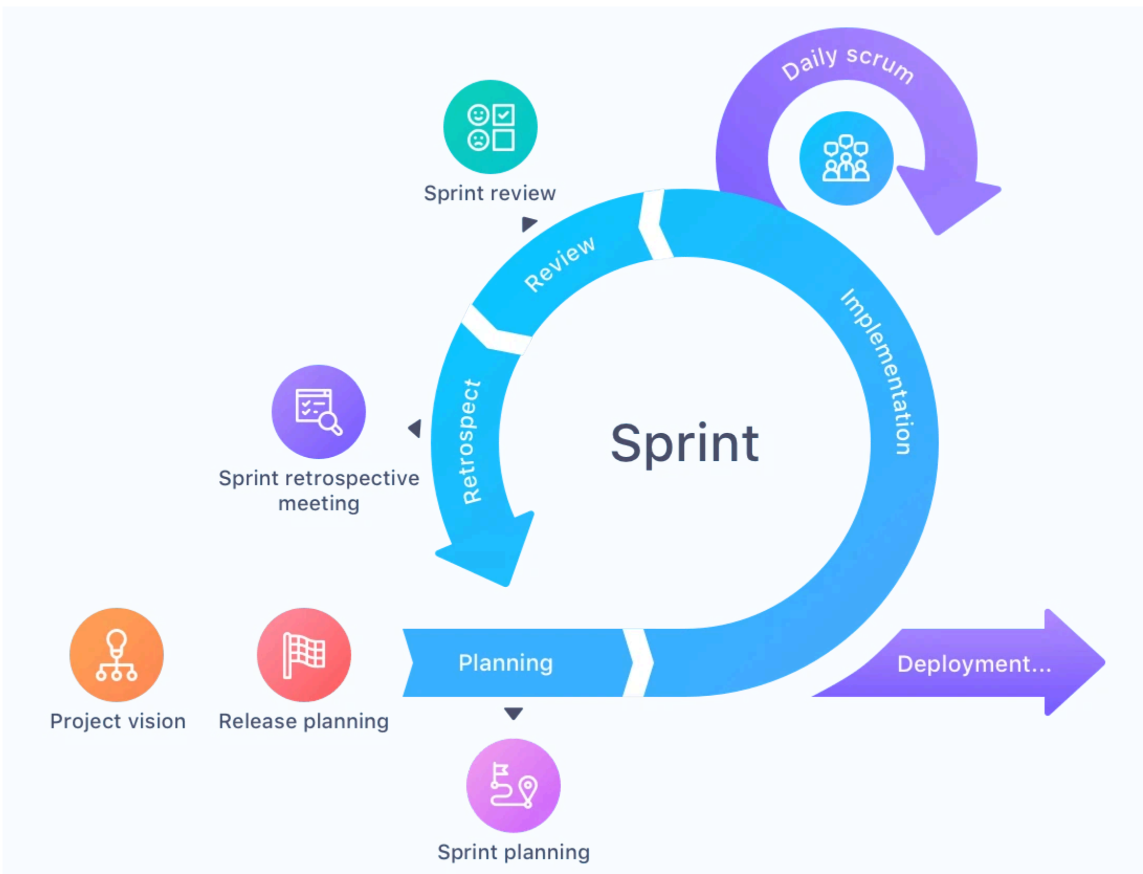

Figure 03. Scrum-based Agile SDLC Diagram.

Note: Hive. (2026, March 31). *The #1 project management software for teams | HIVE.* https://hive.com/blog/kanban-vs-scrum/

Figure 3 illustrates the Scrum-based Agile SDLC as applied to BloodLedger. The diagram represents a structured, sprint-driven cycle governed by the core Scrum ceremonies: Sprint Planning, Daily Scrum, Sprint Review, and Sprint Retrospective, with development activities occurring within each Sprint. Each cycle produces a working system increment, which is evaluated during the Sprint Review by key stakeholders, including hospital personnel and PRC Lipa representatives. Feedback gathered is then systematically incorporated into the next Sprint's backlog during Sprint Planning. This iterative and incremental structure ensures that the system continuously adapts to the evolving operational requirements of Lipa City's hospital network while maintaining alignment with real-world clinical workflows and constraints.

Agile Scrum Phases for BloodLedger

Phase	BloodLedger Activity	Primary Stakeholders	Key Deliverables
Planning (Sprint 0a from June 1–14, 2026)	Stakeholder identification; requirements elicitation through hospital interviews; Sprint Backlog construction	Hospital laboratory staff, PRC Lipa, DOH representatives, hospital administrators	System requirements document, stakeholder register, project scope
Design (Sprint 0b from June 15–28, 2026)	Blockchain topology, IoT scanning workflow, API contract, UI/UX wireframes; smart contract endorsement policy; security architecture	Blood bank heads, medical technologists, system architects	System architecture diagram, use case diagrams, data flow diagrams, database schema
Development (Sprints 1–5 from June 29–	Sprint Planning → Chaincode implementation	Development team (JoBuLat Tech)	Functional BROA and RPS modules, working dashboard

September 6, 2026)	(infrastructure, inventory, transfer, IoT pipeline, React dashboard) → Sprint-level unit and integration testing → Sprint Review and Retrospective per Sprint		prototype, IoT scan integration
Testing (September 7–27, 2026)	System Testing across all six functional modules; User Acceptance Testing with blood bank personnel; defect remediation before deployment sign-off	Medical technologists, laboratory staff, blood bank heads	Test reports, bug-fix logs, validated chaincode, UAT feedback
Deployment (September 28–October 11, 2026)	Phased parallel rollout beginning with primary blood bank nodes; BloodLedger operates concurrently with existing manual systems during a 3–6 month validation period	Hospital administrators, IT personnel, PRC Lipa	Deployed consortium network, onboarded hospital nodes, validation period reports
Maintenance (October 12, 2026 onwards)	Chaincode upgrades via Hyperledger Fabric versioning; network expansion to additional hospitals; ongoing performance monitoring via Hyperledger Explorer	System administrators, consortium network members, DOH	Updated chaincode versions, network expansion documentation, performance dashboards

Table 1. Agile Scrum Phases for BloodLedger.

Requirements Analysis

This section defines the specific problems BloodLedger is designed to resolve, the functional capabilities required from each system module, the non-functional quality attributes the system must uphold, and the stakeholders whose operational needs govern the system's design.

The requirements of BloodLedger were derived directly from the operational failures documented in Chapter 1 and validated through primary consultations with healthcare personnel at Lipa City's participating hospitals. Four specific problems constitute the basis for the system's functional scope.

A key weakness identified through the interview analysis and document review is manual inventory recording and ghost stock discrepancies. Blood inventory at primary supplier hospitals is maintained through excel files and google sheets that are updated manually after each blood unit is used or transferred. Key informant interviews with Ms. Hannah Mae Atienza, former Section Head of the Blood Bank at Mary Mediatrix Medical Center, revealed that staff frequently fail to deduct utilized units during peak operational hours, creating dangerous discrepancies between physical stock and recorded inventory—a phenomenon referred to as "ghost stock". These mismatches expose hospitals to the risk of administering blood based on inaccurate availability data, a risk that the system is designed to eliminate.

Compounding this is the absence of real-time inter-hospital blood visibility. When a hospital is unable to fulfill an internal blood request, staff must contact other facilities through phone calls or consumer messaging applications such as Facebook Messenger. Ms. Jessica Concha, Chief Medical Technologist at N.L. Villa Memorial Medical Center, noted during the interview that urgent requests broadcasted through group chats frequently go unnoticed because laboratory personnel are engaged in critical clinical tasks and cannot continuously monitor mobile devices. This informal coordination model results in delayed blood sourcing during emergencies, with no guarantee that availability information relayed verbally or through chat reflects actual stock levels at the time of the inquiry.

Document review of inter-hospital transfer workflows also uncovered the fragmentation of inter-hospital transfer custody records. Once a blood unit is dispatched from a supplying hospital or the PRC Lipa hub, digital tracking ceases entirely. No mechanism exists to confirm whether the unit arrived safely at the receiving facility, was successfully transfused, or expired in transit. This post-release blind spot prevents the PRC from exercising visibility over its distributed supply and limits the ability of regulatory bodies such as the DOH to audit blood chain-of-custody – a gap that directly motivated the inclusion of geo-verified transfer tracking in the proposed system.

Finally, the SWOT analysis identified as a critical internal weakness the system's passive and reactive blood wastage management. Near-expiry detection depends entirely on physical inspection of individual blood bags or periodic review of manual logbooks. Without an automated monitoring system, blood units expire in low-demand facilities while adjacent hospitals face critical shortages. Staff at N.L. Villa Memorial Medical Center confirmed instances of preventable wastage directly attributable to this gap, validating the need for the Blood Redistribution Optimization Algorithm (BROA) embedded in the proposed system.

Based on the identified operational gaps, the functional requirements of BloodLedger defines the specific behaviors and operations the system shall perform. They are organized by system modules to maintain traceability between each operational problem and its corresponding technical solution.

ID	Module	Requirement Description
FR-01	IoT Scan and Ingest	The system shall automatically register a blood unit into the digital inventory upon barcode or QR code scan, capturing blood type, component, unit ID, collection date, and expiration timestamp without manual data entry.
FR-02	IoT Scan and Ingest	The system shall reject any dispatch transaction where the scanned blood unit violates FEFO ordering, automatically surfacing the earliest-expiring qualifying unit of the same blood type.
FR-03	Real-Time Consortium Dashboard	The system shall provide a city-wide live inventory matrix displaying current blood stock levels per blood type per hospital, updated in real time upon

		each scan event.
FR-04	Real-Time Consortium Dashboard	The system shall display low-stock and overstock indicators per blood type per hospital, with configurable minimum and safe stock thresholds.
FR-05	Inter-Hospital Transfer Request and Approval	The system shall allow authorized hospital staff to submit a digital blood requisition specifying blood type, quantity, and clinical urgency level.
FR-06	Inter-Hospital Transfer Request and Approval	The system shall compute an RPS priority score for concurrent requisitions and present them to the supplying hospital in ranked order for fulfillment.
FR-07	Inter-Hospital Transfer Request and Approval	The system shall execute the BROA algorithm to generate redistribution recommendations when near-expiry units or overstock conditions are detected.
FR-08	Proactive Expiry Alert	The smart contract shall automatically flag blood units whose age reaches or exceeds 35 days and generate a redistribution recommendation via BROA.
FR-09	Proactive Expiry Alert	The system shall display expiry warnings on the dashboard alert center with the unit ID, blood type, current hospital, and days remaining until expiration.
FR-10	Geo-Verified Transfer Tracking	The system shall capture and permanently record the GPS coordinates and timestamp of a blood bag at outbound dispatch and inbound receipt scans.
FR-11	Geo-Verified Transfer Tracking	The system shall provide a transfer status tracker visible to all consortium nodes, displaying Dispatch, In Transit, and Receipt states for each active inter-hospital transfer.
FR-12	Role-Based Access	The system shall enforce four access tiers: Medical Technologist (scan and view local inventory), Hospital Admin (approve transfers, full local view), DOH/PRC (read-only city-wide), and System Administrator (full access).
FR-13	Offline Resilience	The system shall buffer IoT scan events in the local PostgreSQL database during internet connectivity loss and synchronize all buffered events to the Hyperledger Fabric ledger upon reconnection.

Table 2. Functional Requirements.

The IoT Scan and Ingest module (FR-01, FR-02) will directly address the ghost stock problem by replacing manual logbook entries with automated scan events. Every physical movement of a blood bag, whether an inbound receipt or an outbound dispatch, is immediately digitized and committed to the local database, eliminating the human memory dependency that produces inventory discrepancies during peak hours.

The Real-Time Consortium Dashboard (FR-03, FR-04) will resolve the visibility gap by aggregating live inventory data across all hospital nodes into a single interface. Stock levels will be updated in real time upon each scan event, making phone-based availability checks unnecessary.

The Inter-Hospital Transfer Request and Approval module (FR-05, FR-06, FR-07) will replace informal group chat coordination with a structured, blockchain-mediated digital requisition workflow. The RPS mechanism will ensure that when multiple hospitals simultaneously request the same blood type, the most clinically urgent and longest-waiting request is prioritized.

The Proactive Expiry Alert module (FR-08, FR-09) will shift blood wastage management from reactive disposal to proactive redistribution. Smart contract chaincode continuously monitors unit ages and autonomously triggers BROA-driven recommendations before expiration occurs.

The Geo-Verified Transfer Tracking module (FR-10, FR-11) will close the post-release tracking blind spot by binding GPS coordinates from the dispatch and receipt scans to the immutable blockchain record, providing chain-of-custody verification for every inter-hospital transfer.

Role-Based Access (FR-12) and Offline Resilience (FR-13) are cross-cutting functional requirements that will ensure the system remains secure and operable across the varied infrastructure conditions of Lipa City hospitals.

Beyond the functional capabilities, the non-functional requirements specify the quality attributes, constraints, and performance standards the system must satisfy independently of any particular function. These properties are essential for ensuring BloodLedger is trustworthy, legally compliant, and practically deployable within the healthcare environment of Lipa City.

ID	Category	Requirement	Description
NFR-01	Privacy	**Zero PHI Collection**	The system shall not collect, store, or process any patient medical records, donor names, or personally identifiable health information. All tracked data shall be restricted to blood unit identifiers, component type, and inventory lifecycle events.
NFR-02	Security	**Immutability and Tamper-Proof Audit Trail**	All inventory transactions and inter-hospital transfer records shall be committed to the Hyperledger Fabric ledger as immutable, append-only blocks that cannot be altered or deleted post-commitment.
NFR-03	Usability	**Low-Barrier Hardware**	The system shall operate on existing hospital computer terminals without requiring specialized hardware upgrades; only a 2D CMOS barcode scanner (USB or Bluetooth) compatible with ISBT-128 labels is required as additional hardware.
NFR-04	Compatibility	**ISBT-128 Label Standard Compliance**	The IoT scan module shall be capable of reading blood bag labels encoded under the ISBT 128 international

			standard to ensure compatibility with externally labeled units originating from PRC or other accredited blood centers.
NFR-05	Reliability	**Offline Resilience**	The local PostgreSQL database shall maintain uninterrupted scan event capture during connectivity loss; no data shall be lost during outages; all buffered transactions shall be synchronized to the blockchain ledger upon restoration of network connectivity.
NFR-06	Performance	**Dashboard Latency**	The real-time inventory dashboard shall reflect scan-triggered inventory changes within 5 seconds under normal network conditions across the consortium.
NFR-07	Compliance	**Data Residency**	All operational data shall be stored on-premise within the hospital's local area network, fully compliant with the data residency and minimization requirements of the Philippine Data Privacy Act of 2012 (RA 10173).

Table 3. Non-Functional Requirements

The zero PHI collection constraint (NFR-01) ensures the system does not create patient privacy risks and remains outside the scope of the more stringent medical records regulations that would otherwise complicate institutional adoption. The immutability requirement (NFR-02) is enforced architecturally through Hyperledger Fabric's append-only distributed ledger, guaranteeing that the audit trail cannot be manipulated after the fact. The low-barrier hardware requirement (NFR-03) is satisfied by designing the scan ingestion layer around widely available USB and Bluetooth barcode scanners compatible with ISBT-128 labels, enabling adoption without capital expenditure on specialized medical devices.

BloodLedger serves a tiered stakeholder network composed of primary blockchain nodes, secondary recipient institutions, and regulatory observers. The operational role and access privileges of each stakeholder are defined by their position within Lipa City's blood supply chain.

Category	Institution	Role in BloodLedger	Access Level
Primary Node	Mary Mediatrix Medical Center	Hospital with active internal blood bank; functions as supplier and recipient within the consortium network	Full local inventory management; transfer request and approval
Primary Node	Medix Medical Hospital	Hospital with active internal blood bank; functions as supplier and recipient within the consortium network	Full local inventory management; transfer request and approval
Primary Node	N.L. Villa Memorial Medical Center	Hospital with active internal blood bank; functions as supplier and recipient within the consortium network	Full local inventory management; transfer request and approval
Secondary Node	Metro Lipa Medical Center	Hospital without dedicated blood bank; functions strictly as a recipient/consumer node that submits digital requisitions and tracks transfer status	Dashboard view; blood request submission; transfer tracking
Secondary Node	Lipa City District Hospital	Hospital without dedicated blood bank; functions strictly as a	Dashboard view; blood request submission; transfer tracking

		recipient/consumer node	
Secondary Node	San Antonio Medical Center	Hospital without dedicated blood bank; functions strictly as a recipient/consumer node	Dashboard view; blood request submission; transfer tracking
Secondary Node	Divine Love Medical Center	Hospital without dedicated blood bank; functions strictly as a recipient/consumer node	Dashboard view; blood request submission; transfer tracking
Secondary Node	Ospital ng Lipa	Hospital without dedicated blood bank; functions strictly as a recipient/consumer node	Dashboard view; blood request submission; transfer tracking
Regulatory	Department of Health (DOH)	National regulatory body mandated by RA 7719; requires access to inventory summaries and audit data for compliance monitoring and policy oversight	Read-only city-wide dashboard; automated compliance report export
Regulatory	PRC Lipa Chapter	Oversee blood collection and distribution hub; observes outbound blood dispatches and supply monitoring	Read-only city-wide dashboard; view blood stock levels across all consortium hospitals; automated compliance report export

Table 4. BloodLedger Stakeholder and User Analysis

Primary nodes such as the Mary Mediatrix Medical Center, Medix Medical Hospital, and N.L. Villa Memorial Medical Center, are the consortium's inventory-holding institutions. Each operates a peer node within the Hyperledger Fabric network, maintains a synchronized ledger copy, and participates in transfer validation. Their medical technologists and laboratory staff will serve as the primary end-users of the IoT scan module and the dashboard's inventory management functions.

Secondary nodes are recipient hospitals without dedicated internal blood banks. These institutions will rely entirely on the real-time dashboard to identify available blood across the consortium and the digital requisition workflow to formally request transfers. The absence of internal blood bank infrastructure means secondary nodes will interact with the system exclusively as consumers, never as suppliers.

Both the Department of Health (DOH) and Philippine Red Cross (PRC) Lipa Chapter serves as the primary regulatory observer. As the agency mandated by Republic Act No. 7719 to oversee national blood services, the DOH requires verified, auditable blood inventory data. BloodLedger's automated report generation, which replaces the hospitals' current manual Google Sheet submissions at 9:00 AM and 4:00 PM, will directly address this operational burden, as confirmed by Ms. Atienza: "DOH CALABARZON has a google sheet that needs to be updated every 9 am and 4 pm. It includes the number of blood units per blood type per blood component." Similarly, PRC Lipa requires real-time visibility into hospital blood stock levels across the city to anticipate supply shortfalls and coordinate timely dispatch. Because both agencies currently lack a live window into hospital inventories, they depend on periodic or informal communications to assess where blood is available. The consortium dashboard will resolve this by granting both the DOH and PRC Lipa read-only access to live, city-wide stock data and an immutable audit trail, enabling evidence-based oversight without requiring direct operational involvement in hospital workflows.

Republic Act No. 7719. (1994). An act promoting voluntary blood donation, providing for an adequate supply of safe blood, regulating blood banks, and providing penalties for violation thereof. Official Gazette of the Republic of the Philippines.

Design of Software and Processes

The BloodLedger system operates as a four-layer pipeline that transforms physical blood bag movements into immutable, consortium-wide blockchain records. At the input layer, USB- or Bluetooth-connected barcode scanners at each member institution decode ISBT-128 labels to extract blood type and expiration identifiers, immediately logging each scan event to a local PostgreSQL buffer. This off-chain buffer serves as the system's resilience layer, the Node.js middleware relays buffered events to the Hyperledger Fabric network via the Fabric SDK, preserving every transaction even during connectivity outages and synchronizing deferred records upon restoration (NFR-05). At the blockchain core, three chaincode modules enforce the blood unit lifecycle, the Inventory Chaincode governs state transitions from Available through InTransit to Received or Expired; the Transfer Chaincode mediates inter-hospital custody via a dual-signature protocol; and the Expiry Alert Chaincode continuously monitors unit ages and autonomously triggers the Blood Redistribution Optimization Algorithm (BROA) at the expiry threshold. BROA operates as a two-tier pipeline, its first tier selects the soonest-expiring qualifying unit via First-Expired, First-Out (FEFO) logic, while its second tier applies Simple Additive Weighting (SAW) across urgency, stock shortage, and distance criteria to score and recommend the optimal recipient hospital within the Lipa City consortium. All state changes are committed as immutable ledger transactions, which are then surfaced through a React dashboard enforcing four role-based access tiers for Medical Technologist, Blood Bank Head, DOH/PRC Regulator, and System Administrator, giving each stakeholder precisely scoped visibility over the city-wide blood supply in real time.

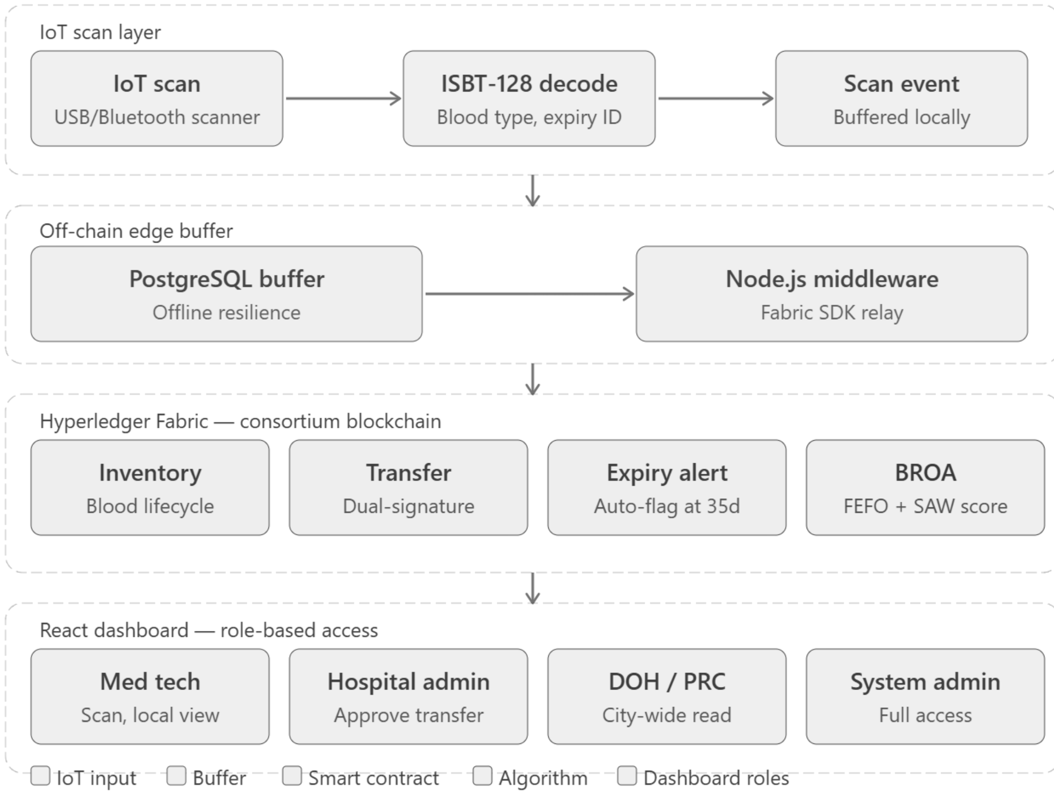

Figure 04. Operational Framework of BloodLedger.

To support this operational flow, the system architecture of BloodLedger will employ a multi-layered, distributed operational framework designed to seamlessly integrate physical blood inventory tracking with a secure, decentralized digital ledger. Unlike traditional centralized systems, this architecture is composed of five distinct but interconnected layers that work cohesively to facilitate real-time monitoring, secure data transmission, and automated inter-hospital blood redistribution.

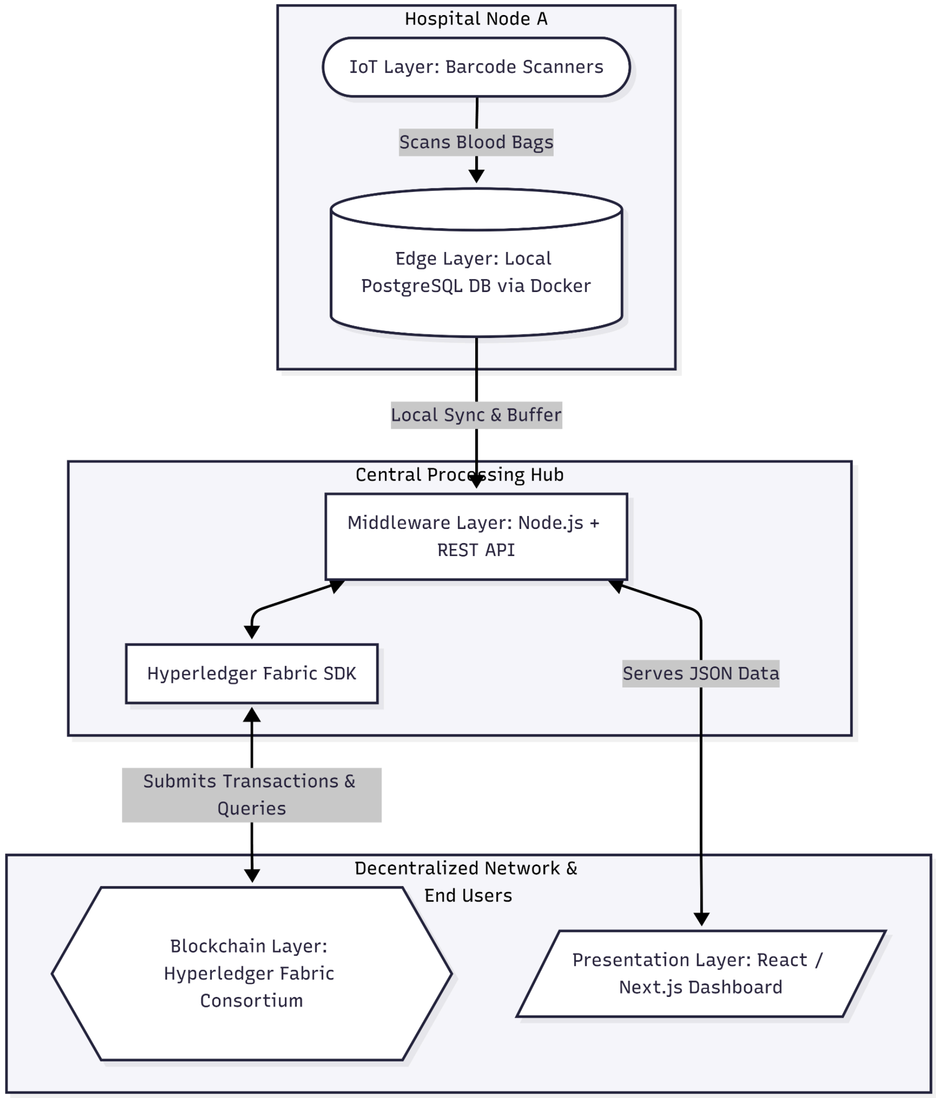

Figure 05. System Architecture of BloodLedger.

At the foundational level, the IoT Layer serves as the primary data entry point at each hospital node. Barcode scanners will be utilized to capture unique identifiers, collection dates, and expiration timestamps directly from the physical blood bags. This hardware-driven approach will eliminate the inaccuracies and delays associated with manual encoding, ensuring that the physical movement of the assets is instantaneously and accurately reflected in the digital space.

The data captured by the IoT devices is immediately transmitted to the Edge Layer, which consists of a PostgreSQL database instance containerized via Docker at each participating facility. Operating as an off-chain local database, this layer will act as a highly resilient data buffer. It will ensure continuous local operations and secure data retention within the hospital's local network, even in the event of external internet connectivity disruptions.

Acting as the critical bridge between localized hospital infrastructure and the decentralized network is the Middleware Layer, which utilizes Node.js, the Hyperledger Fabric Software Development Kit (SDK), and RESTful APIs. This layer will handle concurrent requests from the edge databases, validate the data payloads, and translate them into standardized transactions to be securely submitted to the blockchain network.

The core data integrity and automated business logic of BloodLedger reside in the Blockchain Layer, powered by a Hyperledger Fabric consortium network. As a permissioned distributed ledger, it ensures that all authorized member institutions maintain a synchronized, tamper-evident record of the blood inventory and inter-hospital transfers. Smart contracts deployed within this layer will automatically execute predefined business rules, governing inventory state changes and enforcing algorithms like the Blood Redistribution Optimization Algorithm (BROA).

Finally, the presentation layer features a dynamic React-based web dashboard. This front-end interface will translate the complex blockchain data into an accessible, real-time graphical user interface. It will provide medical technologists, hospital administrators, and regulatory bodies with a centralized portal to monitor city-wide blood stock levels, manage transfer requests, and receive automated alerts for critical stock or impending expirations.

The use case diagram for BloodLedger maps out the behavioral interactions between the system's human and system actors and the core functionalities provided by the decentralized platform.

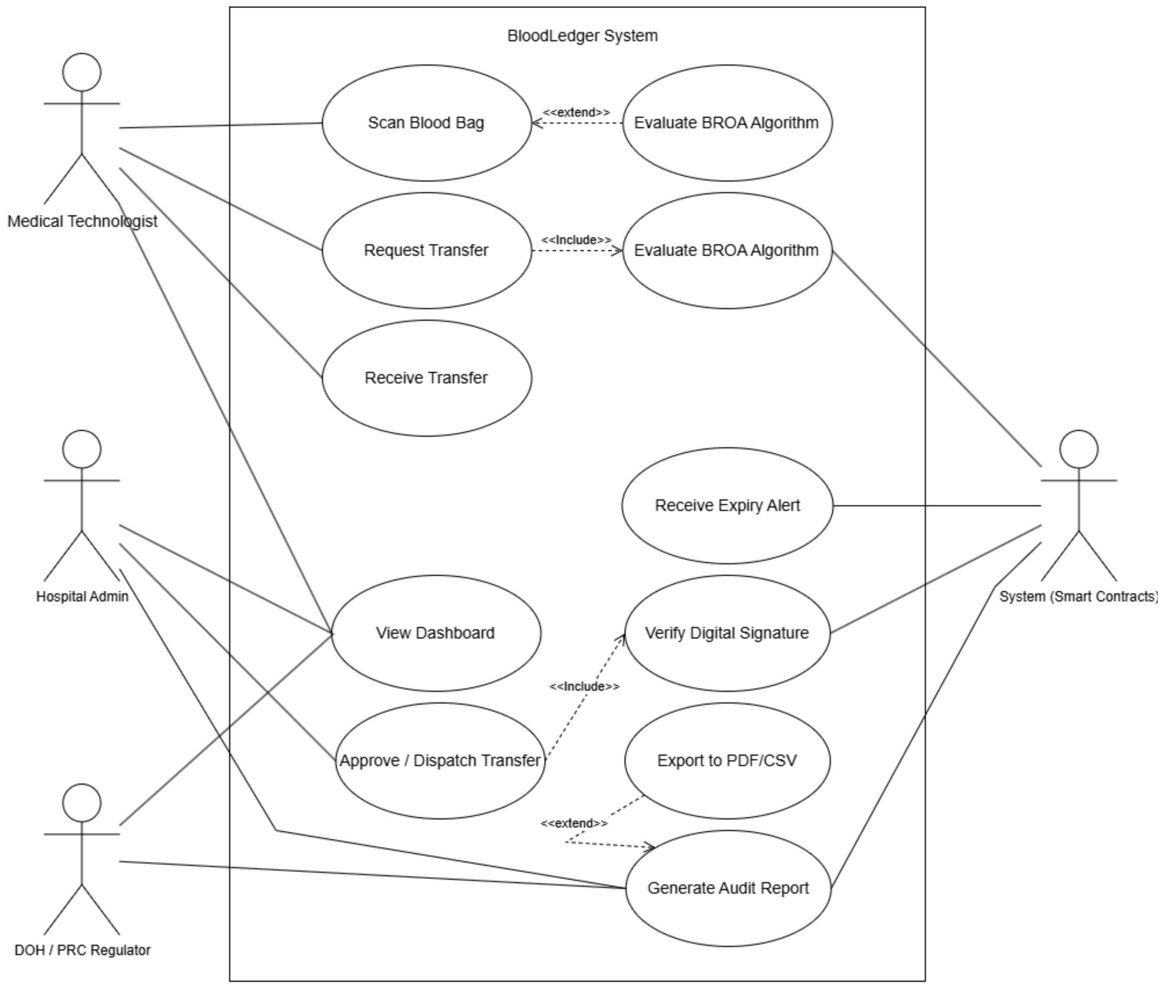

Figure 06. Use Case Diagram of BloodLedger.

This diagram establishes the boundaries of the system and defines role-based access controls to ensure secure and efficient operations. The primary human actors include the Medical Technologists, who handle the physical scanning and receiving of blood inventory; Hospital Administrators, who oversee stock levels and approve outbound transfers; and DOH/PRC Lipa Regulators, who require read-only oversight for city-wide auditing. Additionally, the System itself, powered by Smart Contracts, acts as an automated secondary actor, responsible for triggering expiration alerts, enforcing transfer protocols, and logging immutable audit trails. Figure 6 illustrates these actors and their respective authorized interactions within the BloodLedger environment.

The Data Flow Diagram maps the journey of information throughout the BloodLedger platform, visualizing how data payloads enter the system, how they are processed, and where they are permanently stored. At the highest level, the Level 0 Context Diagram establishes the system's macro-interactions, illustrating how external entities—such as IoT Barcode Scanners, hospital personnel, and DOH/PRC regulators—input data and receive outputs from the centralized BloodLedger system.

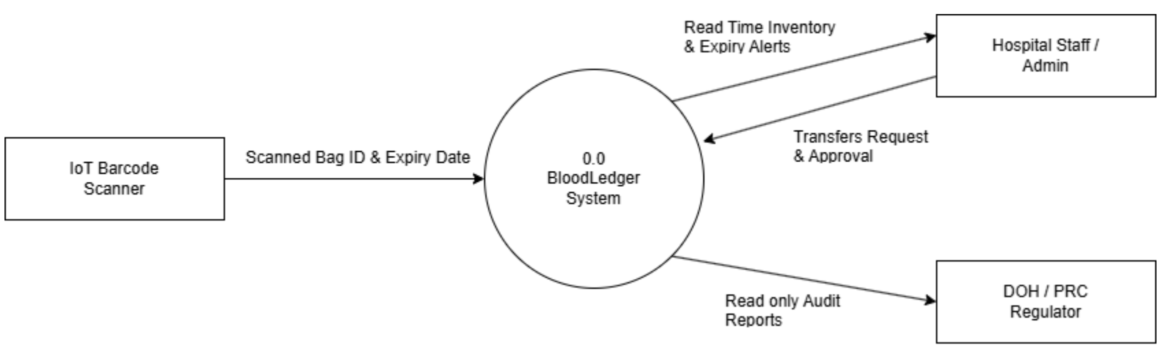

Figure 07. Data Flow Diagram of BloodLedger Level 0.

To provide a granular architectural view, the system is decomposed into a Level 1 Data Flow Diagram. This level outlines four primary functional subsystems: (1) Inventory Management, which processes incoming physical scans; (2) Transfer Management, which routes request and approval data between hospitals; (3) Expiry Monitoring, which continuously reads timestamps to trigger automated alerts; and (4) the Reporting Subsystem, which compiles read-only audits. The Level 1 DFD illustrates the precise data routing between the external users, these four processing nodes, and the dual-storage architecture consisting of the off-chain local PostgreSQL database and the on-chain Hyperledger Fabric distributed ledger.

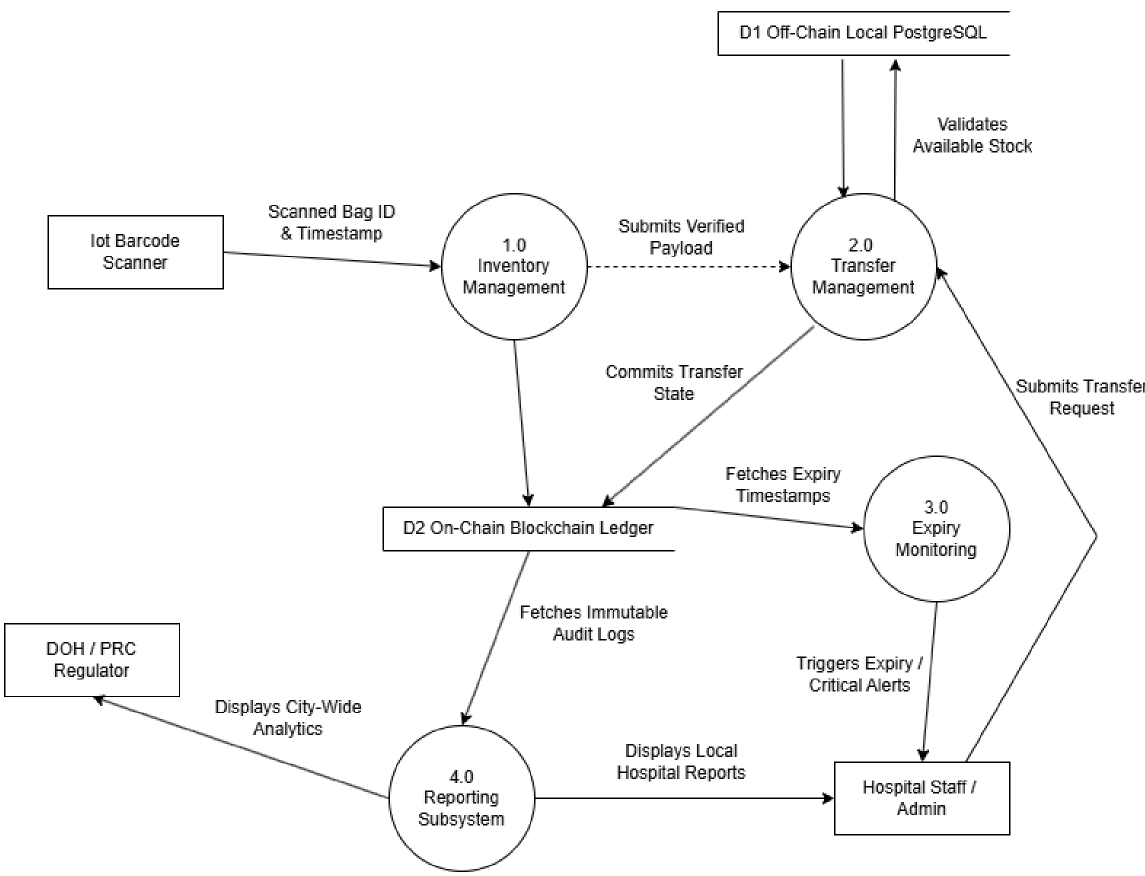

Figure 08. Data Flow Diagram of BloodLedger Level 1.

The Class Diagram for BloodLedger provides a comprehensive visual representation of the platform's static structure, meticulously defining the core data entities, their inherent attributes, functional operations, and the complex relational mappings that bind them together. As the foundational architectural blueprint, this model bridges the gap between physical healthcare workflows and their digital counterparts, detailing the interactions between key domain classes such as hospital nodes, blood units, users, and ledger transactions. Crucially, it serves a dual purpose vital to BloodLedger's hybrid architecture: it dictates the highly structured, relational schema for the off-chain PostgreSQL database optimizing high-speed queries, UI filtering, and local dashboard rendering while simultaneously defining the rigorous, immutable data structures required for the on-chain smart contracts. By clearly delineating system behaviors (such as unit registration, status updates, and inter-hospital transfers) alongside vital attributes (like ISBT-128 IDs, lifecycle statuses, and expiry timestamps), the class diagram ensures strict data consistency, structural integrity, and seamless state synchronization between the centralized database and the decentralized blockchain network.

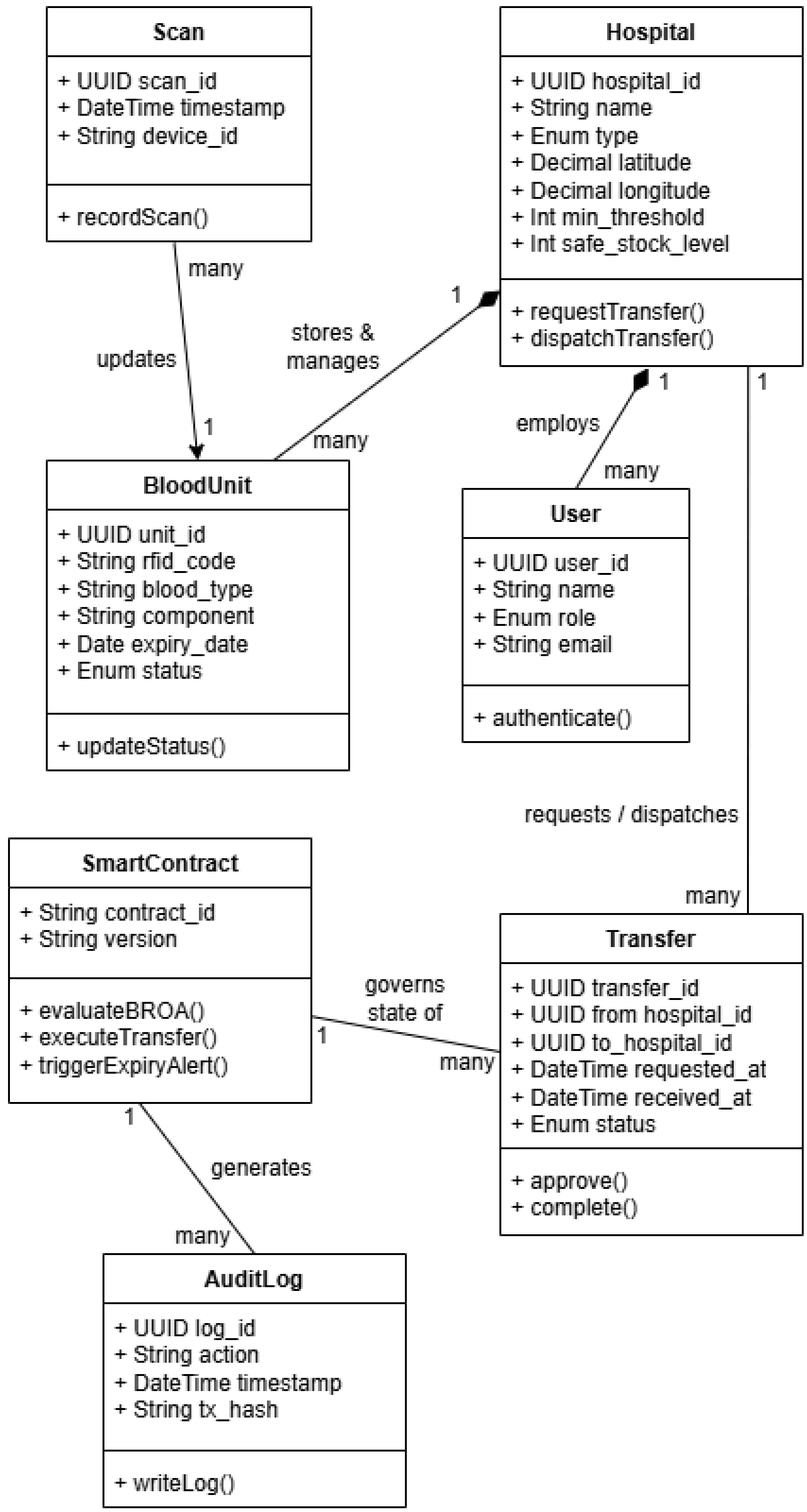
Scan
+ UUID scan_id
+ DateTime timestamp
+ String device_id
+ recordScan()
Hospital
+ UUID hospital_id
+ String name
+ Enum type
+ Decimal latitude
+ Decimal longitude
+ Int min_threshold
+ Int safe_stock_level
+ requestTransfer()
+ dispatchTransfer()
many
updates
1
stores &
manages
1
many
employs
1
many
1
BloodUnit
+ UUID unit_id
+ String rfid_code
+ String blood_type
+ String component
+ Date expiry_date
+ Enum status
+ updateStatus()
User
+ UUID user_id
+ String name
+ Enum role
+ String email
+ authenticate()
requests / dispatches
many
SmartContract
+ String contract_id
+ String version
+ evaluateBROA()
+ executeTransfer()
+ triggerExpiryAlert()
governs
state of
1
many
Transfer
+ UUID transfer_id
+ UUID from hospital_id
+ UUID to_hospital_id
+ DateTime requested_at
+ DateTime received_at
+ Enum status
+ approve()
+ complete()
1
generates
many
AuditLog
+ UUID log_id
+ String action
+ DateTime timestamp
+ String tx_hash
+ writeLog()

Figure 09. Class Diagram of BloodLedger.

The primary classes include the Hospital node, which acts as the central entity managing multiple BloodUnit objects and employing role-based User accounts. Operational classes such as Transfer and Scan model the transactional movement of resources and the IoT hardware data ingestion processes, respectively. Furthermore, the SmartContract class encapsulates the automated business logic that governs transfer state changes (such as enforcing the Blood Redistribution Optimization Algorithm), while the AuditLog class ensures an immutable, cryptographic record of all system events. The multiplicity and associations among these entities—such as a single transfer connecting a requesting hospital and a dispatching hospital—are visually represented to ensure strict data integrity, referential tracking, and seamless integration between the physical inventory and the blockchain ledger.

BloodLedger will implement a robust off-chain database architecture utilizing a self-hosted PostgreSQL instance containerized via Docker to manage the high-velocity data generated by IoT barcode scanners and ensure system resilience during network outages as seen in Figure 10.

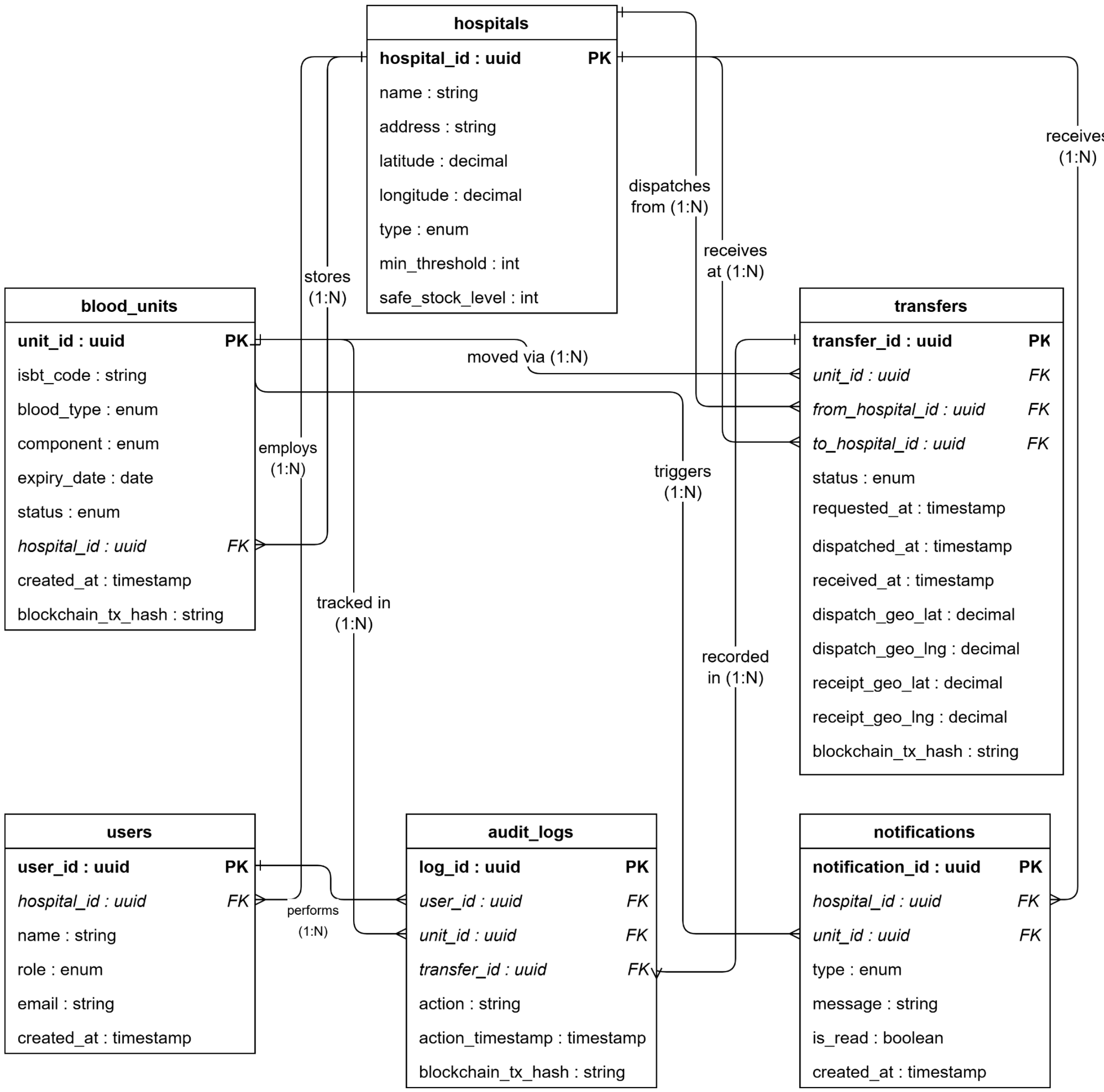

Figure 10. Database Architecture of BloodLedger.

PostgreSQL will serve as the primary off-chain Relational Database Management System (RDBMS), selected for its strict ACID-compliant transactional guarantees and advanced native support for JSON columns, allowing the seamless ingestion of semi-structured JSON payloads alongside structured relational tables. The schema will consist of six core tables—blood_units, hospitals, transfers, users, audit_logs, and notifications—secured by strict foreign key constraints to enforce absolute referential integrity. By containerizing this database and the Node.js middleware via Docker, the entire off-chain system will be launched on the hospital's Local Area Network (LAN), removing third-party cloud dependencies and ensuring uninterrupted operation during internet outages while fulfilling the data residency mandates of the Philippine Data Privacy Act of 2012 (RA 10173). Crucially, this local PostgreSQL database will act as a high-speed buffer, establishing a "Sync Boundary" where data is only pushed to the permanent Hyperledger Fabric ledger once a transaction is verified and ready for immutable commitment.

The proposed framework will utilize Hyperledger Fabric to establish a permissioned consortium topology. Unlike public blockchains, this architecture will restrict access to authenticated healthcare institutions. The network will be structured so that each participating institution will host one peer node. To ensure rapid and fault-tolerant consensus across this decentralized architecture, the network will utilize a single ordering service powered by the RAFT consensus algorithm.

To maintain strict access control and cryptographic security, a Membership Service Provider (MSP) will be implemented. The MSP will issue X.509 digital certificates to each institution, guaranteeing that only verified entities can join the network or submit transactions. Once authenticated, all participating hospitals will communicate over a single shared consortium channel. This unified channel design will ensure that all nodes maintain synchronized, real-time visibility of the city blood supply. The underlying ledger structure will consist of immutable, append-only blocks, guaranteeing that historical supply chain events can never be altered or deleted.

The operational logic of the network will be enforced entirely by smart contracts, technically referred to as chaincode within the Hyperledger ecosystem. Written in Node.js, these self-executing scripts will handle validation and state changes without requiring intermediary human oversight. The chaincode will be divided into three primary modules:

Inventory Chaincode will manage the foundational tracking of physical assets through functions such as CreateBloodUnit and UpdateStatus. It will securely transition the blood unit's state across its predefined lifecycle sequence (Available → Reserved → InTransit → Received / Expired). Transfer Chaincode will govern the secure lateral movement of blood bags between facilities using functions including InitiateTransfer, DispatchWithGeoTag, and ReceiveWithGeoTag. To finalize any inter-hospital transaction, the ValidateSignatures function will require cryptographic digital signatures from both the dispatching sender and the acknowledging receiver, preventing any unilateral manipulation of the custody chain. Expiry Monitor Chaincode, acting as an automated sentry, will continuously calculate the age of active blood units. It will automatically trigger an alert when a unit's age reaches or exceeds 35 days for example, autonomously invoking the redistribution recommendation algorithms to prevent biological wastage.

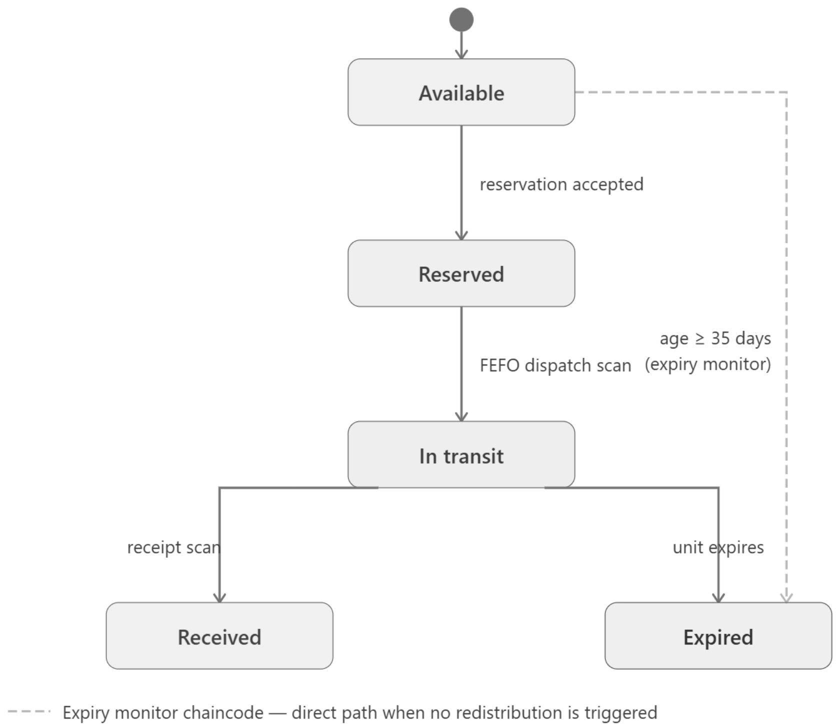

Figure 11. Blockchain state machine for blood unit lifecycle.

The blockchain state machine provides the secure infrastructure and strict lifecycle tracking for these blood units, and it relies on intelligent logic to autonomously determine exactly how and where near-expiry assets should be routed. To drive these logistical decisions, the Blood-Redistribution Optimization Algorithm (BROA) will serve as the unique algorithmic contribution of the BloodLedger system. Operating as a two-tier composite pipeline, BROA will integrate two established decision-making methodologies into a single programmatic workflow. This algorithm will fundamentally shift the consortium from a reactive, emergency-based procurement model to a proactive, data-driven supply management network.

The first tier governs the outbound selection of blood units using the First Expired, First Out (FEFO) protocol. Unlike traditional First In, First Out (FIFO) systems—which sequence items based on their arrival time—FEFO sequences units strictly by their expiration date, a critical distinction in blood management where shelf life directly dictates biological waste risk.

This protocol will be enforced at the IoT scan layer. When a dispatch event occurs, the system will query the available stock in ascending order of expiration dates, autonomously surfacing the earliest-expiring qualifying unit without requiring manual sequencing by laboratory staff. If a medical technologist attempts to scan and dispatch a newer blood unit while a sooner-to-expire unit of the exact same blood type remains available, the smart contract (chaincode) will automatically reject the transaction. This ensures absolute compliance with biological waste reduction protocols.

The second tier governs where the selected blood unit will be sent. When BROA detects a near-expiry unit or an overstocked facility, it will trigger a Multi-Criteria Decision Making (MCDM) method known as Simple Additive Weighting (SAW). Because the criteria rely on different scales of measurement (e.g., unit counts vs. kilometers), SAW will first apply a min-max normalization step to each criterion to ensure they are mathematically comparable. It will then multiply them by assigned weights and calculate a composite score for every candidate recipient hospital in the Lipa City network. The routing decision relies on the following scoring formula: Score = (w1 × NormalizedUrgency) + (w2 × NormalizedStockShortage) − (w3 × NormalizedDistance). To ensure optimal clinical outcomes, the criteria weights are defined and justified as follows.

Urgency criteria with a weight of 0.50 (w1) represents the recipient hospital's stock deficit relative to its absolute minimum critical threshold. It receives the highest weight because preventing an immediate life-threatening shortage takes absolute clinical precedence. Then, StockShortage criteria weighing 0.30 (w2) represents the number of units the recipient hospital needs to reach its general "safe" operational stock level. This holds medium weight, as balancing the overall network prevents future emergencies. Also, Distance criteria that weighs 0.20 (w3) is referenced from a preloaded static distance table (in kilometers) between consortium hospitals in Lipa City. As a subtractive penalty, it holds the lowest weight because while geographical proximity reduces transit time, Lipa City's relatively small area makes distance a secondary concern compared to clinical urgency.

The hospital yielding the highest composite score will receive the redistribution recommendation. Upon approval, the smart contract will validate the route and permanently log the recommendation as an immutable transaction on the blockchain.

```Go
FUNCTION ExecuteBROA(bloodType, dispatchingHospital):
    // TIER 1: FEFO Unit Selection
    availableUnits = QueryBlockchainInventory(bloodType, dispatchingHospital)
    SortAscendingByExpiryDate(availableUnits)
    selectedUnit = availableUnits[0] // Selects the soonest-to-expire unit

    IF selectedUnit.age >= 35_DAYS OR dispatchingHospital.status == OVERSTOCKED:
        // TIER 2: SAW Destination Scoring
        bestScore = -1
        targetHospital = NULL

        FOR EACH hospital IN LipaConsortiumNetwork:
            IF hospital != dispatchingHospital:
                urgencyScore = CalculateUrgency(hospital, bloodType)
                shortageScore = CalculateShortage(hospital, bloodType)
                distanceScore = LookupStaticDistance(dispatchingHospital, hospital)

                // Min-Max Normalization step
                normalizedUrgency = urgencyScore / MAX
                normalizedShortage = shortageScore / MAX
                normalizedDistance = distanceScore / MAX

                // SAW Formula implementation
                currentScore = (0.50 * normalizedUrgency) + (0.30 *
normalizedShortage) - (0.20 * normalizedDistance)

                IF currentScore > bestScore:
                    bestScore = currentScore
                    targetHospital = hospital

        RETURN RouteRecommendation(selectedUnit, targetHospital)
    ELSE:
        RETURN "No redistribution required."
```

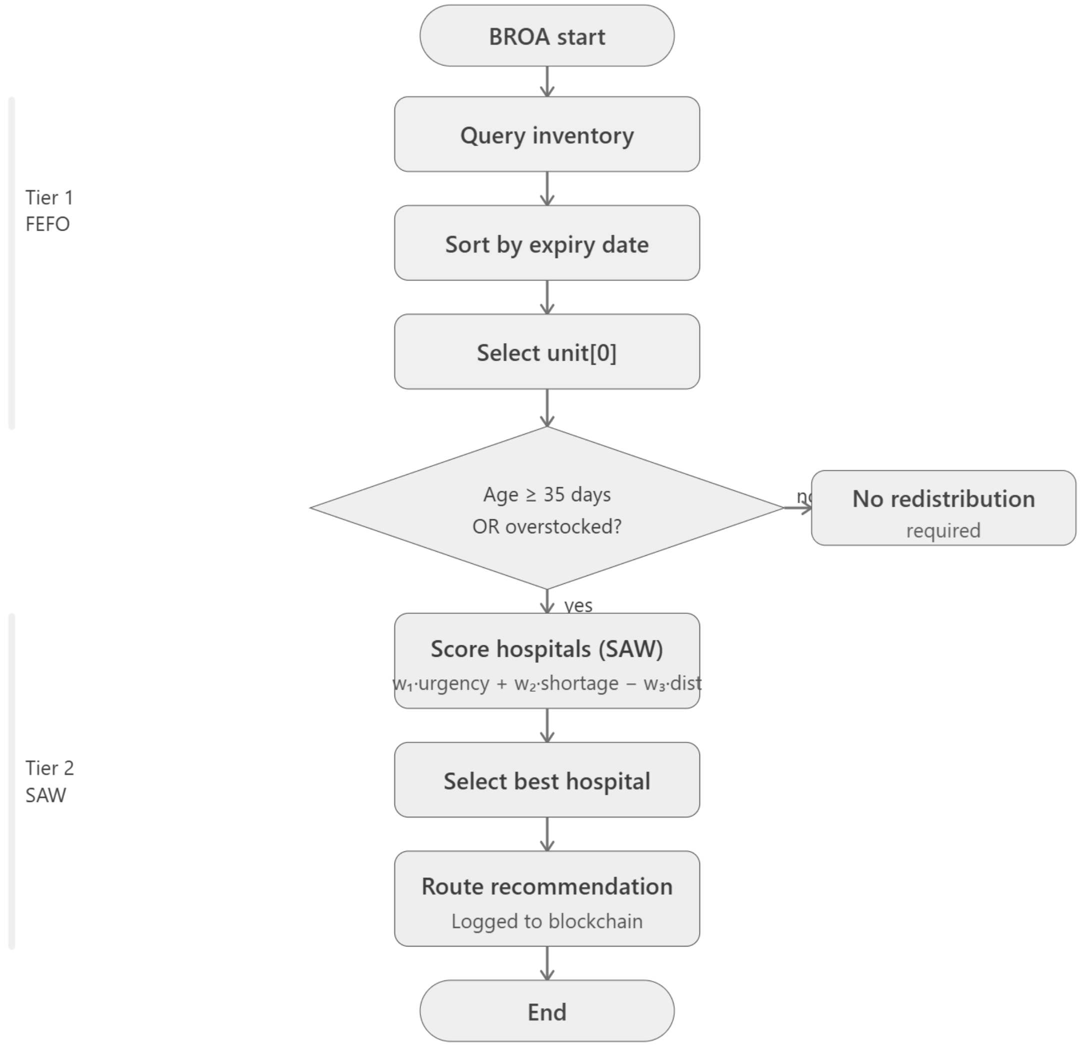

Figure 12. Logical pipeline of the Blood-Redistribution Optimization Algorithm (BROA).

While BROA governs the proactive outbound redistribution of blood supplies, the Request Priority Scoring (RPS) algorithm will manage inbound demand contention. RPS will be designed specifically to resolve conflicts that arise when multiple requestor hospitals, those without dedicated internal blood banks, simultaneously compete for the exact same limited blood stock at a supplying node.

To establish a fair, data-driven queue for fulfilling these simultaneous requests, the smart contract will compute a priority score for each request using the following static weighted formula: Score = (w_1 × ClinicalUrgency) + (w_2 × CappedWaitTime). To ensure the algorithm prioritizes life-saving interventions while remaining fair to older requests, the variables and weights will be defined and quantified as follows.

ClinicalUrgency weighing 0.70 (w_1) will be a staff-declared urgency level submitted alongside the digital requisition. This will be quantified mathematically: Critical is equivalent to 3 (i.e., immediate life-saving transfusion required), Urgent is equal to 2 (i.e., transfusion required within hours), and Routine is 1 (i.e., elective or stable procedure). Meanwhile, CappedWaitTime that weighs 0.30 (w_2) will be the elapsed duration measured in hours since the request was immutably logged on the blockchain. To prevent mathematical overriding of new Critical requests by older Routine requests, this value is strictly capped at a maximum of 24 hours. This acts as a controlled mechanism to prevent the starvation of lower-urgency requests.

The requestor hospital that yields the highest RPS score will be prioritized for fulfillment from the supplier's available stock. Then, it is critical to establish the operational boundaries of the RPS algorithm. Requestor hospitals will function strictly as consumers within the consortium; they will not hold physical inventory and, consequently, will not participate in BROA redistribution decisions. Furthermore, RPS will exclusively govern demand contention. All other requestor-side operations such as checking real-time blood availability across the city, identifying the nearest supplier, and tracking the transit status of a dispatched unit will be handled via direct blockchain ledger queries rather than algorithmic computation.

```go
FUNCTION ExecuteRPS(incomingRequests, availableSupply):
    requestQueue = []

    FOR EACH req IN incomingRequests:
        // 1. Quantify Clinical Urgency
        IF req.urgency == "Critical" THEN urgencyValue = 3
        ELSE IF req.urgency == "Urgent" THEN urgencyValue = 2
        ELSE IF req.urgency == "Routine" THEN urgencyValue = 1

        // 2. Calculate elapsed wait time (in hours) and apply 24-hour cap
        hoursWaiting = (GetCurrentBlockchainTime() - req.timestamp)
        cappedWait = MIN(hoursWaiting, 24)

        // 3. Apply RPS Formula
        req.priorityScore = (0.70 * urgencyValue) + (0.30 * cappedWait)

        // Add to evaluation queue
        requestQueue.push(req)

    // 4. Sort the queue from highest score to lowest score
    SortDescendingByScore(requestQueue)

    // 5. Output the ranked queue for the supplier to fulfill
    RETURN requestQueue
```

Figure 13. Pseudocode for the Request Priority Scoring (RPS) mechanism governing demand contention.

Following the algorithmic resolution of supply and demand, the system must translate these digital decisions into real-world action. The Inter-Hospital Transfer Protocol will act as the execution layer of BloodLedger, bridging the algorithmic decision-making with the physical logistics of moving blood between Lipa City medical facilities. This protocol will operate as a strict, blockchain-enforced state machine (Dispatch → Transit → Receipt) to guarantee the secure, traceable, and compliant transfer of biological assets. The physical transfer of a blood unit will follow a sequential, tamper-proof workflow governed entirely by smart contracts.

To begin the sequence, the process will initiate proactively when the Expiry Monitor chaincode flags a near-expiry unit, and the BROA algorithm successfully recommends an optimal recipient hospital. Upon acceptance of this transfer recommendation, the selected blood unit's status will immediately change to a reserved state on the consortium dashboard. This crucial locking mechanism will prevent double-booking and ensures the unit cannot be accidentally claimed by another facility or used locally.

Upon initiation of the dispatch phase, the sending hospital's laboratory staff will submit an outbound scan using an IoT barcode scanner. Before approving this dispatch, the smart contract will strictly validate FEFO compliance, ensuring that if the staff attempts to dispatch a unit while a sooner-to-expire unit of the exact same blood type is available in their inventory, the transaction is rejected. Upon a valid scan, the system will capture a GPS geo-tag, initiate the ValidateSignatures chaincode function to collect the sender's cryptographic digital signature, and permanently record the exact geographic location and timestamp of the dispatch event on the immutable blockchain.

Once the unit is successfully dispatched, its ledger status will officially update to indicate it is in transit. During this critical phase, the physical unit's transit status will become visible in real-time to all participating consortium nodes as well as the DOH and PRC Lipa regulatory dashboard, thereby maintaining absolute city-wide transparency.

Finally, upon physical arrival at the destination hospital, the receiving laboratory staff will perform an inbound scan. This action will capture a final destination geo-tag, verifying the exact physical delivery coordinates. To conclude the workflow, the smart contract will execute the final step of the ValidateSignatures function by collecting the receiver's cryptographic signature, verifying the custody chain, and simultaneously updating both the sender's and receiver's local inventories, officially closing the transaction loop.

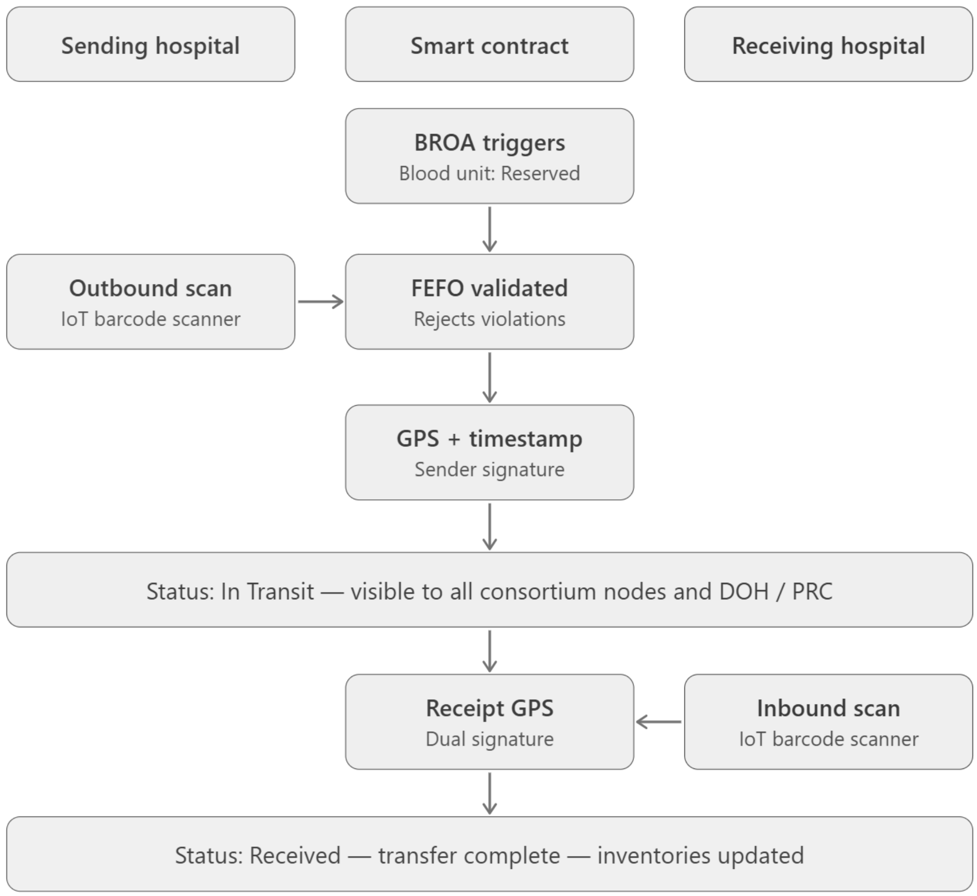

Figure 14. Inter-hospital transfer sequence diagram illustrating the Dispatch, Transit, and Receipt state machine.

User Interface (UI) and User Experience (UX) Design

The User Interface (UI) and User Experience (UX) design of the BloodLedger platform will be developed in accordance with established Human-Computer Interaction principles, specifically prioritizing clarity and efficiency for high-stress emergency environments. Because the system will be utilized by medical technologists, hospital administrators, and regulatory bodies who may not have advanced technical backgrounds, accessibility and ease of use are paramount. The interface will employ strategic color-coding to denote blood types and urgency levels—such as utilizing red for critical shortages, amber for warnings, and blue for informational alerts. These design decisions will ensure that users can rapidly process complex inventory data and execute time-sensitive decisions regarding inter-hospital blood redistribution.

Figure 15 illustrates the Login and Hospital Authentication screen. The interface will employ role-based access control, requiring staff to select their specific hospital node, regulatory agency (such as DOH or PRC Lipa) and membership to authenticate their session. The design will emphasize institutional trust and ensure end-to-end encryption, requiring users to sign in via secure blockchain authentication.

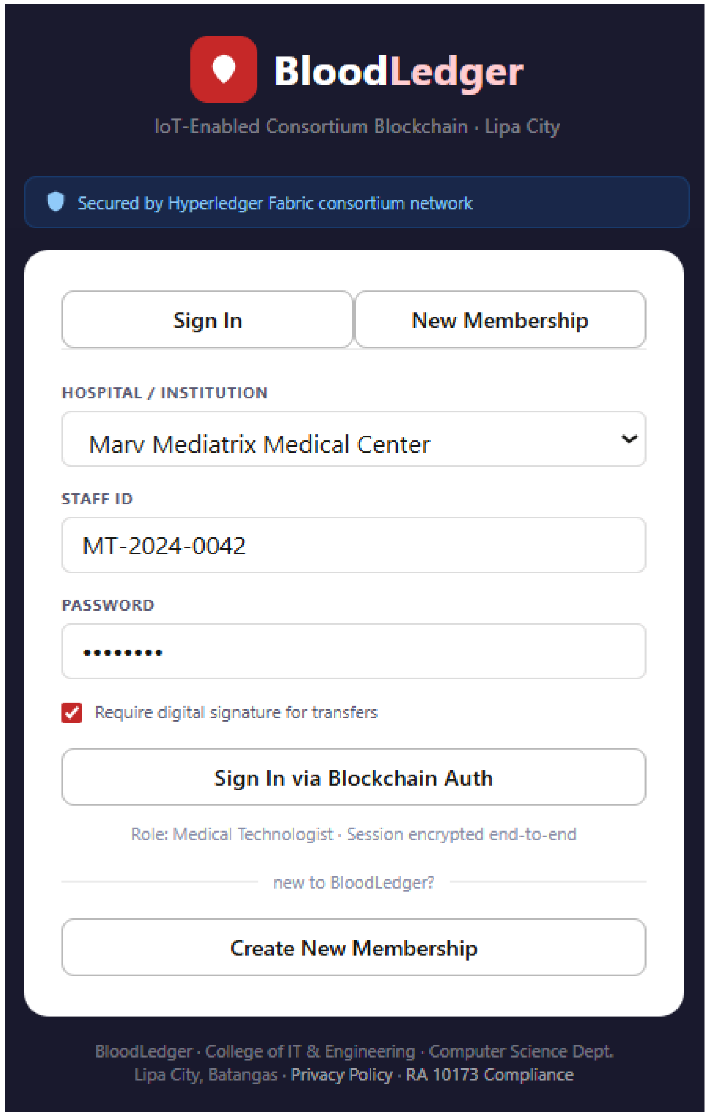

Figure 15. Login and Hospital Authentication Interface.

The core of the BloodLedger interface is the Live Data Matrix Dashboard (Figure 16). This module will feature statistical summary cards displaying total available units, critical stock alerts, impending expirations, and pending transfer requests across the consortium. A city-wide blood type matrix will utilize a color-coded traffic light system (green indicating adequate, amber for low, and red for critical) to instantly convey inventory status. Furthermore, a recent transfers panel will track the real-time status of inter-hospital routing.

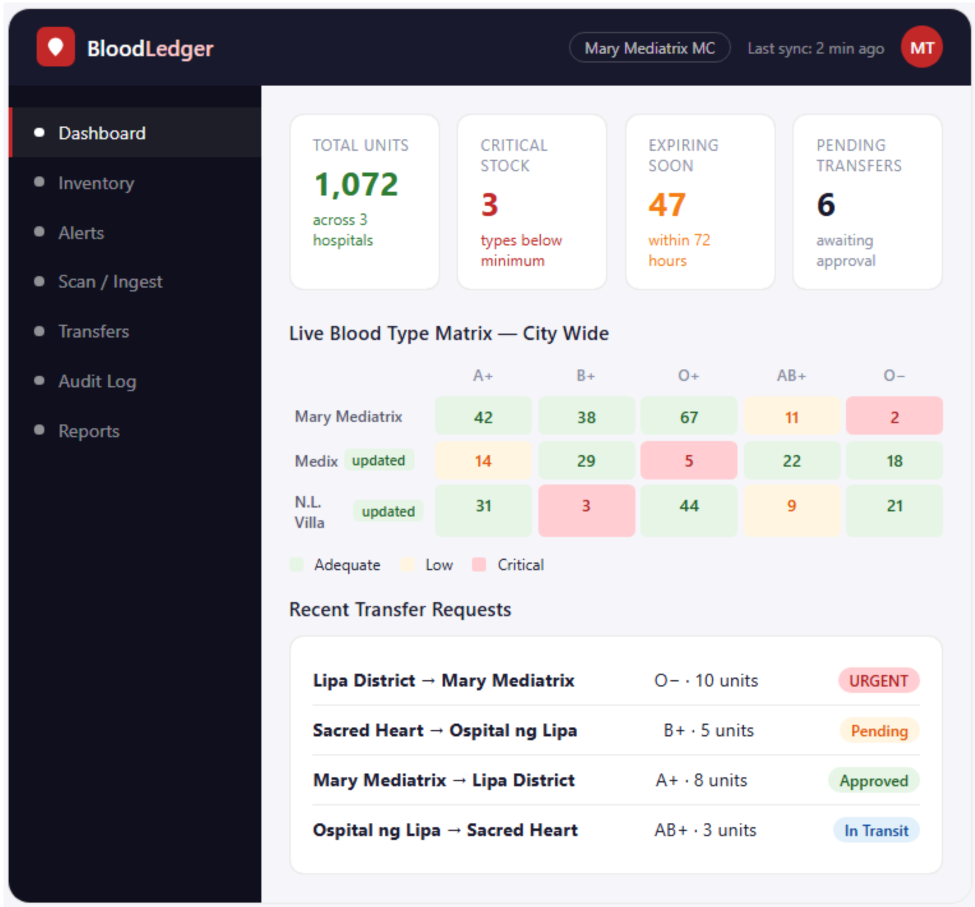

Figure 16. Live Data Matrix Dashboard.

Figure 17 depicts the Alert Center, which will utilize a severity-layered notification system. Critical alerts will be visually highlighted with a red top-bar, while warnings and informational notices use amber and blue, respectively. Each notification card will explicitly state the triggering cause, present the smart contract's recommended action (such as an optimal redistribution source identified by the BROA algorithm), and provide quick-action buttons for rapid response.

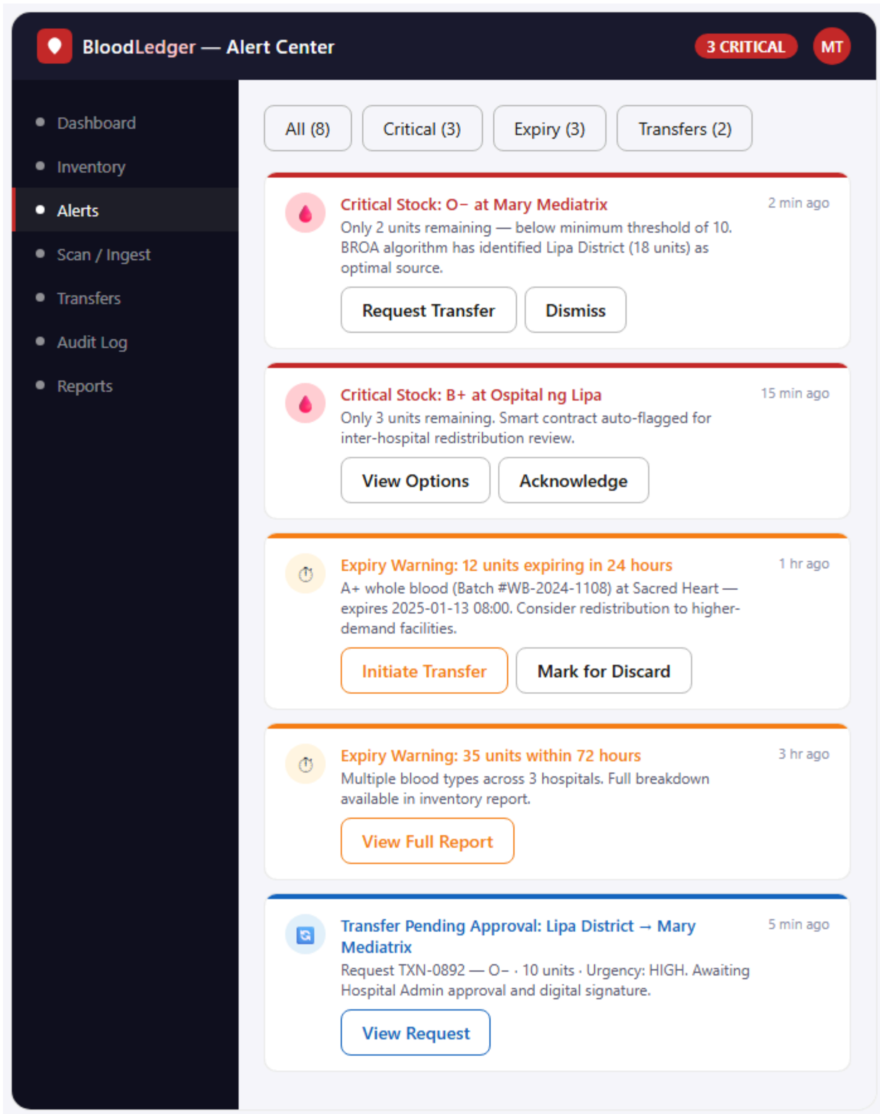

Figure 17. Alert Center Interface.

The Blood Unit Inventory screen will enable Blood Bank Representatives and Administrators to rapidly prioritize redistribution and prevent wastage by displaying detailed, blockchain-traceable blood bag data including ISBT-128 IDs, blood types, and expiration metrics within a dynamically filterable, color-coded interface automatically sorted by First Expired, First Out (FEFO) to enforce smart contract sequencing rules.

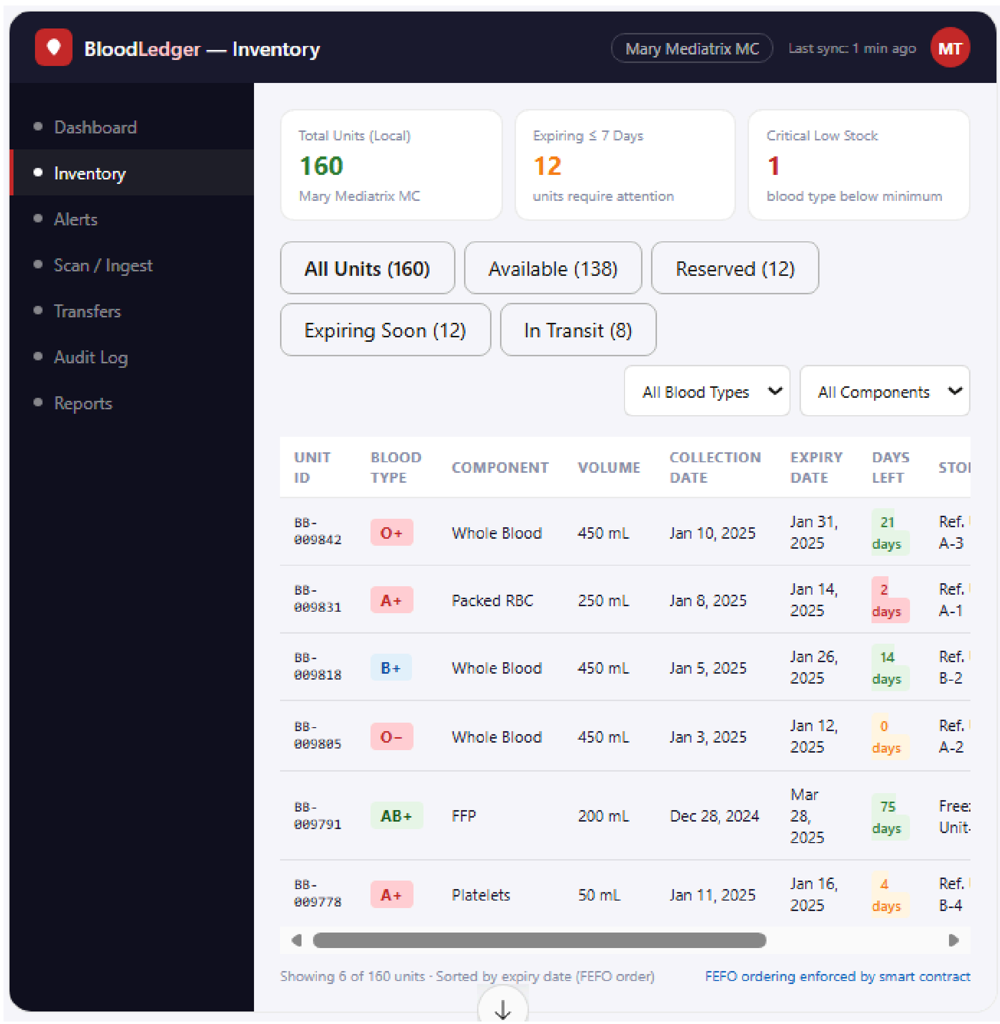

Figure 18. Blood Unit Inventory Interface.

The Scan and Ingest Interface (Figure 19) is designed for seamless data entry at the hospital edge layer. It will feature a live scanner viewport for IoT barcode reading, accompanied by manual entry fallbacks. Upon scanning, a confirmation panel will display the parsed blood type, component details, expiration date, and donor screening status. This will ensure the medical technologist can visually verify the data before committing it as an immutable transaction to the blockchain ledger.

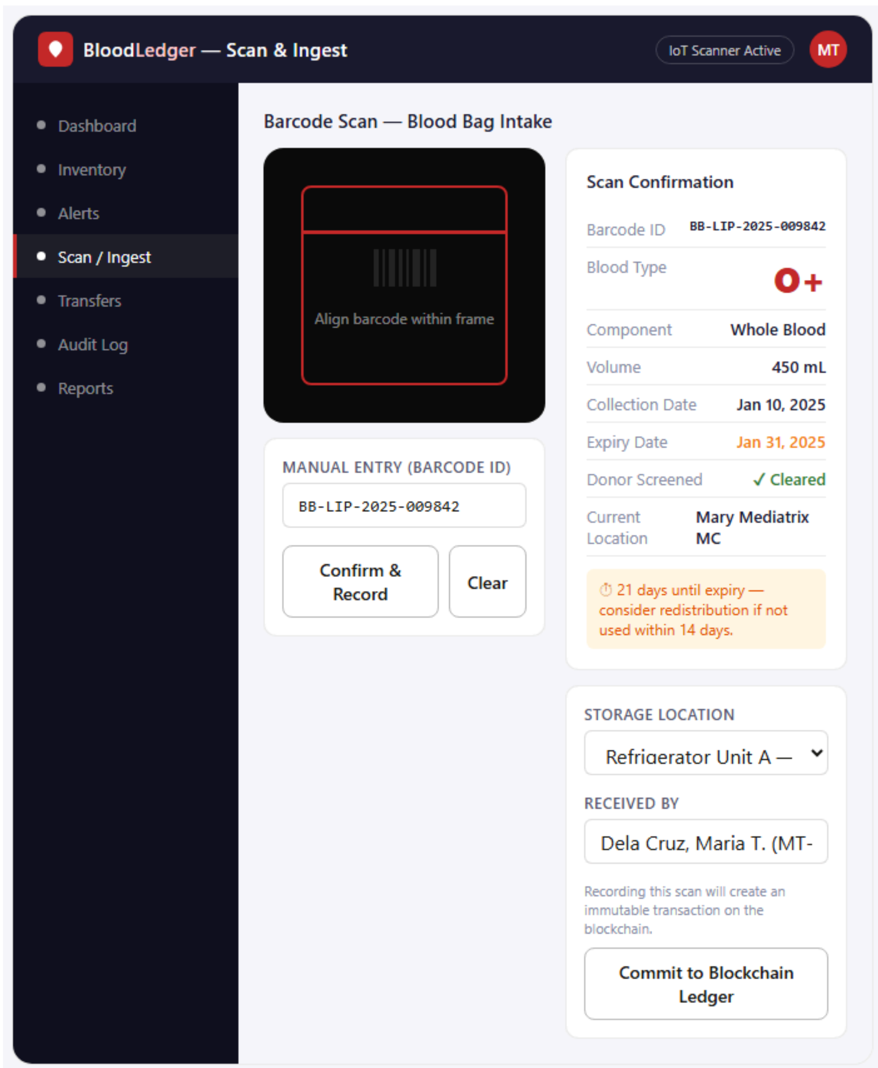

Figure 19. Scan and Ingest Interface.

Figure 20 shows the Transfer Request and Approval Workflow, which breaks down complex redistribution logistics into a guided six-step progression.

The interface will incorporate a blood type selector and prominently display the Blood Redistribution Optimization Algorithm (BROA) scoring card, recommending the optimal source hospital based on stock levels, proximity, and expiration urgency. An integrated authorization panel will be provided for Blood Bank Head to securely approve or reject requests using cryptographic digital signatures.

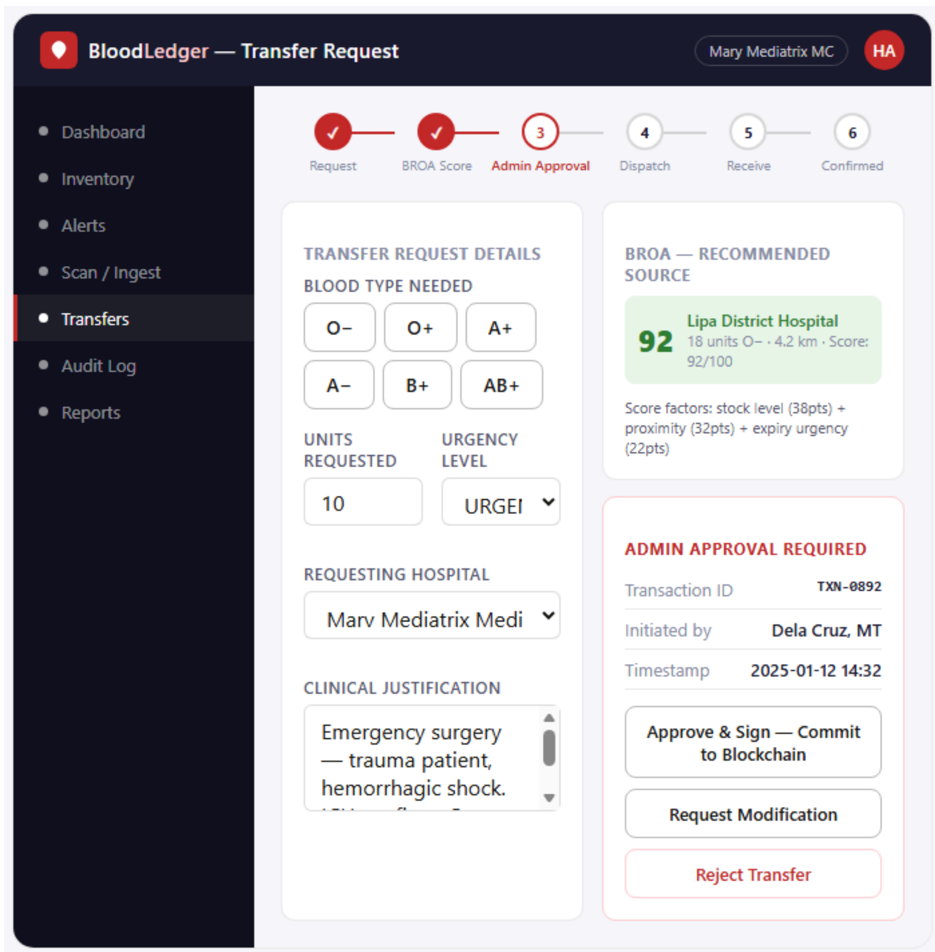

Figure 20. Transfer Request and Approval Workflow.

To ensure transparency and accountability, the Audit Log Viewer (Figure 21) will present a comprehensive, immutable history of system transactions. The interface will feature a clearly labeled non-editable table displaying timestamps, cryptographic transaction hashes, and color-coded action types, supplemented by filtering chips and search functionalities for efficient auditing.

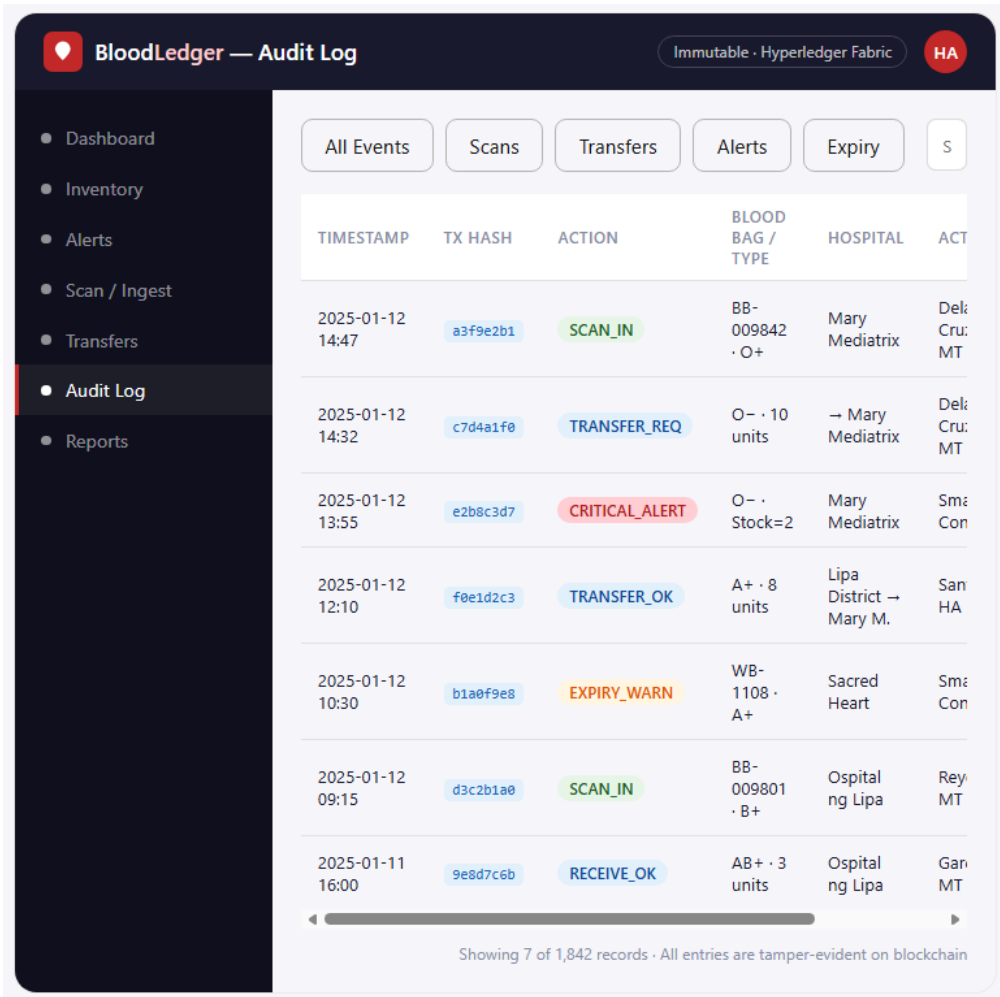

Figure 21. Audit Log Viewer.

Finally, the DOH/PRC Lipa Reporting View (Figure 22) provides regulatory bodies with a read-only oversight dashboard. This interface will highlight macro-level Key Performance Indicators (KPIs) alongside horizontal bar charts detailing inter-hospital blood distribution and transfer completion rates. This comprehensive view will facilitate data-driven policy making and compliance tracking, with built-in functionalities to export data to PDF or CSV formats.

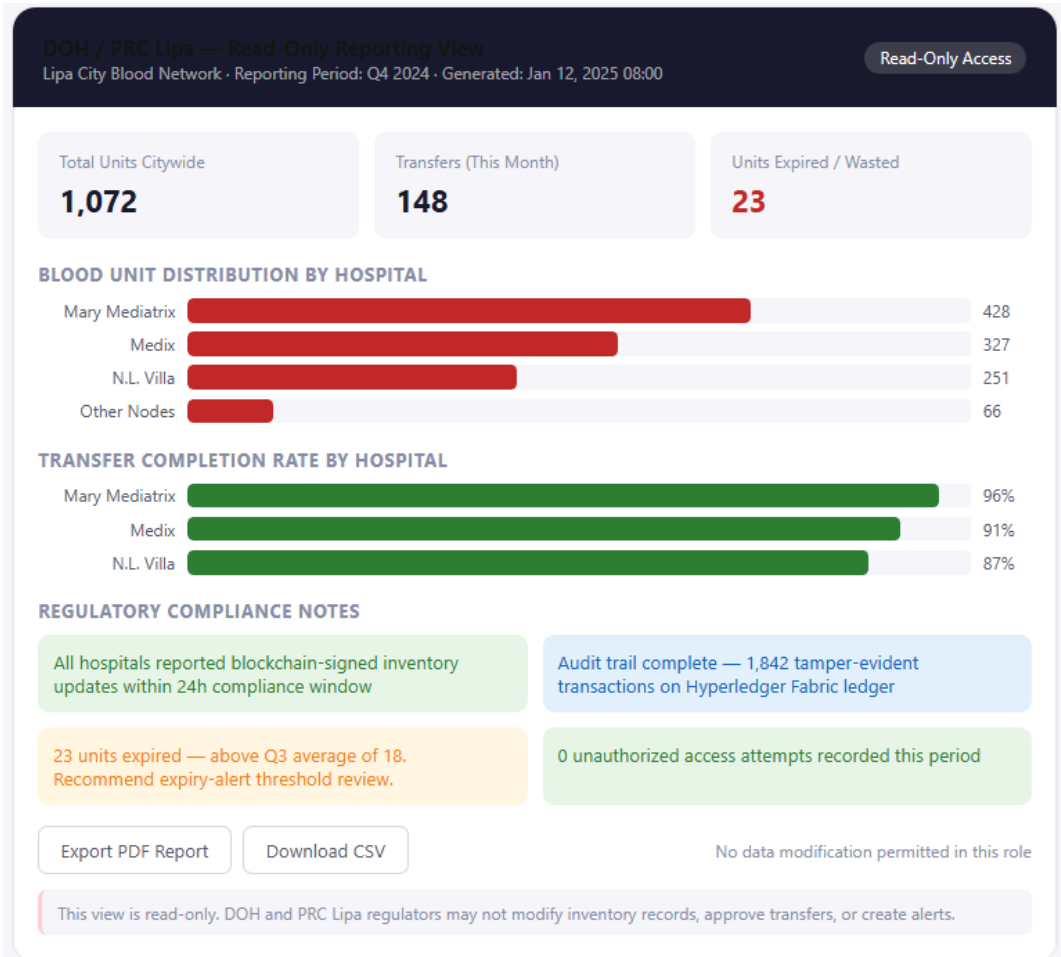

Figure 22. DOH/PRC Lipa Reporting View.

Implementation Plan

This section outlines the technical requirements, development tools, and deployment strategy necessary to transition BloodLedger from a conceptual architecture into a fully functional prototype. The project follows a Scrum-based Agile Software Development Life Cycle (SDLC), an iterative approach that organizes work into time-boxed Sprints, each producing tested, working software reviewed by stakeholders at its close. The lifecycle is structured into six sequential stages: Planning, Design, Development (i.e., executed across five focused Sprints), Testing, Deployment, and Maintenance. The Planning and Design stages collectively constitute Sprint 0, the pre-development foundation that defines the project's scope, architecture, and backlog before iterative development begins. Each of the five development Sprints (Sprint 1 through Sprint 5) integrates its own Sprint Planning, development, sprint-level testing, and Sprint Review within its two-week timeframe, ensuring that stakeholder feedback from blood bank personnel is continuously incorporated throughout the build. Formal system-level validation, comprising System Testing and User Acceptance Testing (UAT), follows after all development Sprints are complete, after which the system proceeds to phased parallel deployment and long-term maintenance.

Technology Requirements

Technology	Description and Purpose
Hyperledger Fabric	Selected over public blockchains to provide a permissioned consortium environment, ensuring enterprise-grade privacy, restricted access, and regulatory compliance.
React	Utilized for the frontend presentation layer. Its component-based architecture and virtual DOM guarantee low-latency dashboard rendering, allowing the UI to update instantly during regional emergencies.

Node.js	Serves as the core backend infrastructure. Its event-driven, non-blocking architecture efficiently processes concurrent transfer requests and IoT scan events from multiple hospitals simultaneously.
Fabric SDK	Integrated within the Node.js middleware to abstract complex cryptographic and network operations, streamlining how the server interacts with the blockchain chaincode.
PostgreSQL	Provides an ACID-compliant off-chain buffer. It securely holds incoming IoT scan events locally before they are permanently synchronized with the blockchain network.
Docker and Compose	Utilized to package the database, middleware, and blockchain peers into isolated containers, allowing the entire full-stack environment to be launched uniformly on any hospital's local server.
RESTful APIs	Facilitates stateless, resource-oriented HTTP communication, acting as the standard data bridge across the IoT scanners, backend server, and frontend dashboard.
JSON Web Tokens (JWT)	Strictly employed to encrypt session management and secure all API endpoints against unauthorized access by unauthenticated personnel.

Table 5. Technology Requirements

<u>Software Requirements</u>

Software	Requirement
Hyperledger Fabric	Version 2.5.x (LTS)
Node.js	Version 18.x (LTS)
Hyperledger Fabric SDK (Node.js)	Version 2.5.x
React	Version 18.x or higher
PostgreSQL	Version 15.x or higher

Docker Engine	Version 24.x or higher
Docker Compose	Version 2.20.x or higher
Git	Version 2.30.x or higher
Modern Web Browser	Google Chrome (Version 115+) or Microsoft Edge (Version 115+)

Table 6. Software Requirements

Hardware Requirements

Hardware	Specifications
IoT Edge Device (Scanner)	2D CMOS Barcode Scanner (e.g., Zebra DS series) with USB 2.0/3.0 or Bluetooth 4.0+ HID connectivity.
Decoding Capability	ISBT-128 Omnidirectional Support (firmware must be capable of decoding global standard blood bag labels).
Host Terminal (Hospital Node)	Standard PC Workstation, Minimum 4GB RAM, running Windows 10/11 or macOS.
Development Computer	Windows 10/11 or macOS, Minimum 16GB RAM, Intel Core i5 / Ryzen 5 or higher (Required for simultaneously running Docker, PostgreSQL, and Hyperledger Fabric nodes locally).
Network Connectivity	Standard hospital Intranet/LAN via Ethernet or Built-in WiFi for local database buffering, with internet access for blockchain ledger synchronization.

Table 7. Hardware Requirements

Development Tools

Tools	Functional Application
Visual Studio Code	The primary Integrated Development Environment (IDE) for code authorship. It will provide syntax highlighting, intelligent code completion, and integrated terminal support across the entire Node.js and React stack.
Figma	For rapid UI/UX prototyping and wireframing. It will allow the team to visually design and refine the real-time consortium dashboard interfaces before committing to frontend code.
Postman	For API endpoint testing and validation. It will allow developers to construct, inspect, and organize HTTP requests into automated test collections prior to frontend or IoT integration.
Git	The distributed version control system. It will track the full change history, support concurrent developer branching, and maintain a strict, auditable commit log per author and timestamp.
DBeaver	A universal database management GUI that will be used to visually inspect, query, and manage the local PostgreSQL off-chain buffer, ensuring data is captured correctly before blockchain synchronization.
Jest	A JavaScript testing framework that will be utilized to write and execute automated unit tests for both the React frontend and the Node.js middleware, ensuring the BROA and RPS algorithms compute accurately.
Hyperledger Explorer	A blockchain graphical user interface (GUI) that will be deployed locally to monitor network health, inspect ledger blocks, and verify that smart contract transactions are successfully committing to the chain.

Table 8. Development Tools

Planning Phase: Sprint 0a from June 1 – June 14, 2026 (2 Weeks)

At the beginning of the project, the Planning phase, designated as Sprint 0a and scheduled for June 1 through June 14, 2026, will establish the scope and resource baseline for BloodLedger development. During this two-week period, the development team will conduct a comprehensive inventory of all technology, software, hardware, and tooling requirements necessary to build and deploy the system, as enumerated in the tables above. The complete technology stack will be finalized, and hardware procurement planning covers the selection of 2D CMOS barcode scanners with ISBT-128 decoding capability and the configuration of host terminals at each participating hospital node. Stakeholder identification will be completed through interviews and surveys with blood bank heads, medical technologists, nurses, and PRC Lipa personnel, producing the primary data that supports all subsequent development decisions. This phase will conclude with the delivery of a finalized system requirements document, a validated technology stack specification, a stakeholder register, and a comprehensive Sprint Backlog, a prioritized list of functional and non-functional requirements that will govern all five subsequent development Sprints.

Design Phase: Sprint 0b from June 15 – June 28, 2026 (2 Weeks)

Building upon the established requirements, the Design phase, designated as Sprint 0b and scheduled for June 15 through June 28, 2026, will translate requirements into technical blueprints before any code is written. The consortium network topology and Hyperledger Fabric channel configuration will be defined with all stakeholder institutions. The database schema for the off-chain

PostgreSQL buffer will be formalized across its six core tables: blood_units, hospitals, transfers, users, audit_logs, and notifications. The RESTful API contract between the Node.js middleware and the React dashboard will be fully documented, specifying all endpoint routes, HTTP methods, and JSON payload structures. The smart contract endorsement policy and the sequential state machine for blood unit lifecycle management are specified in detail, defining the permissible state transitions: Available → Reserved → InTransit → Received / Expired. Concurrently, the system's security architecture will be formalized during this phase, encompassing the MSP-governed network access model, the RBAC tier structure across the four user roles, and the PHI data isolation policy stipulating that no patient records, names, or diagnosis histories shall be stored within the system, ensuring structural compliance with RA 10173 and RA 7719 before implementation begins. UI/UX wireframes for all major dashboard screens such as the city-wide inventory matrix, transfer request panel, expiry alert center, and regulatory reporting interface will be prototyped in Figma and reviewed with blood bank representatives for usability validation. The primary deliverables of Sprint 0b include the finalized system architecture diagram, database schema, API specification document, smart contract design, security architecture specification, and validated UI wireframes.

Development Phase: Sprints 1–5 from June 29 – September 6, 2026 (10 Weeks)

With all technical blueprints finalized in Sprint 0, the Development phase will gradually deliver the BloodLedger system across five two-week Sprints spanning June 29 through September 6, 2026. In accordance with the Scrum framework, each Sprint is not a pure coding interval but a complete iterative cycle: it opens with a Sprint Planning session that selects and scopes the backlog items for that Sprint; progresses through active Development of the Sprint's deliverables; runs sprint-level unit and integration tests in Jest and Docker against every component built during that Sprint; and closes with a Sprint Review and Retrospective in which working software is demonstrated to designated hospital stakeholders and lessons learned are logged for the next Sprint. Any defects surfaced during sprint-level testing are remediated within the same Sprint before the review. Formal system-wide validation — comprising System Testing and UAT with actual blood bank personnel — is a distinct, dedicated phase that follows after all five Sprints are complete. Each Sprint corresponds to one of the system's five core architectural layers, as detailed below.

Sprint 1: Infrastructure Provisioning from June 29 – July 12, 2026 (2 Weeks)

Sprint 1 focuses on infrastructure provisioning. The Hyperledger Fabric consortium network will be set up using Docker Compose, configuring one peer node per participating hospital, a RAFT-based ordering service, and a single shared consortium channel. The Membership Service Provider (MSP) will be initialized to issue X.509 digital certificates to each institution, establishing the cryptographic identity infrastructure required for permissioned access. The off-chain PostgreSQL database will be containerized alongside the network, and schema migrations for all six core tables are applied and validated using DBeaver. The Sprint Review at the close of Week 2 will demonstrate a live, multi-node Fabric network with verified peer connectivity and a correctly seeded PostgreSQL schema to the advisory stakeholders.

Sprint 2: Core Chaincode Modules from July 13 – July 26, 2026 (2 Weeks)

Sprint 2 delivers the core chaincode modules. The Inventory Chaincode will be developed in Node.js and deployed to the Fabric network, implementing the CreateBloodUnit and UpdateStatus functions along with the full blood unit lifecycle state machine (Available → Reserved → InTransit → Received / Expired). The Expiry Monitor Chaincode, developed concurrently, introduces the automated threshold detection logic that flags blood units approaching or exceeding the component-specific expiry window and invokes the BROA redistribution pipeline. All chaincode functions will be written with deterministic logic to satisfy Fabric's endorsement requirements. Sprint-level unit tests in Jest will validate every chaincode function against valid inputs, boundary conditions (e.g., a unit expiring exactly on the threshold day), and invalid inputs (e.g., duplicate unit IDs). The Sprint Review will demonstrate successful chaincode deployment, a verified state machine on the live Fabric test network, and passing Jest unit test results.

Sprint 3: Transfer Chaincode & Algorithm Implementations from July 27 – August 9, 2026 (2 Weeks)

Sprint 3 delivers the Transfer Chaincode and algorithm implementations. The Transfer Chaincode will implement the InitiateTransfer, DispatchWithGeoTag, ReceiveWithGeoTag, and ValidateSignatures functions, enforcing the dual-signature protocol that permanently closes each inter-hospital custody transfer on the ledger. The BROA two-tier pipeline, FEFO unit selection followed by SAW-based destination scoring, will be encoded into the chaincode and validated against simulated hospital transfer scenarios. The RPS mechanism will be implemented within the transfer request workflow to rank concurrent requisitions by composite clinical urgency and capped wait time scores. Sprint-level smart contract tests will deploy the Transfer Chaincode onto a local Fabric network to verify that transfer state transitions, FEFO enforcement, dual-signature finalization, and automated expiry triggers behave correctly under live network conditions, a critical gate, given that deployed chaincode is immutable. The Sprint Review will demonstrate a complete simulated inter-hospital transfer recorded immutably on the ledger, with BROA and RPS scoring verified against test scenarios.

Sprint 4: IoT Scan Ingestion Pipeline & Node.js Middleware from August 10 – August 23, 2026 (2 Weeks)

Sprint 4 delivers the IoT scan ingestion pipeline and Node.js middleware. The scan parsing module will be developed to decode ISBT-128 barcode payloads into structured blood unit records and persist them to the local PostgreSQL buffer. The Node.js middleware will integrate the Hyperledger Fabric SDK to translate buffered scan events into signed blockchain transactions and submit them to the consortium channel. Offline buffering and deferred synchronization logic will be implemented to satisfy NFR-05, ensuring no scan events are lost during connectivity interruptions and that all buffered transactions are synchronized upon network restoration. Sprint-level integration tests will be executed within the local Docker Compose environment to verify the full pipeline, from barcode scanner input through PostgreSQL buffering to blockchain ledger commitment, including offline buffering and deferred synchronization under simulated network latency. The Sprint Review will demonstrate a live scan-to-ledger pipeline, including a successful offline-then-sync scenario.

Sprint 5: React Dashboard & RESTful API Layer from August 24 – September 6, 2026 (2 Weeks)

Sprint 5 delivers the React dashboard and the complete RESTful API layer. The Node.js backend will expose endpoints for inventory queries, transfer request submission, and alert retrieval, with all routes protected by JWT-based session authentication. The React dashboard will be built as a component-based application rendering the real-time city-wide inventory matrix, the inter-hospital transfer management panel, the expiry alert center, and the regulatory compliance reporting interface. Role-based rendering logic enforces the four access tiers: Blood Bank Representative, Blood Bank Head, DOH/PRC Regulator, and System Administrator. All API endpoints will be validated using Postman prior to frontend integration, and blockchain network health is monitored throughout using Hyperledger Explorer. The Sprint Review, the final development Sprint demo, will present the fully integrated BloodLedger prototype to hospital stakeholders, with all six functional modules operating end-to-end in the Docker environment.

Testing Phase: Formal Validation from September 7 – September 27, 2026 (3 Weeks)

Distinct from the sprint-embedded unit, integration, and smart contract testing conducted continuously within each Sprint, the formal Testing phase, scheduled for September 7 through September 27, 2026, validates the fully integrated BloodLedger system as a complete, working whole. This phase escalates through two dedicated validation levels. System Testing (Weeks 1–2, September 7–20) executes full end-to-end functional scenarios across all six modules — Blood Bag Scan and Ingest, Inter-Hospital Transfer Request and Approval, Proactive Expiry Alert and BROA Redistribution, Geo-Tag Verification, Offline Sync and Resilience, and Role-Based Access Control, against the simulated consortium environment in Docker, maintaining direct traceability to each functional requirement. Any defect surfaced during System Testing triggers a defect log entry and returns the affected component to the development team for remediation before the UAT gate proceeds. User Acceptance Testing (Week 3, September 21–27) is then conducted with actual blood bank heads, medical technologists, and laboratory staff from Mary Mediatrix Medical Center, Medix Medical Hospital, N.L. Villa Memorial Medical Center, and the Philippine Red Cross Lipa Chapter. UAT respondents interact with the BloodLedger prototype through guided workflows and complete a five-point Likert-scale survey evaluating Functionality, Usability, Reliability, Security, and Performance. Qualitative feedback gathered during UAT sessions is documented and used to inform final system refinements prior to deployment.

Deployment Phase: Parallel Implementation from September 28 – October 11, 2026 (2 Weeks)

To mitigate operational risks and ensure zero disruption to life-saving healthcare services, the deployment of BloodLedger, commencing September 28, 2026, will utilize a phased, parallel implementation strategy. Rather than executing a sudden network-wide transition, the rollout will be divided into strategic tiers. The onboarding process will begin exclusively with primary blood bank nodes — Mary Mediatrix Medical Center, Medix Medical Hospital, and N.L. Villa Memorial Medical Center — to establish a stable, secure core network. Only after this foundational network achieves consensus and operational stability will the secondary recipient hospitals, which function strictly as consumers, be systematically integrated into the consortium channel.

At the individual facility level, the introduction of the new technology will be governed by a rigorous 3 to 6-month parallel validation period beginning October 12, 2026. During this phase, the blockchain-driven dashboard and IoT barcode scanners will operate concurrently with the hospitals' existing manual logbooks and legacy phone-based protocols. This dual-entry approach serves as a critical safety mechanism, guaranteeing that blood bank representatives can manually cross-reference the automated smart contract states — such as inventory counts, expiry triggers, and inter-hospital transfer statuses — against their trusted physical records. This parallel execution allows system administrators to comprehensively validate data integrity, algorithm accuracy (BROA and RPS), and off-chain database resilience in a live clinical environment before fully deprecating the legacy manual systems.

The strategic deployment timeline is visually represented in the parallel implementation diagram below. The figure demonstrates the overlap between conventional manual operations and the integration of the decentralized BloodLedger network, ensuring a seamless, risk-free transition from reactive procurement to automated, proactive supply chain management.

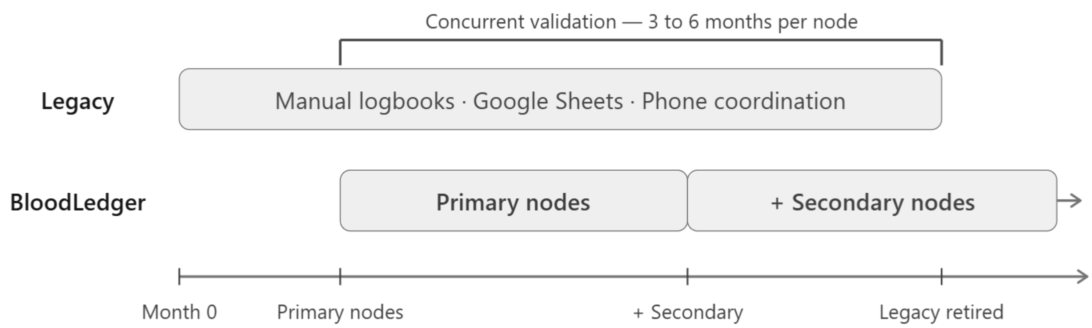

Figure 23. Parallel Implementation Diagram illustrating the concurrent operation of legacy manual logbooks and the BloodLedger system during the validation phase.

<u>Maintenance Phase: Ongoing from October 12, 2026 onwards</u>

The Maintenance phase, commencing October 12, 2026 following the completion of the initial parallel deployment validation, addresses post-deployment system health through three primary operational activities. First, chaincode upgrades are managed using Hyperledger Fabric's built-in chaincode versioning mechanism, which allows updated business logic, such as revised BROA weighting criteria, expanded blood component classifications, or adjusted RPS urgency thresholds, to be deployed to the consortium network without disrupting active ledger operations or invalidating historical transaction records. Second, network expansion procedures are executed to onboard additional hospitals in Lipa City beyond the initial consortium, following the same phased parallel implementation strategy established during deployment. Third, ongoing performance monitoring is maintained through Hyperledger Explorer dashboards and system administrator access to application logs, ensuring that transaction throughput, synchronization latency, and offline buffer health remain within the non-functional performance thresholds defined in the system requirements. Node-level maintenance is the responsibility of each participating hospital's IT personnel, while the development team retains responsibility for consortium-level chaincode governance and network configuration updates.

Software Tests

Software testing for BloodLedger is conducted across five hierarchical levels—Unit, Integration, Smart Contract, System, and User Acceptance Testing (UAT)—to verify that each functional requirement is correctly implemented and that the system performs reliably across the consortium's distributed architecture.

Unit Testing

Unit testing validates individual system components in complete isolation from external dependencies. For BloodLedger, unit tests target three primary categories of isolated logic: individual chaincode functions (e.g., CreateBloodUnit, UpdateStatus, DispatchWithGeoTag), RESTful API endpoint handlers, and the IoT scan parsing module responsible for decoding ISBT-128 barcodes into structured blood unit records.

All unit tests are implemented using Jest, a JavaScript testing framework, and executed within a mock environment that simulates Hyperledger Fabric peer responses using stubs. Each chaincode function is tested against valid inputs, boundary conditions (e.g., a unit expiring exactly on the threshold day), and invalid inputs (e.g., duplicate unit IDs, malformed barcodes) to confirm deterministic behavior. API handler tests verify that correct HTTP status codes and JSON response structures are returned for both successful and error-condition requests.

Integration Testing

Integration testing validates the end-to-end data pipeline that spans from the IoT scan device at the hospital edge to the distributed blockchain ledger. The pipeline under test covers four consecutive layers: (1) the barcode scanner's USB/Bluetooth input to the scan parsing module, (2) the parsing module's write operation to the local PostgreSQL buffer, (3) the Node.js middleware's submission of the buffered event to the Hyperledger Fabric SDK, and (4) the blockchain ledger's successful commitment of the transaction and propagation to consortium peers.

Integration tests are executed using a local Docker Compose environment that instantiates a two-peer Hyperledger Fabric test network alongside a PostgreSQL container. Test scenarios include normal pipeline execution under stable connectivity, pipeline behavior under simulated network latency, and offline buffering with deferred synchronization. The primary integration test criteria require that no scan events are lost during connectivity interruptions and that the sequence and content of blockchain-committed transactions exactly match the locally buffered events.

Smart Contract Testing

The smart contract testing phase validates the autonomous chaincode logic (Inventory lifecycles, BROA, and RPS) deployed on Hyperledger Fabric. Unlike unit tests that isolate functions in a mock environment, this phase deploys chaincode onto a local Fabric network to verify ledger state transitions, enforcement rules, and alerts under real conditions. This is critical because smart contracts are immutable post-commit; any defect missed cannot be corrected without redeploying an entirely new version, making this testing level the final safety net for rejecting invalid or fraudulent transactions.

Two categories are tested: First, transfer state machine transitions (Available → Reserved → InTransit → Received → Expired), ensuring only authorized actors trigger valid steps, out-of-sequence transitions are rejected, and dual cryptographic signatures finalize transfers. Second, expiry trigger logic, where the Expiry Monitor automatically generates near-expiry alerts, invokes BROA for redistribution, and commits an Expired status to the immutable ledger once component-specific thresholds (e.g., 35 days for RBCs) are exceeded—eliminating the manual oversight gaps documented in current hospital workflows.

System Testing

System testing validates BloodLedger as a complete, integrated whole by executing end-to-end functional scenarios across all consortium nodes simulated in the local Docker environment. Each test scenario is derived directly from a specific functional requirement to maintain full requirements traceability. System testing covers the six functional modules of BloodLedger: Blood Bag Scan and Ingest, Inter-Hospital Transfer Request and Approval, Proactive Expiry Alert and BROA Redistribution, Geo-Tag Verification, Offline Sync and Resilience, and Role-Based Access Control.

All test cases constitute the BloodLedger system test suite. Pass criteria require that the system's observed behavior matches the expected output as specified in each test case. Any deviation triggers a defect log entry, and the affected component is returned to the development phase for remediation before the next test cycle.

User Acceptance Testing (UAT)

User Acceptance Testing is the final validation phase, conducted with actual healthcare professionals from the participating consortium institutions. UAT respondents are purposely selected from blood bank heads, medical technologists, and laboratory staff at Mary Mediatrix Medical Center, Medix Medical Hospital, N.L. Villa Memorial Medical Center, and the Philippine Red

Cross Lipa Chapter, the same individuals whose operational realities informed the system's requirements.

During UAT, respondents interact with the BloodLedger prototype in a controlled demonstration environment pre-loaded with simulated blood inventory data. Each participant is guided through key workflows: scanning a blood bag, checking the real-time dashboard, submitting a blood requisition, reviewing an expiry alert, and inspecting an audit log entry. Following the demonstration, participants complete the UAT survey, rating the system's Functionality, Usability, Reliability, Security, and Performance using a five-point Likert scale.

UAT results are analyzed using the weighted mean method. Feedback gathered during UAT sessions, including qualitative comments from participants, is documented and used to inform final system refinements before the parallel deployment phase.

Test Case Tables

The following tables document the complete system test suite for BloodLedger. Each test case specifies an identifier, description, input conditions, expected output, and a placeholder Result column to be populated during actual test execution.

Test ID	Test Case Description	Input / Precondition	Expected Output	Result

TC-S01	Valid barcode scan registers blood unit	ISBT-128 barcode scanned; unit not previously registered	Blood unit record created in local DB with blood type, component, expiry date, and timestamp; dashboard updates within 5 seconds	**Pass**
TC-S02	Duplicate scan rejected	Same unit ID scanned twice	System returns error: 'Unit already registered'; no duplicate record created	**Pass**
TC-S03	FEFO violation rejected at dispatch	Lab staff scans a newer unit for dispatch when an earlier-expiry unit of same blood type exists	Smart contract rejects transaction; dashboard alert indicates FEFO violation with earliest-expiry unit highlighted	**Pass**
TC-S04	Scan with malformed barcode	Barcode does not conform to ISBT-128 format	System returns parsing error; no record written to database	**Pass**
TC-S05	Offline scan buffered and synced	Scanner operates while LAN internet is disconnected	Scan event stored in local PostgreSQL buffer; upon reconnection, event syncs to blockchain ledger; dashboard reflects update	**Pass**

Table 9. Blood Bag Scan and Ingest Test Cases

Test ID	Test Case Description	Input / Precondition	Expected Output	Result
TC-T01	Blood requisition submitted with Clinical Urgency = Critical	Authorized staff submits requisition for 2 units O+, urgency = Critical	Requisition logged on blockchain; RPS score computed; transfer request appears in supplier hospital's Pending Requests queue	**Pass**

TC-T02	RPS prioritization of concurrent requests	Two hospitals submit requisitions simultaneously for same blood type; Hospital A: Urgent; Hospital B: Critical	Hospital B's requisition is ranked first in the supplier's queue based on higher RPS score	**Pass**
TC-T03	Transfer approved and unit status updated to Reserved	Supplier hospital admin approves requisition	Blood unit status updates to Reserved on consortium dashboard; unit no longer available for other requests	**Pass**
TC-T04	Dispatch scan triggers In Transit state with geo-tag	Staff scans blood bag at dispatch point	Unit status updates to In Transit; GPS coordinates and timestamp recorded on blockchain; all consortium nodes reflect new status	**Pass**
TC-T05	Receipt scan closes transfer loop	Staff at receiving hospital scans incoming blood bag	Unit status updates to Received; supplier inventory decremented; receiver inventory incremented; transfer event permanently logged on ledger	**Pass**
TC-T06	Unauthorized transfer approval attempt	Medical Technologist (non-admin) attempts to approve a transfer request	System returns authorization error; request remains pending; no state change occurs	**Pass**

Table 10. Inter-Hospital Transfer Request and Approval Test Cases

Test ID	Test Case Description	Input / Precondition	Expected Output	Result
TC-E01	Near-expiry alert triggered at threshold	Blood unit age reaches 35 days as evaluated by expiry	Smart contract flags unit as near-expiry; alert displayed in	**Pass**

		monitor chaincode	dashboard Alert Center with unit ID, blood type, hospital, and days remaining	
TC-E02	BROA generates redistribution recommendation	Near-expiry unit flagged; consortium network has eligible recipient hospitals	BROA executes FEFO selection + SAW scoring; hospital with highest composite score receives redistribution recommendation on its dashboard	**Pass**
TC-E03	BROA SAW scoring — urgency-weighted	Recipient Hospital A has stock at minimum threshold; Recipient Hospital B has adequate stock; both equidistant from supplier	Hospital A receives higher SAW score (higher Urgency weight); redistribution recommendation directed to Hospital A	**Pass**
TC-E04	Expiry alert displayed for all consortium nodes	Unit flagged as near-expiry at Mary Mediatrix	Alert visible on dashboards of PRC Lipa, Medix, and N.L. Villa; secondary node dashboards show city-wide alert status	**Pass**
TC-E05	Expired unit status updated automatically	Blood unit age exceeds component-specific expiry limit (e.g., 35 days for PRBC, 5 days for platelets)	Smart contract transitions unit status to Expired; unit removed from available inventory; event logged immutably on ledger	**Pass**

Table 11. Proactive Expiry Alert and BROA Redistribution Test Cases

Test ID	Test Case Description	Input / Precondition	Expected Output	Result
TC-G01	Dispatch geo-tag captured and	Dispatch scan performed with	Latitude and longitude coordinates appended	**Pass**

	stored	device location services enabled	to dispatch transaction; coordinates stored immutably on blockchain alongside unit ID and timestamp	
TC-G02	Receipt geo-tag captured and stored	Receipt scan performed at receiving hospital	Latitude and longitude coordinates of receiving hospital appended to receipt transaction; dual geo-tag chain-of-custody record complete on ledger	**Pass**
TC-G03	Geo-tag retrieval in audit log	Authorized admin queries audit log for a completed transfer	Audit log displays dispatch coordinates, receipt coordinates, timestamps, and unit ID in chronological order	**Pass**
TC-G04	Geo-tag present without GPS signal (fallback)	Device does not have active GPS lock at time of scan	System uses pre-registered hospital facility coordinates as fixed-location fallback; fallback flag noted in transaction record	**Pass**

Table 12. Geo-Tag Verification Test Cases

Test ID	Test Case Description	Input / Precondition	Expected Output	Result
TC-O01	Scan events captured during full internet outage	LAN internet disconnected; barcode scanner continues operation	Scan events written to local PostgreSQL buffer without error; no data loss; system indicates offline mode to user	**Pass**
TC-O02	Buffered events synchronized upon reconnection	Internet connectivity restored after period of offline buffering	All buffered scan events submitted to Hyperledger Fabric	**Pass**

			ledger in sequence; blockchain ledger reflects all events with original timestamps	
TC-O03	Dashboard reflects offline status indicator	Blockchain sync unavailable; local DB operational	Dashboard displays 'Offline Mode — Local Data Only' indicator; local inventory data remains accessible and accurate	**Pass**
TC-O04	No data duplication upon sync	Hospital buffered 15 scan events offline; sync triggered upon reconnection	Exactly 15 new events appear on blockchain; no duplicates; sequence integrity maintained	**Pass**

Table 13. Offline Sync and Resilience Test Cases

Test ID	Role Tested	Action Attempted	Expected Output	Result
TC-R01	Medical Technologist	Scan blood bag inbound receipt	Access granted; scan registered successfully; inventory updated	**Pass**
TC-R02	Medical Technologist	Approve inter-hospital transfer request	Access denied; error message: 'Insufficient privileges for transfer approval'	**Pass**
TC-R03	Hospital Admin	Approve inter-hospital transfer request	Access granted; transfer moved to Approved state; smart contract executed	**Pass**
TC-R04	Hospital Admin	View inventory of another hospital's blood bank	Access denied to private inventory; only city-wide aggregate view available	**Pass**
TC-R05	DOH / PRC Lipa Regulator	View city-wide inventory dashboard	Access granted; read-only dashboard displayed with all consortium data	**Pass**

TC-R06	DOH / PRC Lipa Regulator	Initiate a blood transfer request	Access denied; regulatory role is read-only; no transactional functions available	**Pass**
TC-R07	Unauthenticated user	Access any dashboard endpoint without valid JWT	All requests rejected with HTTP 401 Unauthorized; no data exposed	**Pass**

Table 14. Role-Based Access Control Test Cases

Data Analysis

This section describes the survey instrument, respondent profile, statistical treatment, and evaluation criteria used to analyze the User Acceptance Testing (UAT) data collected from healthcare personnel who interacted with the BloodLedger prototype.

The primary data collection instrument for evaluating BloodLedger is a structured UAT survey administered to healthcare professionals directly involved in blood bank and transfusion operations at the participating consortium institutions. The survey is modeled on the ISO 9126 software quality characteristics—Functionality, Usability, Reliability, Security, and Performance—adapted to reflect the specific clinical workflows validated during data gathering.

Respondents are purposively selected based on their direct involvement in blood inventory management, blood request coordination, or transfusion administration. Eligible participants include blood bank heads, medical technologists, laboratory staff, and infection control nurses from Mary Mediatrix Medical Center, Medix Medical Hospital, N.L. Villa Memorial Medical Center, and the Philippine Red Cross Lipa Chapter. This selection strategy ensures that evaluation feedback originates from individuals whose operational realities are directly reflected in the system's functional requirements.

To ensure adequate coverage of all five evaluation dimensions, a minimum of one qualified respondent per participating institution is targeted, with additional respondents sought from larger blood bank teams. Participation is voluntary, and respondents complete the instrument after interacting with the BloodLedger prototype during a structured demonstration session, ensuring that all ratings are grounded in direct system experience rather than speculative assessment.

Statistical Treatment

The UAT survey employs a five-point Likert scale for all rating items. This scale quantifies respondents' level of agreement with descriptive statements about each system feature across the five evaluation criteria areas. The numeric values and their corresponding verbal interpretations are defined in Table 15.

Scale Value	Verbal Interpretation	Description
5	Strongly Agree	The feature performs exactly as needed; fully meets the operational requirements of blood bank personnel in a clinical environment.

4	Agree	The feature largely meets operational requirements with only minor limitations that do not significantly impact clinical workflows.
3	Neutral	The feature is acceptable but has notable areas for improvement; respondent neither endorses nor rejects its suitability.
2	Disagree	The feature does not adequately meet operational requirements; significant improvements are needed before clinical adoption.
1	Strongly Disagree	The feature fails to meet operational requirements; the respondent would not use or recommend the feature in its current state.

Table 15. Likert Scale for UAT Survey Responses

The primary statistical measure applied to the collected ratings is the Weighted Mean ($\bar{x}$), which computes the average response score for each evaluation item and each criteria area by accounting for the frequency of each Likert response. The Weighted Mean formula is:

$$\textbf{Weighted Mean } (\bar{x}) = \Sigma(f_i \times x_i) / N$$

Where f_i represents the frequency of respondents who selected scale value x_i for a given item, and N is the total number of respondents. The Weighted Mean for each individual survey item is computed first; the area-level Weighted Mean is then calculated as the arithmetic mean of all item-level weighted means within the criteria area. This two-level aggregation ensures that no single item disproportionately influences the area score.

The resulting weighted mean scores are interpreted against the five-range scale presented in Table 16. This interpretation table translates the numeric score into a verbal descriptor and its corresponding implication for system readiness.

Weighted Mean Range	Verbal Interpretation	Implication for System Adoption
4.51 – 5.00	Strongly Agree	The system fully meets the operational needs of blood bank personnel; recommended for deployment without modification.
3.51 – 4.50	Agree	The system adequately meets operational needs; minor refinements recommended prior to parallel deployment.
2.51 – 3.50	Neutral	The system partially meets needs; moderate revisions required; further validation cycles recommended.
1.51 – 2.50	Disagree	The system does not adequately meet operational needs; significant redesign required before deployment.
1.00 – 1.50	Strongly Disagree	The system fails to meet operational requirements; fundamental rearchitecting required.

Table 16. Interpretation of Weighted Mean for BloodLedger UAT

A system is considered to have achieved acceptable performance across a criteria area if its area-level weighted mean score falls within the Agree range (3.51–4.50) or higher. Criteria areas scoring below 3.51 are flagged for targeted refinement and a subsequent validation cycle before the system proceeds to the parallel deployment phase.

Evaluation Criteria Areas and Survey Items

The UAT survey is organized into five evaluation criteria areas, each containing two to three specific assessment items. Table 17 presents all evaluation criteria, their corresponding sub-items, and sample survey statements as they appear in the UAT instrument. Each statement is rated on the five-point Likert scale.

Criteria Area	Sub-Item	Sample Survey Statement
Functionality	Blood bag scan accuracy	The barcode scan correctly records the blood type, component, and expiration date of each blood unit.
	Transfer request and approval flow	The digital blood requisition process accurately communicates my request to the supplying hospital without errors.
	Proactive expiry alert accuracy	The expiry alert correctly identifies blood units approaching the end of their shelf life and recommends redistribution in a timely manner.
Usability	Ease of scan and ingest operation	Scanning and recording a blood bag entry or exit is straightforward and requires minimal training.
	Dashboard clarity	The real-time inventory dashboard presents blood stock information in a clear and immediately understandable format.
	Alert center legibility	Expiry warnings and transfer notifications are prominently displayed and easy to distinguish from routine information.
Reliability	Offline scan buffering	The system reliably captures and stores scan data during internet connectivity loss without any loss of records.
	Data consistency across nodes	The inventory information displayed on my hospital's dashboard is consistent with the

		information visible to other consortium hospitals.
Security	Data privacy confidence	I am confident that patient information is not exposed through the BloodLedger system.
	Access control effectiveness	The system correctly restricts access to functions beyond my assigned role (e.g., I cannot approve transfers if I am not an authorized administrator).
Performance	Dashboard response speed	The inventory dashboard updates quickly after a scan event, allowing me to act on current stock data without noticeable delays.
	Emergency scenario responsiveness	During simulated emergency scenarios, the system provided blood availability information and requisition processing at a speed adequate for clinical decision-making.

Table 17. BloodLedger UAT Evaluation Criteria, Sub-Items, and Sample Survey Statements

The Functionality criteria area evaluates whether BloodLedger's core operational features—barcode scan accuracy, transfer request processing, and proactive expiry alerts—perform correctly from the perspective of clinical end-users. This directly tests whether the system fulfills the functional requirements FR-01 through FR-09 in practice.

The Usability criteria area assesses whether the system is accessible to non-technical hospital staff. Given that medical technologists and laboratory personnel are the primary daily users of the scan module and dashboard, usability ratings are critical indicators of whether the system can achieve practical adoption without extensive training overhead.

The Reliability criteria area examines the system's behavior under conditions of connectivity loss, a realistic operational scenario in healthcare facilities with variable network infrastructure. Respondents evaluate whether the offline buffering mechanism operates transparently and whether the data they observe is consistent with what is visible to other institutions in the consortium.

The Security criteria area captures respondents' confidence in the system's privacy controls and access management. Because BloodLedger involves the sharing of institutional inventory data across multiple organizations, user confidence in data isolation and role enforcement is essential for institutional trust and voluntary adoption.

The Performance criteria area evaluates whether the system's response speed meets the temporal demands of clinical decision-making. Blood transfusion scenarios are time-critical, and any perceptible delay between a scan event and a dashboard update, or between a requisition submission and a supplier notification, directly impacts the clinical utility of the system.

References

Access Partnership. (2025). *Privacy's hidden price tag: The cost of siloed data in healthcare.* https://accesspartnership.com/opinion/privacys-hidden-price-tag-cost-siloed-data-healthcare/

Agrawal, T. K., Angelis, J., Khilji, W. A., Kalaiarasan, R., & Wiktorsson, M. (2022). Demonstration of a blockchain-based framework using smart contracts for supply chain collaboration. International Journal of Production Research, 61(5), 1497–1516. https://doi.org/10.1080/00207543.2022.2039413

Aguila, E. M., Actub, S. D. D., Adalia, E. M. C., Aguirre, N. N. M., Agulto, A. G. B. F., Alfaro, A. D. M. T., Alvaro, M. R. G., Apiado, L. D. C., & Calacday, C. R. D. (2025). Implementation challenges of smarter and integrated local health information system. Journal of Interdisciplinary Perspectives, 3(10), 285-300. https://doi.org/10.69569/jip.2025.607

Ajayi, et al. (2024). Enhancing blood supply chain management with blockchain technology to improve diagnostic precision and strengthen health information security. International Journal of Innovative Science and Research Technology (IJISRT), 2025. https://ijisrt.com/assets/upload/files/IJISRT25APR214.pdf

Alghamdi, S. Y. (2023). A Review of Blood Delivery for Sustainable Supply Chain Management (BSCM). Sustainability, 15(3), 2757. https://doi.org/10.3390/su15032757

Alqarni, M. A., Alkatheiri, M. S., Chauhdary, S. H., & Saleem, S. (2023). Use of Blockchain-Based smart contracts in logistics and supply chains. Electronics, 12(6), 1340. https://doi.org/10.3390/electronics12061340

American Association of Blood Banks (AABB). (2024). Whole blood and red blood cell components. Regulatory Affairs. https://www.aabb.org/

Asdecker, B., & Felch, V. (2026). New Possibilities, New Responsibilities: Ethical Challenges and Guidance for Location Tracking Device-based Research in Supply Chains.

Atienza, H. M. (2026, January 30). Interview with Blood Bank Section Head [Personal interview]. Mary Mediatrix Medical Center, Lipa City.

Centre for Development of Advanced Computing. (2016). e-RaktKosh: Centralized blood bank management system. Ministry of Health and Family Welfare, Government of India. https://www.eraktkosh.in/

Chafiq, T., Azmi, R., Fadil, A., & Mohammed, O. (2024). Investigating the potential of blockchain technology for geospatial data sharing: Opportunities, challenges, and solutions. Geomatica, 76(2), 100026. https://doi.org/10.1016/j.geomat.2024.100026

Clohessy, T., & Acton, T. (2019). Investigating the influence of organizational factors on blockchain adoption. Industrial Management & Data Systems, 119(7), 1457-1491. https://doi.org/10.1108/IMDS-08-2018-0365

Concha, J. (2026, February 4). Interview with Chief Medical Technologist [Personal interview]. N. L. Villa Memorial Medical Center, Lipa City.

Conduah, A. K., Ofoe, S., & Siaw-Marfo, D. (2025). Data privacy in healthcare: Global challenges and solutions. Digital health, 11, 20552076251343959. https://doi.org/10.1177/20552076251343959

Conjares, M. N. T. M. M., Talamayan, J. T. B., Rey, A. M. T., Wong, J. J. A. V., Pantig, I. M. T., Evangelista, A. L. A., & Ulep, V. G. T. (2025). Strengthening health information systems: Policies, gaps, and a framework for reform (PIDS Discussion Paper Series No. 2025-31). Philippine Institute for Development Studies. https://pids.gov.ph/publication/discussion-papers/strengthening-health-information-systems-policies-gaps-and-a-framework-for-reform

Davis, F. D. (1989). Perceived usefulness, perceived ease of use, and user acceptance of information technology. MIS Quarterly, 13(3), 319-340. https://doi.org/10.2307/249008

De Nerol, I., Kamil, D., Angeline, E., Kharismatunnisaa, F., Nurfadillah, F., Firdayanti, Renaldi, M. A., Darmawan, M. D. M., Octavia, N., Parmana, R.

R., Satria, R. P. S., & Saputra, Y. (2024). IoT-based barcode scanning system for production and warehouse management. Journal of Applied Science, Technology & Humanities, 1(1), 72–86. https://doi.org/10.62535/0zv1h756

Department of Health. (2005). Rules and regulations for the establishment of the Philippine National Blood Services amending pertinent provisions of Admin. Order No. 9, s. 1995 (Administrative Order No. 2005-0002). Supreme Court E-Library. https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/10/50961

Dillon, M., Oliveira, F., & Abbasi, B. (2017). A two-stage stochastic programming model for inventory management in the blood supply chain. International Journal of Production Economics, 187, 27–41. https://doi.org/10.1016/j.ijpe.2017.02.006

Eder, R. J. C. (2024). Philippine Red Cross online blood bank management information system. Central Philippine University Repository. https://repository.cpu.edu.ph/handle/20.500.12852/23

Epic Systems Corporation. (n.d.). About Epic. Retrieved February 25, 2026, from https://www.epic.com/about/

Fishburn, P. C. (1967). Additive utilities with incomplete product sets: Application to priorities and assignments. Operations Research, 15(3), 537–542. https://www.jstor.org/stable/168461

Garcia, H. A. C., Mendoza, G. P., Roxas, J. H. B., Vergara, V. D. D., & Santos, A. S. (2025). CrOptimize: Integrated agri-tech solution for optimized farm lot utilization and resource management. TENCON 2025 - 2025 IEEE Region 10 Conference (TENCON), 317-321. https://doi.org/10.1109/TENCON66050.2025.11374919

Girish, B. J., Tiwari, A., Gupta, M. K., & Vasnani, H. (2025). Barcode technology in healthcare systems enhancing medication safety accuracy efficiency and error reduction. International Journal of Science and Research Archive, 17(3), 633–649. https://doi.org/10.30574/ijsra.2025.17.3.3302

Haemonetics Corporation. (2024). The only blood bank vendor with an Epic global cooperation agreement. Haemonetics Hospital Solutions. https://hospital.haemonetics.com/-/media/files/hospital/transfusion-management/safetrace-tx-and-epic-integration-col-copy-002259-us-aa.pdf

Heeres, T. J., Tran, T. M., & Noort, B. A. (2023). Drivers and Barriers to Implementing the internet of Things in the health care supply Chain: Mixed Methods multicase study. Journal of Medical Internet Research, 25, e48730. https://doi.org/10.2196/48730

Herrera, G. A. R., Florencondia, N., & Aduna, J. R. (2025). Geotagging for transparent infrastructure delivery: An information systems success model assessment of the Philippine Rural Development Project. International Journal of Innovative Science and Research Technology (IJISRT), 10(7), 3349-3354. https://doi.org/10.38124/ijisrt/25jul1085

Hive. (2026, March 31). The #1 project management software for teams | HIVE. https://hive.com/blog/kanban-vs-scrum/

Home Care Service Application for Patients - Scientific Figure on ResearchGate. Available from: https://www.researchgate.net/figure/Agile-Methodology-13_fig1_357548552

Hosseinpour, H., Magnotti, L. J., Bhogadi, S. K., Anand, T., El-Qawaqzeh, K., Ditillo, M., Colosimo, C., Spencer, A., Nelson, A., & Joseph, B. (2023). Time to Whole Blood Transfusion in Hemorrhaging Civilian Trauma Patients: There Is Always Room for Improvement. Journal of the American College of Surgeons, 237(1), 24–34. https://doi.org/10.1097/XCS.0000000000000715

International Council for Commonality in Blood Banking Automation [ICCBBA]. (2020). ISBT 128 standard technical specification (Version 6.0.0). https://www.iccbba.org/

ISBT Working Party on Blood Supply Management. (2023). International survey of blood component wastage rates. ISBT Science Series. https://www.ovid.com/journals/isbt/fulltext/10.1111/voxs.12236

Ismail, R. D., Hussein, H. A., Salih, M. M., Ahmed, M. A., Hameed, Q. A., & Omar, M. B. (2022). The Use of Web Technology and IoT to Contribute to the Management of Blood Banks in Developing Countries. Applied System Innovation, 5(5), 90. https://doi.org/10.3390/asi5050090

Jiménez-Delgado, G. I., Aslan, I., Hernández-Palma, H., Orozco Bohorquez, M., Acosta Ortega, F., Plazas-Alvarado, J., & Morales-Espinosa, R. (2025). Design of a Framework Based on MCDM and Data Analytics for Improving the Inventory Management of Supplies for Clinical Studies: A Case Study in a Research Center of a High Complexity Clinic. In International Conference on Human-Computer Interaction (pp. 74-92). Springer, Cham. https://doi.org/10.1007/978-3-031-76809-5_6

Kandasamy, J., Vimal, K. E. K., Singh, A. P., Magnani, A., Gokhale, A., & Jagtap, S. (2025). Analysis of key challenges to implementation of FEFO in perishable food supply chain. Journal of Agriculture and Food Research, 21, 101848. https://doi.org/10.1016/j.jafr.2025.101848

Khan, S. a. R., Ahmad, Z., Sheikh, A. A., & Yu, Z. (2022). Digital transformation, smart technologies, and eco-innovation are paving the way toward sustainable supply chain performance. Science Progress, 105(4), 368504221145648. https://doi.org/10.1177/00368504221145648

Kotian, M., Raut, T., Malvia, P., Walunj, N., & Pathak, M. (2024). Blood chain: Blockchain based blood registry and supply system. International Research Journal of Engineering and Technology (IRJET), 11(4). https://www.irjet.net/archives/V11/i4/IRJET-V11I4330.pdf

Kunal, S., Gandhi, P., Rathod, D., Amin, R., & Sharma, S. (2024). Securing patient data in the healthcare industry: A blockchain-driven protocol with advanced encryption. Journal of education and health promotion, 13, 94. https://doi.org/10.4103/jehp.jehp_984_23

Kundu, S., Munoz Valencia, A., Virk, S. K., Kumar, N., Gadgil, A., Mammen, J. J., Roy, N., & Raykar, N. (2024). Defining blood deserts and access to blood products for 660 million people: a geospatial analysis of eight states in Northern India. BMJ global health, 9(10), e015637. https://doi.org/10.1136/bmjgh-2024-015637

Lamoriza, M. J. (2026, February 8). Interview with Staff Nurse regarding blood request coordination [Personal interview]. Medix Lipa.

Manila Bulletin. (2024, July 5). Blood donors are lifesavers. https://mb.com.ph/2024/7/5/blood-donors-are-lifesavers

Mardani, A., Jusoh, A., & Zavadskas, E. K. (2015). Fuzzy multiple criteria decision-making techniques and applications — two decades review from 1994 to 2014. Expert Systems with Applications, 42(8), 4126–4148. https://doi.org/10.1016/j.eswa.2015.01.003

Mishra, T., Sainani, D., Kurup, P., Kumar, H., & Khairnar, D. (2024). Blood Bank Management System and Inventory Optimization. International Scientific Journal of Engineering and Management (ISJEM), 3(3), 1-4. https://doi.org/10.55041/ISJEM01349

Mosca, M. R. (2026, February 3). Interview with Staff Nurse regarding blood transfusion protocols [Personal interview]. Medix Lipa.

National Blood Authority. (n.d.). BloodNet for blood products from Lifeblood. Australian Government. https://www.blood.gov.au/blood-products/access-and-ordering/bloodnet-blood-products-lifeblood

National Blood Services Act of 1994, Rep. Act No. 7719 (1994). https://www.officialgazette.gov.ph/1994/05/05/republic-act-no-7719/

Ndlalane, S., Otegbeye, O., & Ezugwu, A. E. (2024). Strategic optimization of blood supply chain management for efficient blood bank operations. CEUR Workshop Proceedings, 3880. https://ceur-ws.org/Vol-3880/paper5.pdf

Pabellano, M. C., Aliwalas, R., & Perez, P. N. (2023). BBMS: Development of a blood bank management system with descriptive analysis for Philippine Red Cross Quezon-Lucena chapter. Manuel S. Enverga University Foundation Research Repository. https://mseuf.edu.ph/research/read/1033

Pangestika, M. W., Aziz, Z. A., & Razali, R. B. (2024). Management blood using optimization algorithm: A state of the art literature review. Proceedings of the 3rd International Conference on Advanced Information Scientific Development (ICAISD 2023), 107-117. https://doi.org/10.5220/0012444600003848

Philippine Red Cross. (2023, June 13). PH Red Cross provides lifesaving blood products to over 500 patients daily from January to May. https://redcross.org.ph/2023/06/13/ph-red-cross-provides-lifesaving-blood-products-to-over-500-patients-daily-from-january-to-may

Prastacos, G. P. (1984). Blood inventory management: An overview of theory and practice. Management Science, 30(7), 777–800. https://doi.org/10.1287/mnsc.30.7.777

Republic Act No. 7719. (1994). An act promoting voluntary blood donation, providing for an adequate supply of safe blood, regulating blood banks, and providing penalties for violation thereof. Official Gazette of the Republic of the Philippines.

Richard, Tukamuhebwa & University III, Kampala International. (2024). Blockchain in Healthcare: Ensuring Data Security and Integrity. Research Output Journal of Public Health and Medicine. 4. 12-17. 10.59298/ROJPHM/2024/421217.

Saad, M., Ali, K., Javed, M. A., Alvi, A. N., & Alfakeeh, A. (2025). Blockchain enabled healthcare supply chain: Review, case study and future opportunities. International Journal of Advanced Computer Science and Applications, 16(8). https://doi.org/10.14569/IJACSA.2025.0160888

SCC Soft Computer. (n.d.). SCC SoftBank. Retrieved February 25, 2026, from https://www.softcomputer.com/products/blood-services-information-systems-suite/scc-softbank/

Sembiring, Irwan & Aji, Bintang & Bayu, Teguh. (2026). Consortium Blockchain Framework for Secure Digital Medical Record Innovation. Aptisi Transactions on Technopreneurship (ATT). 8. 138-151. 10.34306/att.v8i1.777.

Shokouhifar, M., & Goli, A. (2023). Designing a Resilient–Sustainable Supply Chain Network of Age-Differentiated Blood Platelets Using Vertical–Horizontal Transshipment and Grey Wolf Optimizer. International Journal of Environmental Research and Public Health, 20(5), 4078. https://doi.org/10.3390/ijerph20054078

Suganya, R., Sidharth, P. J., Vinston Jose, C., Lighittha, P. R., Sundara Srivathsan, M., & Prithivraj, S. (2026). Tracking the blood supply chain using blockchain technology and intelligent automation: BloodChain++. In Industry 6.0 for Sustainable Supply Chains in Agriculture, Healthcare, and Asset Management (pp. 269-312). IGI Global. https://doi.org/10.4018/979-8-3373-3191-1.ch009

Taj, S., Imran, A. S., Kastrati, Z., Daudpota, S. M., Memon, R. A., & Ahmed, J. (2023). IoT-based supply chain management: A systematic literature review. Internet of Things, 24, 100982. https://doi.org/10.1016/j.iot.2023.100982

Thamrin, A. (2022). Hierarchical cloud-based consortium blockchains for secure and reliable big data storage service in healthcare systems [Master's thesis, University of Massachusetts Dartmouth]. https://doi.org/10.62791/20233

Tinam-isan, M. A. C., & Naga, J. F. (2024). Exploring the landscape of health information systems in the Philippines: A methodical analysis of features and challenges. International Journal of Computing and Digital Systems, 16(1), 225-237. http://dx.doi.org/10.12785/ijcds/160118

Tornatzky, L. G., & Fleischer, M. (1990). The processes of technological innovation. Lexington Books.

Uche, C. L., Ezirim, E. O., Ogbenn, A. A., Chikezie, K., Owojiyugbe, T. O., & Iwuoha, E. C. (2024). Blood usage by speciality in health institutions in Abia State: Documentation, a major challenge. Clinics in Nursing, 3(1), 1-8. https://doi.org/10.31579/2835-8147/061

Vargas, P. M. D. (2025). House Bill No. 1283: An act mandating the interoperability of health information systems, and for other purposes. House of Representatives, 20th Congress of the Republic of the Philippines. https://docs.congress.hrep.online/legisdocs/basic_20/HB01283.pdf

Varghese, A., Thilak, K., & Thomas, S. M. (2024). Technological advancements, digital transformation, and future trends in blood transfusion services. International Journal of Advances in Medicine, 11(2), 147–152. https://doi.org/10.18203/2349-3933.ijam20240368

Vista Angkasa Apartment Management System - Scientific Figure on ResearchGate. Available from: https://www.researchgate.net/figure/Agile-Model-shows-the-Agile-Model-which-includes-the-phases-involves-in-the-methodology_fig1_379456763

Wagner, S. M., & Postel, C. (2022). Using the right digital technologies for supply chain visibility. ISE Magazine, 54(8), 32-37.

World Health Organization. (2023, June). Blood safety and availability. https://www.who.int/news-room/fact-sheets/detail/blood-safety-and-availability

World Health Organization. (2024, December 1). A new system of possibilities: Digitalizing blood bank information in Yemen. WHO Regional Office for the Eastern Mediterranean. https://www.emro.who.int/yemen/news/a-new-system-of-possibilities-digitalizing-blood-bank-information-in-yemen.html

Yang, J., Ali, M. A., Kodera, Y., & Nogami, Y. (2023). A proposal of medical information management system based on consortium blockchain. 2023 International Technical Conference on Circuits/Systems, Computers, and Communications (ITC-CSCC), 1–6. https://doi.org/10.1109/ITC-CSCC58803.2023.10212878

Yap, J. C., & Fedoc-Minguito, J. (2023). Impact of the COVID-19 pandemic on blood supply: A comparative cross-sectional study of the pre-pandemic and pandemic era. Philippine Journal of Pathology, 8(2), 12–17. https://doi.org/10.21141/PJP.2023.11

Yigit, E., & Dag, T. (2024). Improving Supply Chain Management Processes Using Smart Contracts in the Ethereum Network Written in Solidity. Applied Sciences, 14(11), 4738. https://doi.org/10.3390/app14114738

Yli-Huumo, J., Ko, D., Choi, S., Park, S., & Smolander, K. (2016). Where is current research on blockchain technology? A systematic review. PLOS ONE, 11(10), e0163477. https://doi.org/10.1371/journal.pone.0163477

Zuniga, P. C., Carpio, J. J., Tayam, Q. A., Garcia, P. L., Lacsina, A., Calderon, J., Santiago, J. F., & Batangan, D. (2026). Development and implementation of an interoperable local health information exchange platform for the attainment of universal health coverage in the Philippines. Frontiers in Digital Health, 7, 1650273. https://doi.org/10.3389/fdgth.2025.1650273